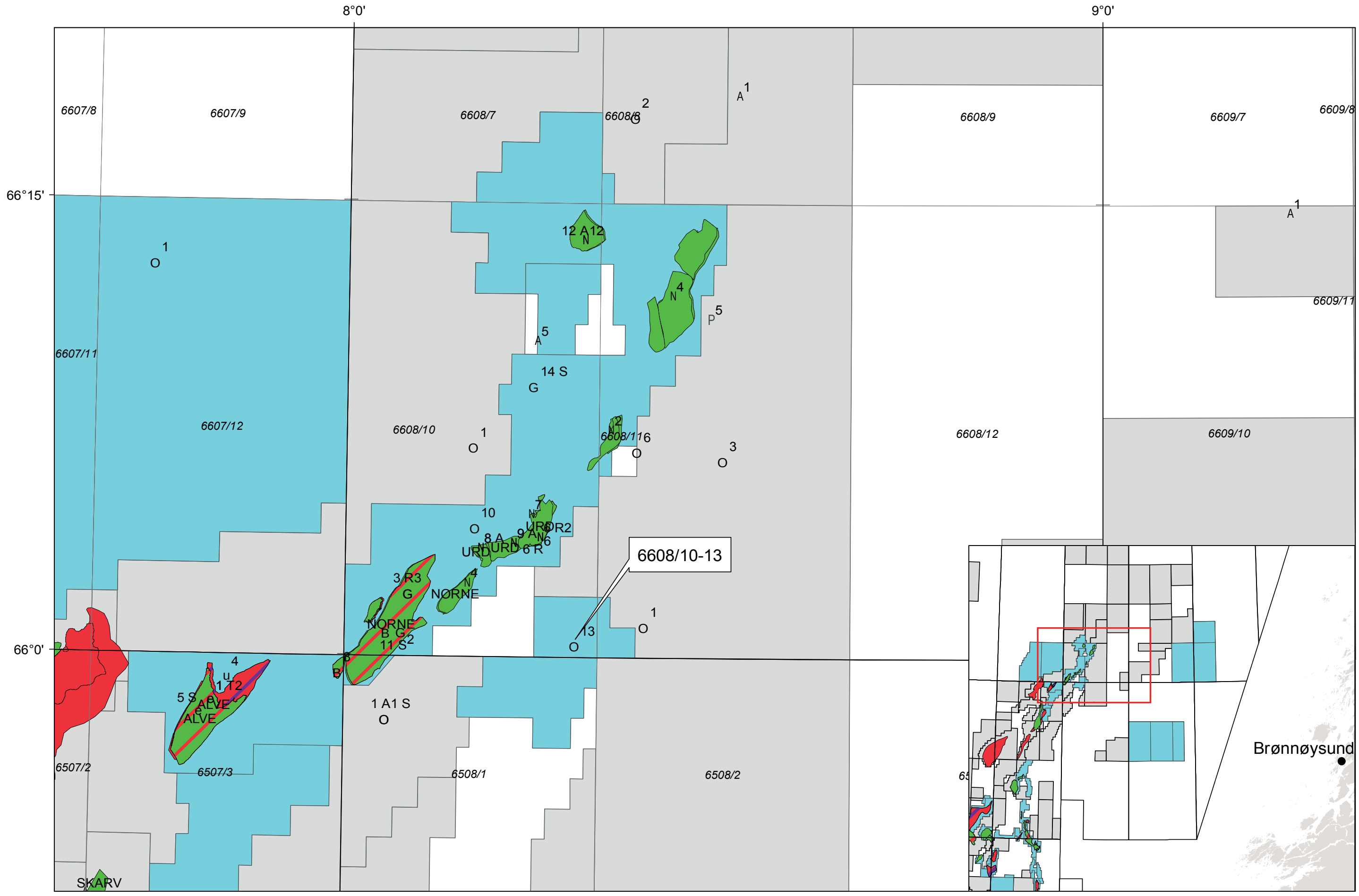
Final Well Report

Well 6608/10-13, Fløien

PL 437



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Statoil

Final Well Report,
Exploration Well 6608/10-13
PL 437

EPDS-6608/10-13-006

Title:
Final Well Report, **Exploration Well 6608/10-13** **PL 437**

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Tamara Kuklina, Anne Marit Løland Larsen, Svein Horn
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Techn. responsible (Organisation unit):	Techn. responsible (Name):	Date/Signature: for Anne Marit
Operation Geologist	**Anne Marit L. Larsen**	4/5-10
Responsible (Organisation unit):	Responsible (Name):	Date/Signature:
GEOP Manager	**Viggo Larsen**	4/5-2010
Recommended (Organisation unit):	Recommended (Name):	Date/Signature:
TNE SST GEOP STA GEO/ **EPN D&W EXNC MO**	**Jørn-Atle Erichsen and Thor Emil Bensvik**	4/5-2010 5/5-2010
Approved by (Organisation unit):	Approved by (Name):	Date/Signature:
EPN EXP ESE	**Dagfinn Alm**	4/5-10

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1 Introduction

1.1 Well data record

Well name	6608/10-13	
Type of well	Exploration	
Prospect	Fløien	
Country	Norway	
Area	Norwegian Sea	
License	PL 437	
Licences	Statoil ASA	100%
Drilling Unit	Ocean Vanguard	
Type	Semi-Submersible	
Water Depth	384 mMSL	
Air Gap	22.0 m	
On licence	12.10.2010	
Rig released	07.11.2010	
TD of the well	Drillers depth	1442 mMD RKB 1442 mTVD RKB
	Loggers depth*	1444.4 mMD RKB 1444.4 m TVD RKB
Formation at TD	Åre Formation	
Surface coordinates	Geographic coordinates	
	Latitude	66° 00′ 25.5″N
	Longitude	08° 18′ 10.2″E
	Datum/ Spheroide	ED50/ Int. 1924
	UTM coordinates	
	Northing	7 321 060 m
	Easting	468 366 m
	Zone	32N, C.M. 9°E
Seismic reference (3D survey)	ST04M14	
	Nearest Inline	1812
	Nearest Crossline	2164

Table 1-1 Operation overview

Section	Start time	End time	Rig name
NO 6608/10-13 MOVE	12.Oct.2009 08:45	17.Oct.2009 12:00	OCEAN VANGUARD
NO 6608/10-13 PRESPUD	17.Oct.2009 12:00	18.Oct.2009 00:00	OCEAN VANGUARD
NO 6608/10-13 36"	18.Oct.2009 00:00	21.Oct.2009 13:45	OCEAN VANGUARD
NO 6608/10-13 9 7/8"	21.Oct.2009 13:45	22.Oct.2009 11:30	OCEAN VANGUARD
NO 6608/10-13 12 1/4"	22.Oct.2009 11:30	27.Oct.2009 21:15	OCEAN VANGUARD
NO 6608/10-13 8 1/2"	27.Oct.2009 21:15	29.Oct.2009 22:30	OCEAN VANGUARD
NO 6608/10-13 PERM P&A	29.Oct.2009 22:30	02.Nov.2009 00:00	OCEAN VANGUARD
NO 6608/10-13 MOVE	02.Nov.2009 00:00	07.Nov.2009 11:30	OCEAN VANGUARD

1.2 Well objectives

The Fløien North prospect is located in Block 6608/10 and 11 within license PL437. PL437 is part of the former PL313 and has been awarded in the APA 2006 to Statoil ASA (60%) and Hydro (40%), with a commitment well within 3 years. In the merge between Statoil and Hydro in 2007, the license became 100% StatoilHydro owned. The Fløien well, 6608/10-13, was the first exploration well to be drilled in the license.

The 6608/10-13 well was planned to penetrate the Åre reservoir Formation of early Jurassic age. The Åre reservoir is heterolithic alternating sands, shales with some coal. The reservoir was expected to be of good quality, according to nearby wells like Falk (6608/11-2) and Linerle (6608/11-4). The main objective for the well was to prove hydrocarbons in Åre 2 Formation. A secondary objective was to test the potential in a deeper reservoir (Åre 1).

1.3 Result of the well

The well 6608/10-13 Fløien was spudded in a water depth of 384 m MSL and drilled vertical to a total depth of 1442 m TVD. No shallow gas was observed by the ROV at det wellhead or by the MWD while drilling the 36" hole or the pilot hole. The well penetrated rocks of Quaternary, Tertiary, Cretaceous and Jurassic age. TD of the well is in the Åre 1 Formation, see Figure 4-3 Well stratigraphy

The well did not penetrate the Åre 2 as prognosed, but penetrated the Åre 1 81 m shallow. The formation was water filled so no cores were cut in the well.

Water gradient of 1.02 g/cc in Åre Formation evaluated from MDT recorded pressure measurements.

1.4 Data acquisition summary

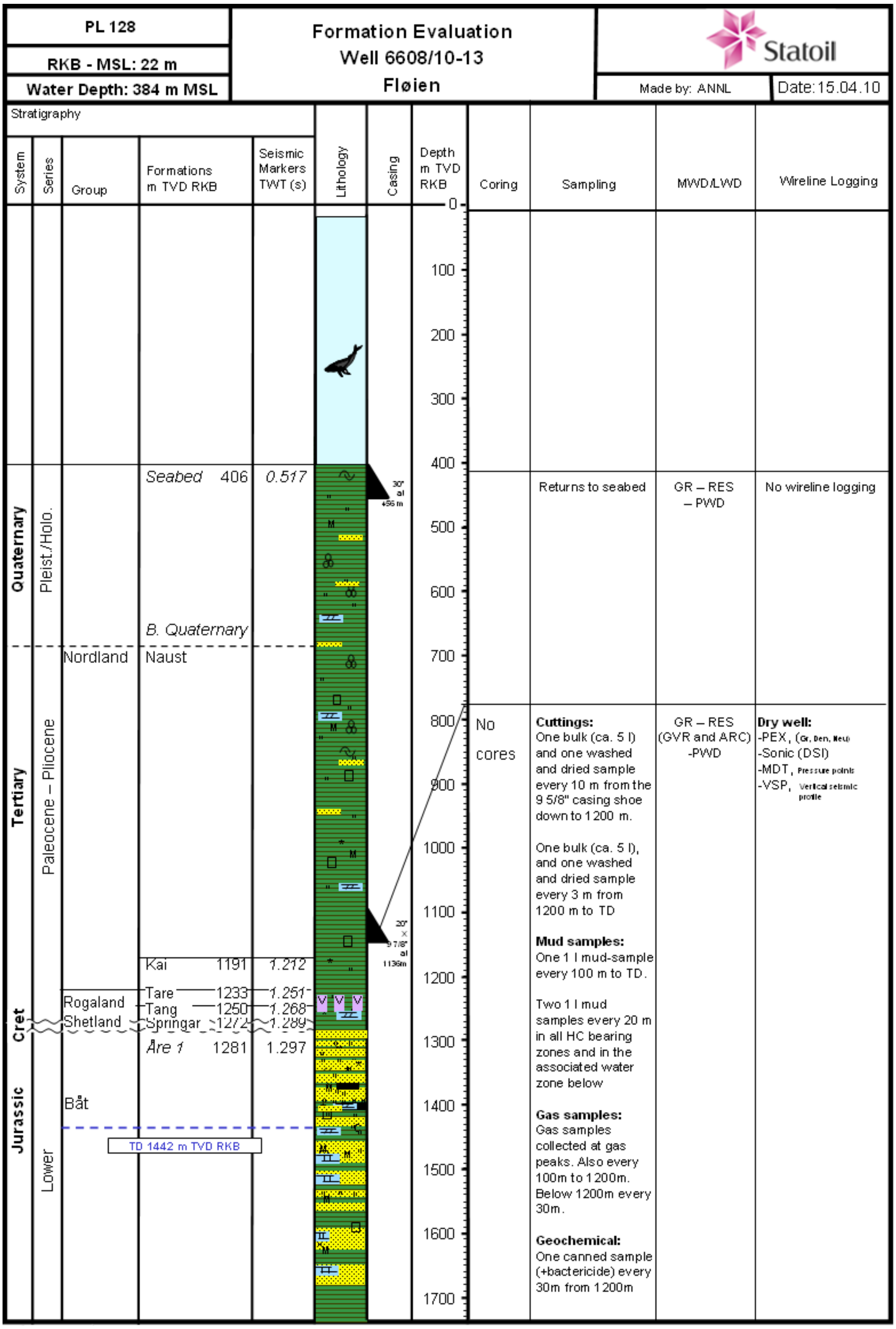


Figure 1-1 Data acqusition

2 Exemptions and non-conformances

Disp no	Title	Doc no	Status
84078	Use of CBL log to evaluate top of cement ouside casing.	General well barrier acceptance criteria in UBO and MPO	Closed

3 Health, safety, environment and quality (HSE&Q)

3.1 RUH

Type (colour code)	Synergi code	Number of
Green	5	3
Green	4	2
Sum		5

3.2 Comments to RUH

Synergi no	Hazard	Description
1115000	4	Synergi 1115000 (Closed) - 17.Oct.2009 - Other non-conformities - Extender cap was loose and had fallen inside Telescope tool After making up the Telescope tool (also referred to as PowerPulse), MWD engineer requested to quickly check inside the tool to ensure the extender cap was properly fitted on the uphole extender before the NMDC was made up to the top of the Telescope. After removing the lifting sub, he was able to look into the tool and immediately noticed that the extender cap was in fact missing and quickly saw that it had fallen off and was inside the tool. With the help of the drill crew the uphole cross over sub was removed and the extender cap was fished out from inside the tool. This process took approximately 15 mins to resolve before the extender cap could be securely fitted and the cross over sub made up again before resuming making up the BHA. The most likely scenario for this occurring is that the extender cap was not properly fitted when it left the shop and it since came loose during transit and fell inside the tool. An internal reporting procedure has been used to bring this to the attention of those concerned in town and appropriate actions will be taken to prevent such an event from re-occuring.
1117102	4	Synergi 1117102 (Closed) - 27.Oct.2009 - Near miss - Left hammer on beam Noticed someone had been working on a pop off valve and left all the tools on top of the pulstion damper and left a hammer on the beam above the mud pump. Possible dropped object. Removed hammer and tools.

3.3 Experience summary

Subject: Pump liner setting in the mud pumps
Section: NO 6608/10-13, MOVE
Rep date: 13.Oct.2009

Keywords: RIG EQUIPMENT
Downtime: **Pot time improvements:**
Comp inv: Statoil
References:
Synergi no: **Cost:**
Synergi desc:

Description:
We normally change the mud pump liners due to pump pressure and flow rate requirements. In the Fløien well TD is shallow. The plan was to use 6 1/2" liners in the 36" section and change to 5 1/2" liners for the remainder of the well.

Changing liners in all the mud pumps requires 2-3 people and 12-18 hours. This is no challenge during a tow, but it does take away critical resources if liners are to be changed during operation (example: during running BOP), thereby having the potential to slow down the operation on the critical time line.

Immediate solution:
Decided on 6" liners.

Future recommended solution:
Pump liner setting should be chosen based on the concept of optimizing the hydraulics. However, if we have good margins on flow and pressure, please evaluate to keep the same liner setting for several sections (or the entire well if the well is shallow). This will reduce the risk exposure to pinch points and optimize the operational efficiency in the main operation (critical time line)

Subject: Optimizing the towing speed
Section: NO 6608/10-13, MOVE
Rep date: 13.Oct.2009
Keywords: ANCHOR OR SUPPLY BOAT
Downtime: **Pot time improvements:**
Comp inv: Statoil
References:
Synergi no: **Cost:**
Synergi desc:

Description:
The towing vessel's work wire is connected to the bridle during a rig move. The spooled length from the vessel to the rig is limited by the water depth. In 100 m water depth, the length of work wire is limited to ~500 m to prevent the wire to touch the sea bed.

During the start of the tow (in shallow water) the pulling tension from Olympic Poseidon was 90 T. As we got into deeper water, the work wire could be elongated, tension and thereby speed increased.

Immediate solution:

Future recommended solution:
Good planning of the rig move can optimize the towing speed and save critical time during long rig moves.

Subject:	Optimize well design		
Section:	NO 6608/10-13, MOVE		
Rep date:	13.Oct.2009		
Keywords:	PLANNING		
Downtime:		**Pot time improvements:**	5
Comp inv:	Statoil		
References:			
Synergi no:		**Cost:**	
Synergi desc:			

Description:
The Fløien well design covers 36" hole, 30" conductor, 9 7/8" pilot hole, 12 1/4" hole, 9 5/8"liner and 8 1/2" reservoir section (4 BHAs). We have contingency for a 17 1/2" section and 13 3/8" casing in case of shallow gas.

An alternative well design is 36" hole, 30" conductor, 9 7/8" pilot hole, 12 1/4" hole, 9 5/8"liner and 9 7/8" reservoir section (3 BHAs).

Reducing the number of BHAs would save critical time (pickup/breakdown and handling on the rig floor) and rental cost (run charge vs re-run charge).

Immediate solution:

Future recommended solution:
In wells with few sections and requirement for pilot hole, please evaluate to optimize the well design with regards to number of BHAs. Having a washed out 9 7/8" hole may have affect wireline logging/fluid sampling, and a change this close to the start-up is not wise. However, it should be evaluated in the planning phase of future wells.

Subject:	Wrong space out on LaFleur fill up and circulating tool.		
Section:	NO 6608/10-13, 12 1/4"		
Rep date:	25.Oct.2009		
Keywords:	CASING/LINER		
Downtime:		**Pot time improvements:**	1
Comp inv:	Odfjell Well Services		
References:			
Synergi no:		**Cost:**	
Synergi desc:			

Description:
The LaFleur was sent out to the rig with incorrect spaceout to fit the 15 ft bails and DDM.

Immediate solution:
Changed out one sub on the LaFleaur with one Diamond sub to achive the correct space out. After the job was done the subs were changed back again so the LaFleur could be backloaded.

Future recommended solution:
It is strange that the spaceout for the LaFleur suddenly turned into a problem. We recently experienced that the LaFleur was sent out to fit 12 ft bails, which it didn`t. This time is was sent out to fit a Varco DDM, but we still got the old MH DDM.

Subject:	RU and tested casing tong		
Section:	NO 6608/10-13, 12 1/4"		
Rep date:	25.Oct.2009		
Keywords:	CASING/LINER		
Downtime:		**Pot time improvements:**	2
Comp inv:	Odfjell Well Services		
References:			
Synergi no:		**Cost:**	
Synergi desc:			

Description:
While circulating hole clean prior to POOH with BHA, the casing tong was lifted in on drill floor and installed/hooked up.
The tong was then tested out while tripping out prior to casing job

Immediate solution:
Tested out the casing tong in advance. That gave us opportunity to fix any possible problems prior to the casing job and without using rig time. This time the testing showed that the tong was working as expected.
Need to remember to take neccessary equipment out of the garage before installing the casing tong due to narrow space.

Future recommended solution:
To be included in next DOP.

Subject:	Displace cement with mud.		
Section:	NO 6608/10-13, 12 1/4"		
Rep date:	28.Oct.2009		
Keywords:	DISPLACEMENT		
Downtime:		**Pot time improvements:**	0

Comp inv: Halliburton
References:
Synergi no: **Cost:**
Synergi desc:

Description:
In an attempt to improve efficiency a change in procedures was suggested such that during cementing of the 9 5/8”casing the cement would be displaced with 1,21 g/cm3 Performadril mud in stead of seawater. This would potentially save time as only the riser then required displacement to mud after being run. Thereafter the shoetrack could be drilled with mud.

Immediate solution:
On displacing the riser to Performadril mud, a larger than expected interface between sea water and mud was observed (15 m3). This can be attributed to the mixing of sea water and mud at well head prior to the riser being run.
Additionally the mud required significant level of treatment with citric acid and sodium bicarbonate after drilling the cement even though a 2 kg/m3 pre-treatment with sodium bicarbonate was performed.
As operation show it once again took in excess of 12 hours to drill trough the plugs and cement in the shoe track. This would have given us adequate time to displace the entire wellbore from seawater to mud without impact on the rig time.

Future recommended solution:
For future jobs we should displace the cement with sea water, unused weighted HiVis sweep mud or displacement mud from previous section. The plugs and shoe track could then be drilled out with seawater and Hi Vis sweeps and the wellbore displaced to drilling mud prior to drilling out the shoe.
This would minimize the interface and the contamination from cement.

Subject: Drilling out 9 5/8”shoe track and 8 1/2" hole with PDC bit and 5 1/2" DP.
Section: NO 6608/10-13, 12 1/4"
Rep date: 28.Oct.2009
Keywords: BIT
Downtime: **Pot time improvements:** 15
Comp inv: Statoil
References:
Synergi no: **Cost:**
Synergi desc:

Description:
A 9 5/8"casing was set from wellhead at 404 m to shoe depth at 1136,5 m.
The 38 m shoe track was drilled out with a 8 1/2" PDC bit, Smith MDi616LEBPX with 5 x12/32" an 1x11/32" nozzles.
Spent 8,5 hrs drilling out the cement plugs with varying drilling parameters. Then drilled out the cement in in the shoe track in 4 hrs.

The BHA consisted of 6 1/2" DCs, 5" HWDP and 5" DP. This was spaced out so the top of the 5" DP should be at the casing shoe when at TD and 5 1/2" DP from casing shoe to surface.
The short 8 1/2" section had TD at +/- 1442 m. During drilling the ECD readings got very high, up to 1,50 g/cm3with 1,21 g/cm3mud and 2030 lpm. The ECD then became a limiting factor on the penetration.

Immediate solution:
Spent excessive time on drilling out the cemet plugs.
Limited the ROP according to the ECD

Future recommended solution:
For a short section like this a high quality rock bit should be evaluated as opposed to PDC bit. It is believed that a rock bit would drill the cement plugs more efficient than a PDC bit. Also a rock bit should be able to drill an interbedded formation like this just as efficient as a PDC bit in water based mud.
The use of 5 1/2" DP in 8 1/2" hole and in 9 5/8"casing or liner must be thoroughly checked against ECD simulations and prognosed formation strength in the planning phase.

Subject:	Improvements in P&A		
Section:	NO 6608/10-13, PERM P&A		
Rep date:	02.Nov.2009		
Keywords:	OTHER		
Downtime:		**Pot time improvements:**	
Comp inv:	Weatherford Norge AS		
References:			
Synergi no:		**Cost:**	
Synergi desc:			

Description:
The job P&A job on Fløien went very fast and according to plan, but some minor adjustments are recommended.

Spent time on removing two subs to optimize cutting depth.
Spent time on positioning rig above WH.
Very effective cutting operation. Kept tool in tension and used motor. Cut 30 x 20" c g/cm3in 3,5 hrs.

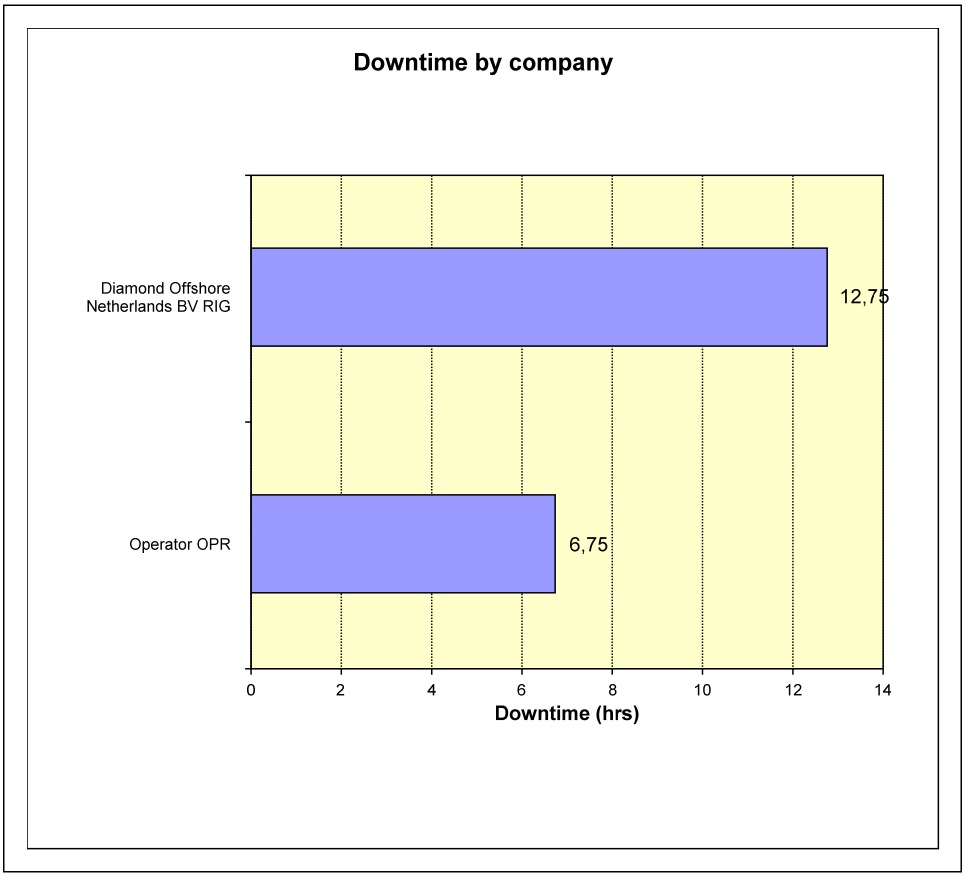
Immediate solution:
Pre made assembly could have been optimised for the cut depth with the option to have enough room to make a second cut if required.

Prior to land and latch Most tool in the guide base the rig could have skidded over location as soon as the BHA had been made up. This would make for a more efficient job.

If the cutting assembly is hanging out side the guide base when assembly is 10 m above the guide base it is more effective to adjust the rig rather than have the ROV attempt to push over the guide base.

Future recommended solution:
1 Optimize cutting depth and spacing before sending offshore
2. Skid rig in position when rih with cutting assy.
3 Position rig in stead of pushing with ROV.
4. Cutting in tension with motor appeared to be an efficient operation.
5. Evaluate skidding the rig off and on location if this is required or not

3.4 Incident by service and company (is not in the original doc)



SERVICE COMPANY: Operator OPR Operator
WELLBORE: NO 6608/10-13

Incident start time	Failure code	Synergi no	Title	Quality cost		Downtime		
				NOK	NOK/d	Total hrs	Comp share %	Comp share hrs
25.oct.2009	RIG-E20 Riser system	1116422	Had to pull out with 4 riser joints. Did not get good pressure test on choke line.	637500	4080000	7,5	50	3,75
06.nov.2009	RIG-E388 General	1118865	Grapple wire pennant parted on anchor #2	510000	4080000	3,0	100	3,0
			Total	1147500				6,75

SERVICE COMPANY: Rig Operations RIG Diamond Offshore Netherlands BV
WELLBORE: NO 6608/10-13

Incident start time	Failure code	Synergi no	Title	Quality cost		Downtime		
				NOK	NOK/d	Total hrs	Comp share %	Comp share hrs
20.oct.2009	RIG-E305 Imported from Bore	1115560	Problems with drawwork in low gear.	1190000	4080000	7,0	100	7,0
25.oct.2009	RIG-E20 Riser system	1116422	Had to pull out with 4 riser joints. Did not get good pressure test on choke line.	637500	4080000	7,5	50	3,75
31.oct.2009	RIG-E20 Imported from Bore	1117501	Broken blue pod reel drive chain	255000	4080000	1,5	100	1,5
01.nov.2009	RIG-E20 Imported from Bore	1117501	Broken blue pod reel drive chain	85000	4080000	0,5	100	0,5
			Total	2167500				12,75

3.5 Time distribution

Table 3-1 Time distribution

		Length	Budget		TL		Actual		Ops
Section	Start time	m	hrs	days	hrs	days	hrs	days	(f)
NO 6608/10-13 MOVE	12.Oct.2009 08:45	408,0	146,0	6,1	102,0	4,3	123,3	5,1	100,0
NO 6608/10-13 PRESPUD	17.Oct.2009 12:00	0,0	32,0	1,3	21,0	0,9	12,0	0,5	100,0
NO 6608/10-13 36"	18.Oct.2009 00:00	66,0	93,0	3,9	65,0	2,7	85,8	3,6	91,8
NO 6608/10-13 9 7/8"	21.Oct.2009 13:45	383,0	41,0	1,7	33,0	1,4	21,8	0,9	100,0
NO 6608/10-13 12 1/4"	22.Oct.2009 11:30	671,0	211,0	8,8	151,5	6,3	129,8	5,4	94,2
NO 6608/10-13 8 1/2"	27.Oct.2009 21:15	297,0	75,0	3,1	62,0	2,6	49,3	2,1	100,0
NO 6608/10-13 PERM P&A	29.Oct.2009 22:30	0,0	162,0	6,8	119,5	5,0	73,5	3,1	97,3
NO 6608/10-13 MOVE	02.Nov.2009 00:00	-408,0	55,0	2,3	36,0	1,5	131,5	5,5	89,1
Sum			815,0	34,0	590,0	24,7	627,0	26,2	

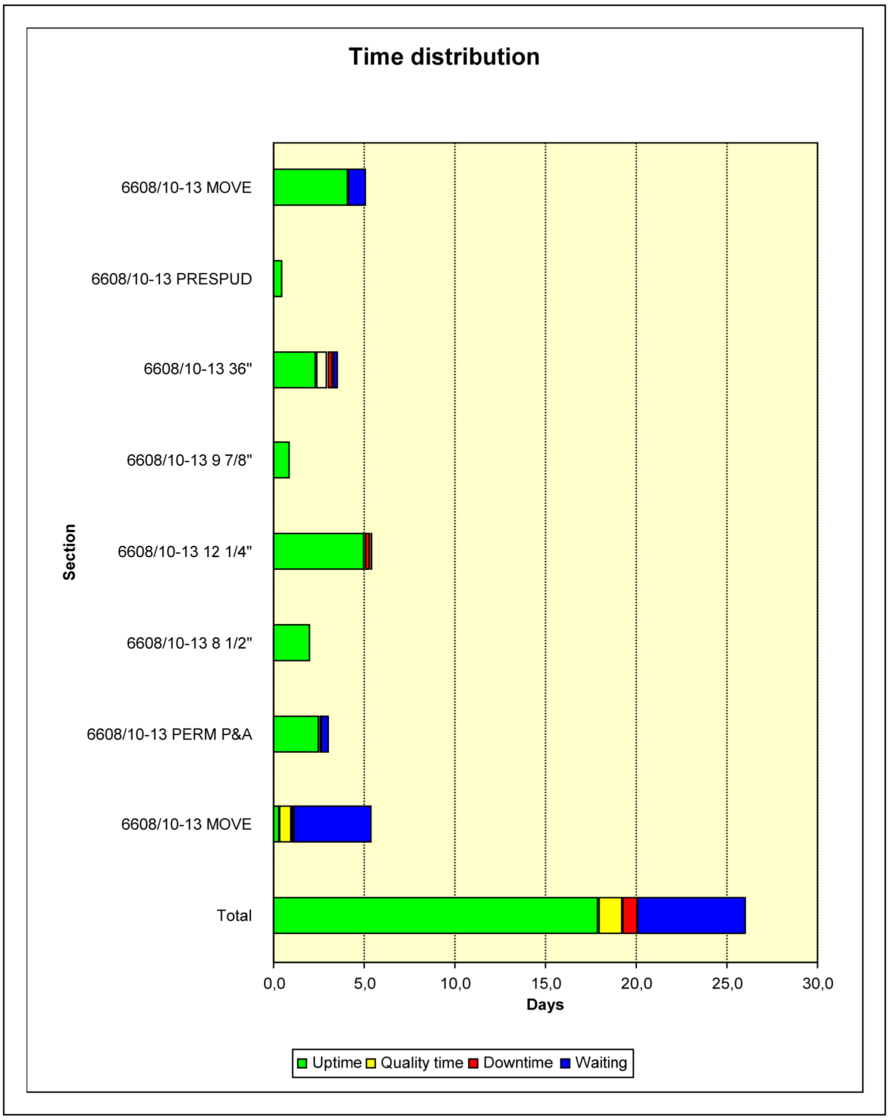


Figure 3-1 Time distribution

The graph above is based on the following details:

Section	Downtime	Uptime	Quality time	Waiting time	Total time
6608/10-13 MOVE	0,0	4,1	0,0	1,0	5,1
6608/10-13 PRESPUD	0,0	0,5	0,0	0,0	0,5
6608/10-13 36"	0,3	2,4	0,6	0,3	3,6
6608/10-13 9 7/8"	0,0	0,9	0,0	0,0	0,9
6608/10-13 12 1/4"	0,3	5,1	0,0	0,0	5,4
6608/10-13 8 1/2"	0,0	2,1	0,0	0,0	2,1
6608/10-13 PERM P&A	0,1	2,6	0,0	0,4	3,1

Section	Downtime	Uptime	Quality time	Waiting time	Total time
6608/10-13 MOVE	0,1	0,3	0,7	4,3	5,5
Total	0,8	18,0	1,3	6,0	26,1

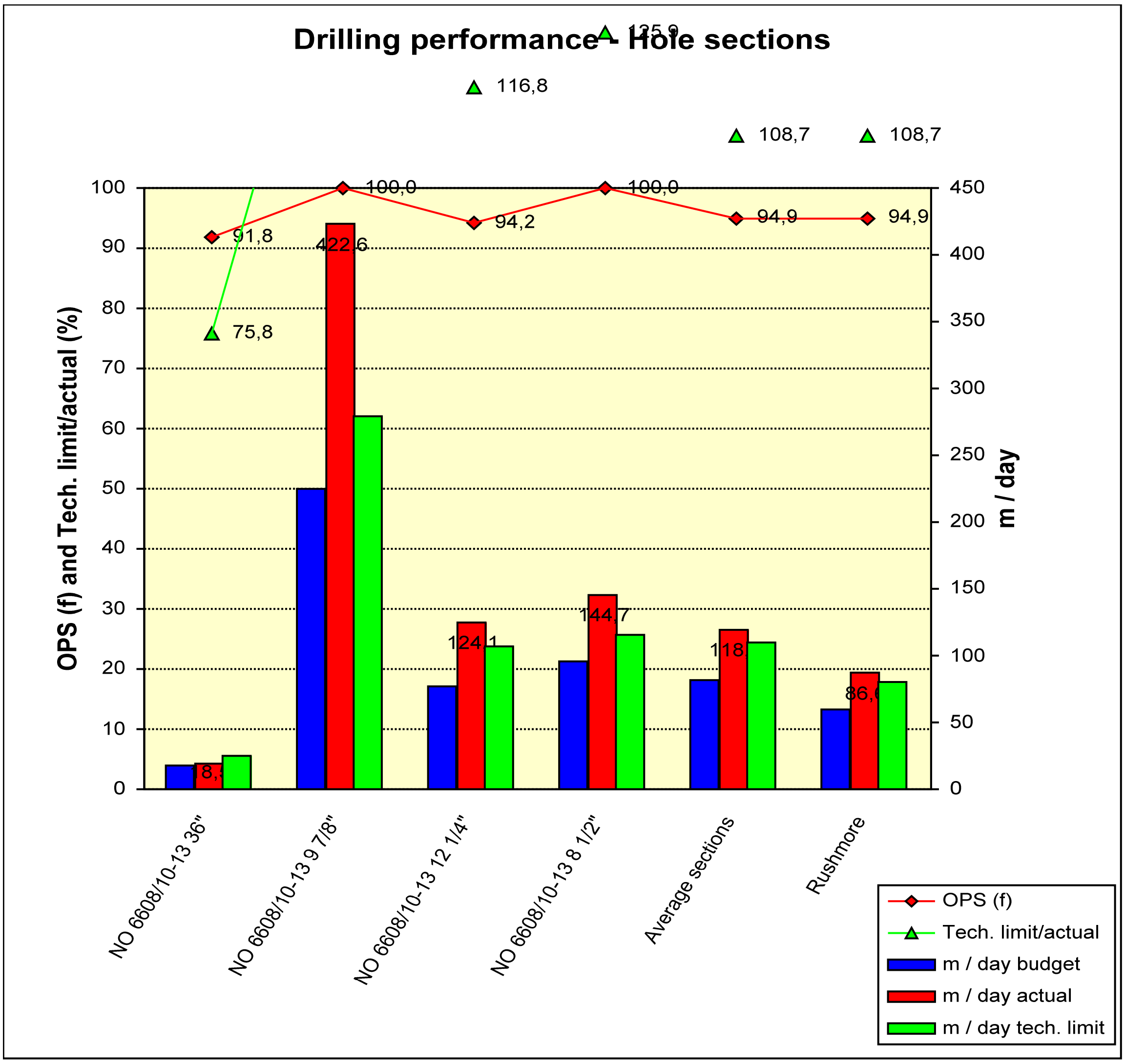


Figure 3-2 Drilling performance

4 Geology and formation data report

4.1 Geological setting and results

The Fløien prospect is located on the Nordland Ridge, which separates the Helgeland Basin and the Dønna Terrace (Figure 4-1 Regional structural map.). The Norne well 6608/10-4 is located 7.6 km northwest of Fløien.

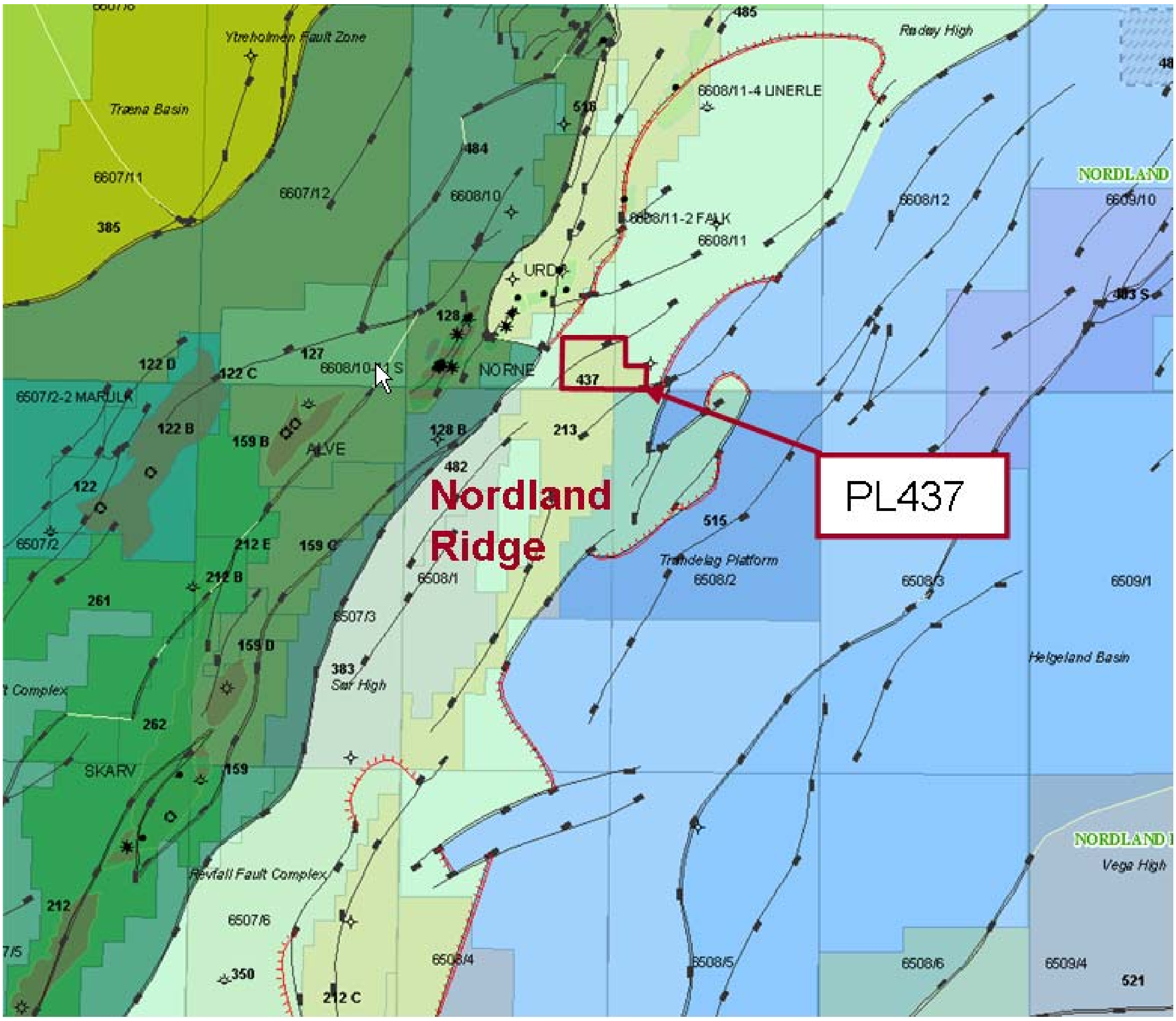


Figure 4-1 Regional structural map.

Fløien consists of 2 segments, Fløien North and Fløien South of which only Fløien North was tested by the 6608/10-13 well. Fløien North was before drilling defined as a tilted trap, downfaulted from Fløien South. The structure was truncated by Base Tertiary/BCU. Fløien North was depended on a sealing fault towards Fløien South and the dry 6608/11-1 well to the north and was based on a base seal model at top Aare 1 Fm level. Prognosed reservoir in the well was Aare 2 Fm, however, a minor possibility for Aare 1 Fm was mentioned before drilling. Migration of hydrocarbons into Fløien North was regarded as the main critical factor for the prospect. A flatspot was observed in the reservoir section.

Results of the 6608/10-13 well have shown that the reservoir section consists of Aare 1 Fm of good porosity but with a lower N/G than expected. Prognosed versus observed stratigraphy is shown in figure 4-2. The Fløien North well was dry with no shows in the reservoir section and the gas reading where close to zero. Therefore, it is migration that is considered the failure of the well and the well has not proved whether the trap works. After drilling, the observed flatspot has been explained as the remains of multiple.

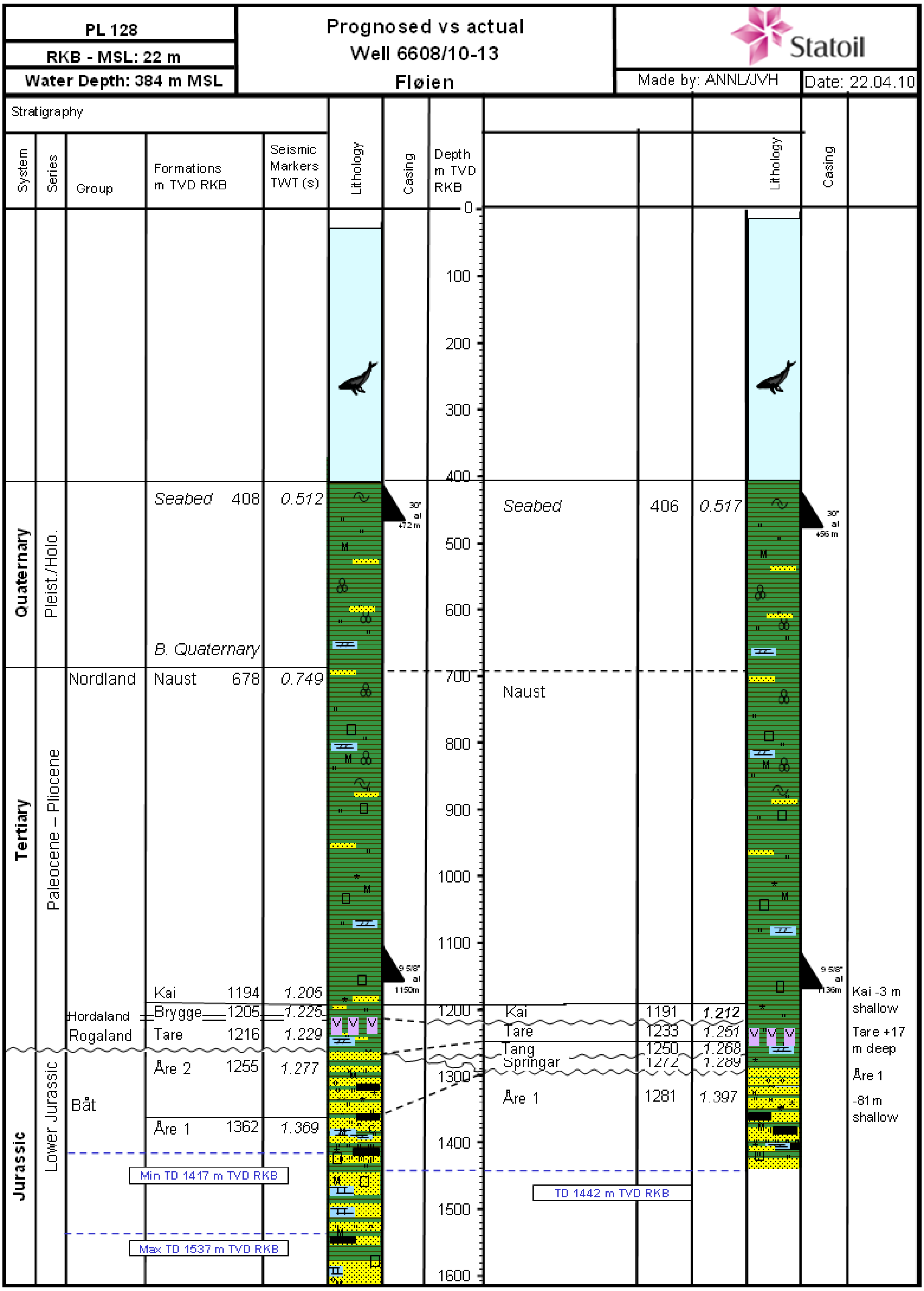


Figure 4-2 Geological prognosis versus well results

4.2 Shallow gas results

A Class 1 shallow gas warning was given at 757 m RKB from site survey. Therefore a 9 7/8" pilothole was drilled from 466 m to 857 m for shallow gas evalution.

No shallow gas was encountered in the well. The pilothole was opened up to 12 1/4" hole and drilled to section TD.

4.3 Stratigraphy

The stratigraphical zonation is based on the interpretation of the biostratigraphic report, log curves and on correlation with nearby wells. The stratigraphy of the entre well is shown in Figure 4-3 Well stratigraphy

The first sediments analysed at 1150 m are of questionable Pleistocene age. Lying conformably below this is an interval of Pliocene age, separated biostratigraphically into Late and Early Pliocene sediments, comprising two biozones. The Middle Miocene is penetrated at 1218 m, with the Late Miocene not identified. This is thought to be a factor of fossil recovery and sampling resolution rather than a stratigraphic break/hiatus.

An unconformity separates the Middle Miocene from the Early Eocene to Late Palaeocene, with Early Miocene, Oligocene and Late to Middle Eocene aged sediments therefore missing. Recognition of this Early Eocene to Late Palaeocene spanned interval is based on a mix of Early Eocene diatom recovery and Late Palaeocene dark green lithologies in the same samples. A thin interval of Middle Palaeocene follows at 1272 m, sitting directly above a base “Tertiary” unconformity.

Cretaceous sediments are first encountered at 1275 m with identification of the Late Campanian. This relatively thin interval is bounded by unconformities above (base “Tertiary”) and below (base Cretaceous).

Penetration of Jurassic sediments is confirmed below the base Cretaceous unconformity at 1290 m. This Early Jurassic interval consists of both the Late Hettangian and Middle to Early Hettangian stages. TD is within the Middle to Early Hettangian aged Early Jurassic.

Lithostratigraphically the well consists of the Nordland, Hordaland, Rogaland, Shetland, and Båt Groups.

The Nordland Group consists of sediments of Early Pleistocene to Middle Miocene age belonging to the Naust and Kai Formations. Hordaland Group sediments are represented by the Early Eocene to Late Paleocene aged Tare Formation, and Rogaland Group proper to the Middle Palaeocene aged Tang Formation.

The Shetland Group consists of sediments of Late Campanian age belonging to the Springar Formation. An unconformity separates this from the overlying Rogaland Group underlying Hettangian aged Åre Formation of the Båt Group. The Åre Formation sediments encountered in this well are interpreted as Åre-1 Formation. TD of the well is in this formation.

4.3.1 Table of chronostratigraphy

Table 4-1 Chronostratigraphy

Age		Top (m MD RKB)	Base (m MD RKB)	Conf.	Comment
Neogene	Early Pleistocene	1150	1150		Top Not Seen
	Late Pliocene	1160	1160		
	Early Pliocene	1170	1215		Alternative top @ 1180m
	Stratigraphic Break				
	Mid Miocene	1218	1221		
Stratigraphic Break					
Paleogene	Early Eocene - Late Paleocene	1224	1269		
	Middle Palaeocene	1272	1272		
Stratigraphic Break					
Cretaceous	Late Campanian	1275	1281		
Stratigraphic Break					
Jurassic	Late Hettangian	1290	1299		
	Early Hettangian	1302	1440		Base Not Seen

Note: this table reflects the interpretation and depths in Stratabugs. * Confidence (green=high, yellow=middle, red=low.)

The biostratigraphic zonation in the biostratigraphic and this report follows the Statoil standard zonation. Interpretations are based on data from the contractor reports (Ichron Ltd, 2010, Microstrat Services, 2010). Chronostratigraphy follows the Geologic Time Scale of Ogg et al. (2008). Lithostratigraphic tops are supplied by the asset and are defined in accordance with Dalland et al. (1988).

Confidence levels are indicated by colours in the tables (confidence: green=high, yellow=middle, red=low).

Note that the tops of biostratigraphic zones which are based on the first downhole occurrences of specimens (LO) may range between this depth and the sample depth above, and that the bases of biostratigraphic zones based on last downhole occurrences (FO) may range between this depth and the sample depth below. On the summary logs such intervals are marked as shaded areas.

Table 4-2 Biostratigraphy

Statoil Biozone (Megaspore zone)	**Top (m MD RKB)**	**Base (m MD RKB)**	*****	**Bioevent**	**Comment**
5	1150	1150		• OCC *C. lobatulus grossa* • OCC *D. laevigata*	Top not seen
6	1160	1160		• LAO *C. lobatulus grossa*	
7	1170	1215		• LCO *C. laevigata* • LO *N. atlantica* (1200m)	Alternative top @ 1180m
10	1218	1221		• LSAO Sponge spicules • LO *Bolboforma* spp.	
19.6 – 20.1	1224	1269		• LO *Coscinodiscus* sp. 1 • LO *R. epigona* • INFLUX Dark green tuffaceous lithology • INFLUX *Cenosphaera* spp. • LSAO *Coscinodiscus* sp. 1 (1245m)	
21.2	1272	1272		• LCO *C. lenticularis*	No younger than 21.2
24.1	1275	1281		• LO red stained microfauna • LO *G. cf. linneiana* • LCO *G. beccariformis* • LO *G. asperus* • LO *D. trochoides* (1278m)	
Indeterminate	1284	1287			Caved microfauna only
45.1 (Ha A-B)	1290	1299		• LO *H. areolatus*	Based on megaspores
45.2 (Nh A?-B)	1302	1440		• LO *N. hopliticus* • LAO *H. areolatus* (1350m) • LSAO *H. areolatus* (1374m) • LCO *N. hopliticus* (1422m)	Based on megaspores. Base not seen

Note: this table reflects the interpretation and depths in Stratabugs. * Confidence (green=high, yellow=middle, red=low.)

<u>*Comment*</u>: Large quantities of reworked Jurassic palynomorphs are present throughout the overburden section

4.3.2 Table of lithostratigraphy

Table 4-3 Lithostratigraphy

Surface name	**MD**	**TVD**	**TWT**
Seabed	406	406	508.69
Kai Fm.	1191	1191	1204.24
Tare Fm.	1233	1233	1243.01
Tang Fm.	1250	1250	1259.90
Springar Fm.	1272	1272	1281.45
Aare Fm.	1281	1281	1289.44

4.4 Lithostratigraphic description

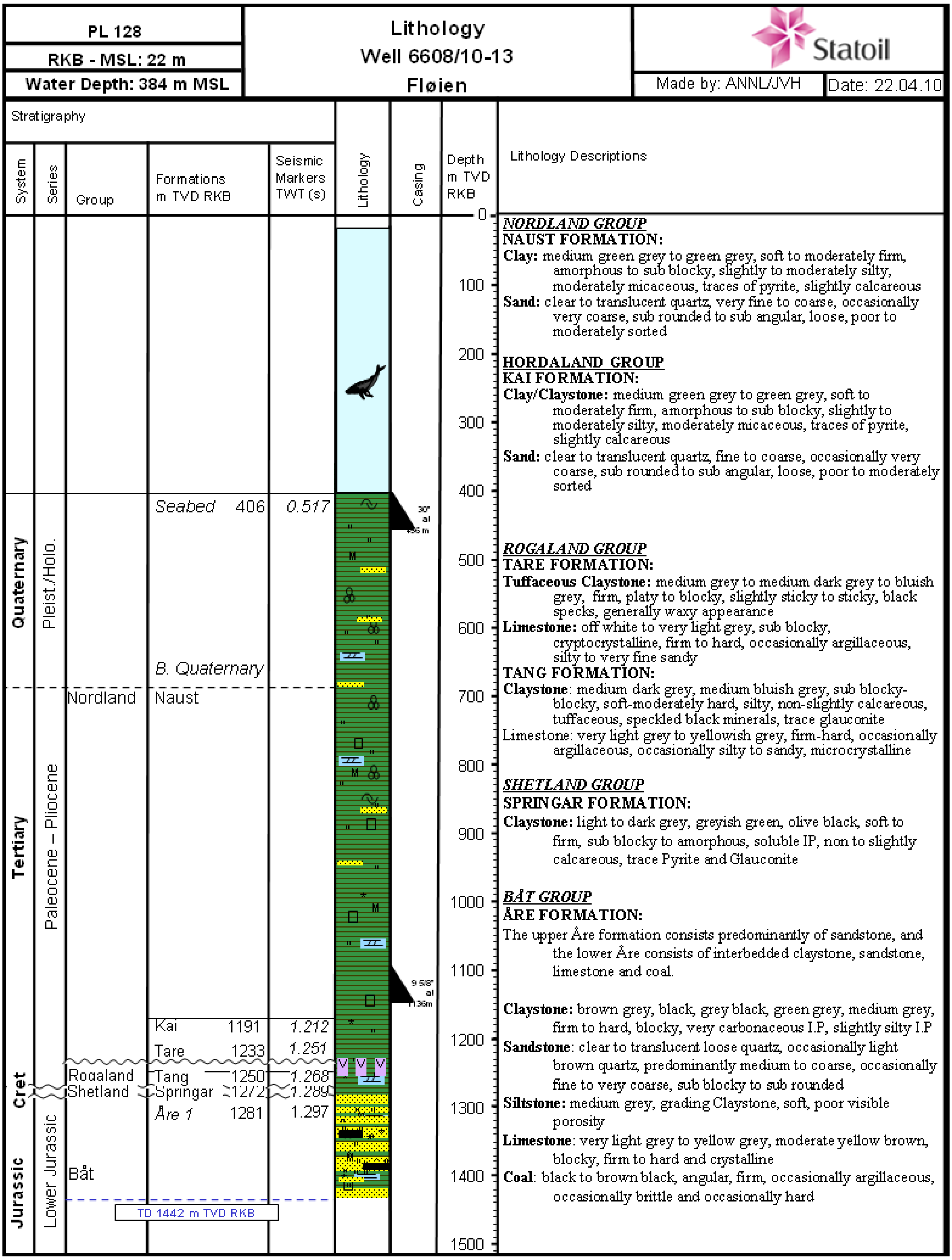


Figure 4-3 Well stratigraphy

4.5 Hydrocarbon indications

Gas readings from this well are only from the 8 ½" section. Generally, little gas was observed during drilling.

Except from occasional very small fragments of coal, no hydrocarbon indications were observed for this section.

4.6 Geophysical results

The well results shows that there is no hydrocarbon induced flatspot at the well position. The flatspot is therefore most likely the reamins of a complex multiple that was not removed during the processing of the ST04M14 3D survey. The seismic tie (Figure 4.4) shows no seismic response at the level of the flatspot, and very little gas was encountered while drilling through the reservoir.

The brighening on top reservoir "downto" the flatspot is most likely lithological as a response to the Åre 1 sands or coals. The post well seismic tie tie is of good quality.

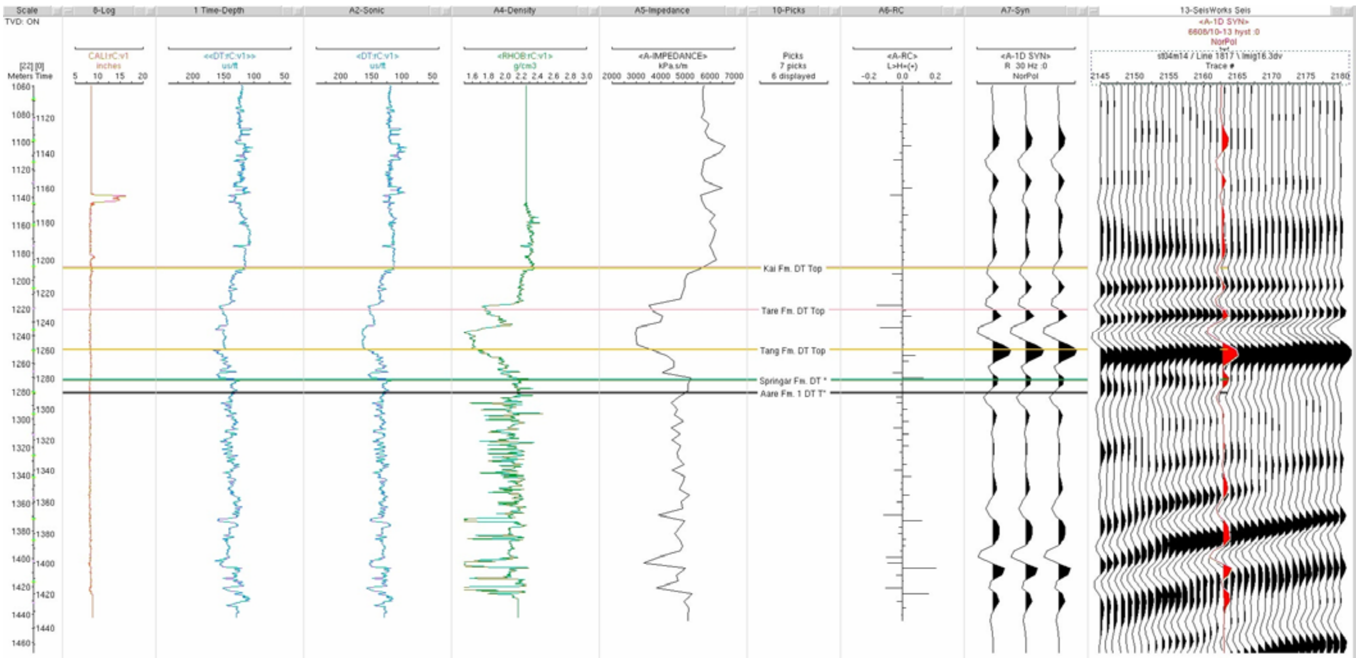


Figure 4-4 Post well seismic tie.

4.7 Data acquisition

4.7.1 Cuttings and mud samples

The sampling commenced according to the Well Programme PL 437, Well 6608/10-13 Fløien in the 8 ½” section from 1150 m MD. The sampling interval was set to 10 m to 1200 m MD and then every 3 m to TD. From 1296 m MD, spot and dry samples were taken every 6 m due to high ROP. Unwashed wet samples were collected in 5 litre buckets. 1 kg of biostrat samples were also collected in bags as well as washed and dried samples were bagged in plastic bags. 1 litre mud samples were retrieved from the active pits every 100 m. Geochemical samples were collected in tin cans every 30 m from 1200 m MD to TD and gas samples were collected in the same interval. Spot samples were taken for description and analysis under microscope.

4.7.2 Conventional coring

No cores were taken in the well.

4.7.3 MWD/LWD

Section	Log no	Report date	Logging company	Interval Upper mMD	Interval Lower mMD	Tools
36"	1	20.10.09	Schlumberger	408.0	464.0	POWERPULSE~Schlumber
		Remarks: DIR only				
9 7/8"	2	22.10.09	Schlumberger	466.0	857.0	ARCVRES8
		Remarks: 9 7/8" pilothole for shallow gas evalution				
12 1/4"	3	24.10.09	Schlumberger	466.0	1142.0	ARCVRES8
		Remarks: Open up 9 7/8"pilothole to 12 1/4" and drill 12 1/4" hole				
8 1/2"	4	29.10.09	Schlumberger	1142.0	1442.0	GVR6 - ARCVRES6 - TELESCOPE
		Remarks:				

4.7.4 *Wireline logging*

Openhole wireline logging was performed by Schlumberger in the 8 ½ in section for this well. An overview of the logging program is shown below.

Table 4-4 Wireline logging

NO	TOOL COMBINATION	RUN	INTERVAL m MD RKB
1	PEX-DSI	1A	1444.4 – 900 m
2	MDT pretest	1A	1413 – 1238 m
3	VSP	1A	1432 – 344 m

4.7.5 *Data quality*

The data quality from wireline logging was of good quality.

4.8 Pore pressure

Pressure estimation for well 6608/10-13 Fløien is based on evaluation of sonic and resistivity data from below 900 m TVD in this well. Above 900 m TVD estimation is based on sonic data from reference well 6608/11-1.

The porepressure and overburden pressure gradients together with the XLOT conducted in the well are presented graphically in Figure 4-8 in Equivalent Mud Weight (EMW) and referenced to TVD KB.

Pore Pressure follow up was carried out with Predict software both offshore and onshore during entire drilling operation. Actual formation pressure measurements were recorded in Åre1 and Tare formations with wireline logging tools.

The pore pressure profile shows a normal trend down to top Kai at 1191 m TVD. The highest pore pressure is estimated in the Kai/Tare formations, with a maximum gradient of 1.18 SG. The pore pressure gradient is then interpreted to decrease rapidly in Tang and Springar formations. At base Springar it decreases to Åre1 formation pressure, which is measured to be 1.02 SG at 1285.6 m TVD. This follows a water gradient. The formation pressure in Tare is measured to 1.1 SG at 1237.9 m TVD, which is comparable with measured pressures in reference well (6608/11-3).

The overburden gradient has been recalculated by using the information from the well and the neighbouring reference wells.

4.8.1 Reservoir pressure summary

Six pressure measurements were performed with the MDT single probe in the reservoir. Of which one was unstable and two were tight pretests.

The pressure measurements were used to define the gradient in water zone in the Åre 1 Formation.

Water gradient is evaluated to be 1.02 g/cc in Åre Formation.

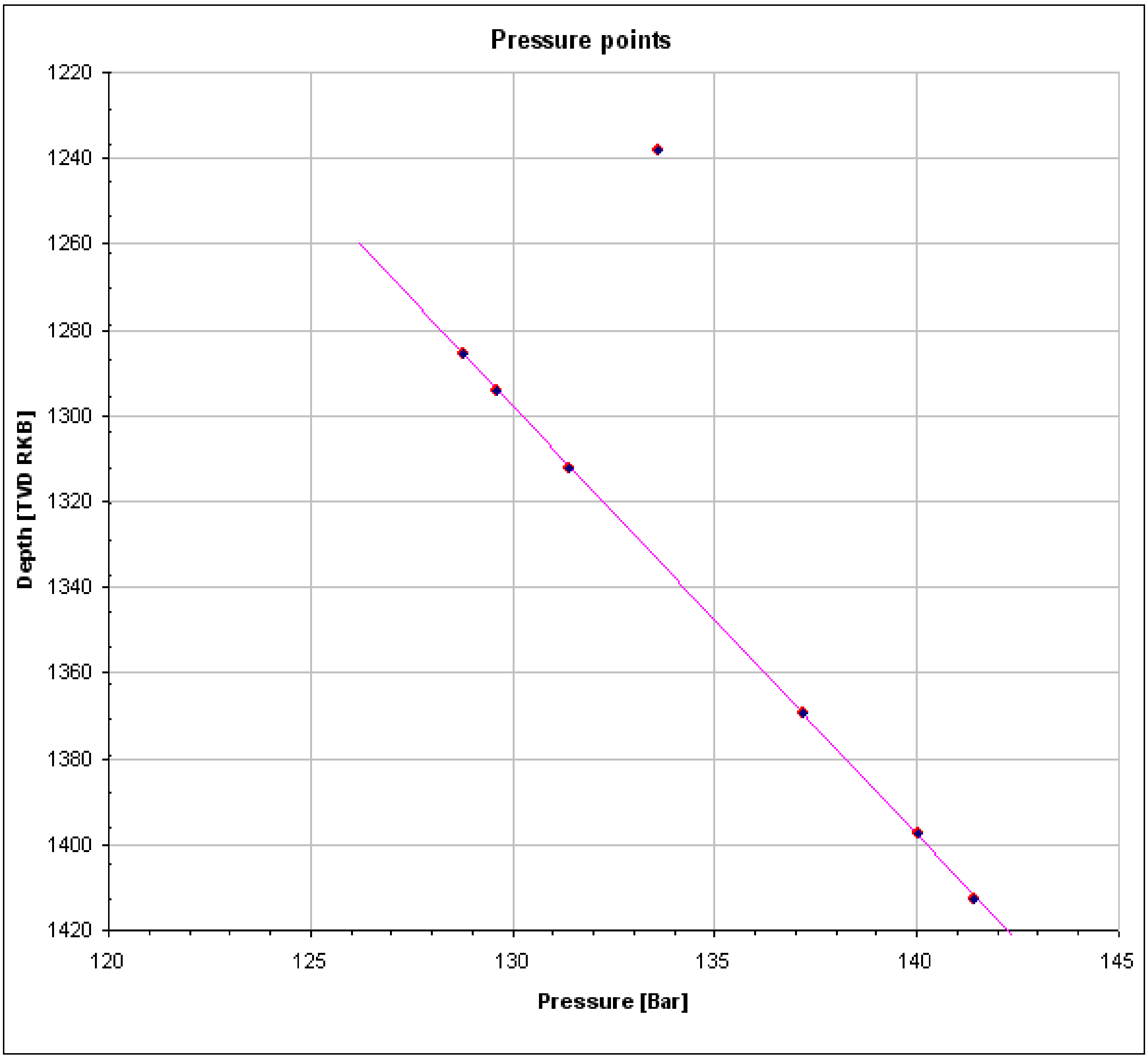


Figure 4-5 MDT pressure measurements in Tare and Åre1 formations

Table 4-5 MDT Pressure points

Run	Test	Formation	Wireline depths			Hydrostatic P before		Formation P		Hydrostatic P after		Temp	Mobility	Comments
#	#		m MD	m TVD RT	m TVD MSL	Bar	g/cc	Bar	g/cc	Bar	g/cc	ºC	md/cP	(tight/no seal/poor/good)
1A	1	Tare	1238.0	1237.9	1215.9	150.894	1.243	133.586	1.100	150.888	1.243	30.50	0.3	Not stable
1A	2	Åre 2	1285.7	1285.6	1263.6	156.729	1.243	128.769	1.021	156.702	1.243	30.16	1175.9	good
1A	3	Åre 2	1294.2	1294.1	1272.1	157.763	1.243	129.620	1.021	157.744	1.243	30.79	1015.6	good
1A	4	Åre 2	1312.2	1312.1	1290.1	159.956	1.243	131.403	1.021	159.938	1.243	31.35	830.0	good
1A	5	Åre 2	1369.5	1369.4	1347.4	166.977	1.243	137.197	1.021	166.973	1.243	33.05	466.1	good
1A	6	Åre 2	1397.7	1397.6	1375.6	170.460	1.243	140.039	1.021	170.456	1.243	34.88	1200.5	good
1A	7	Åre 1	1413.0	1412.9	1390.9	172.444	1.244	141.434	1.020	172.445	1.244	36.02	656.8	good
1A	8	Åre 1	1248.7	1248.6	1226.6	152.209	1.243			152.198	1.243	33.67	0.04	Tight
1A	9	Åre 1	1242.0	1241.9	1219.9	151.370	1.242			151.417	1.243	28.90	0.09	Tight

4.9 Leak off test

An extended leak off test (XLOT) with two cycles was performed at 1136.5 mTVD below the 9 5/8”casing shoe, giving a result of 1.87 SG with the use of 1.2 SG mud weight. The MinHor value is interpreted to be 1.42 SG. Both cycles in the extended leak off test are plotted in Figure 4-6 XLOT below 9 5/8”shoe. (The time scale is seconds since 2009-10-27 17:08:00). The XLOT values are listed in Table 4-6 XLOT 9 5/8”shoe.

For more information about the XLOT interpretation please see “XLOT NO 6608_10-13 Fløien 9-5-8 October 2009” report.

Table 4-6 XLOT 9 5/8”shoe

	Cycle	Pressure, surface [bar]	Pressure downhole [bar]	Equivalent mud weight [g/cm³]	Comments
Leak-off pressure	1	74	208	1.87	
Breakdown pressure	1	84	214	1.92	
Closure pressure	1	24	159	1.43	
	2	23	158	1.42	
Opening pressure	1	85	214	1.92	
	2	41	172	1.54	
Propagation pressure	1	39	171	1.53	
	2	38	170	1.52	

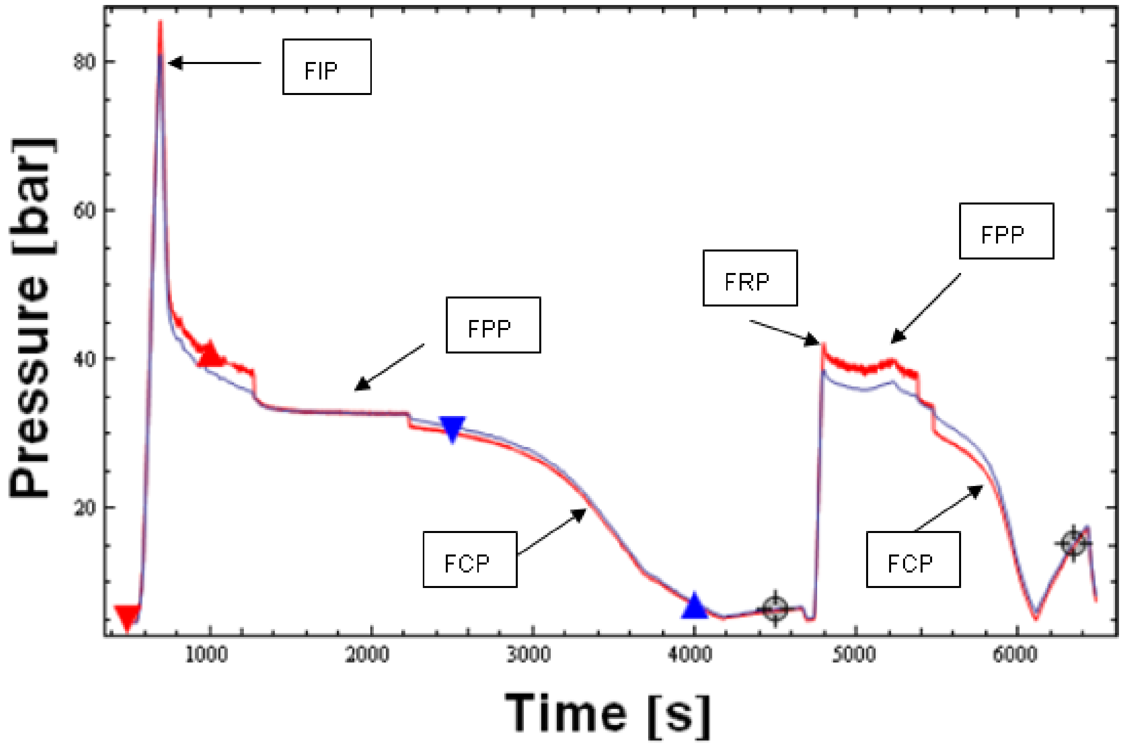


Figure 4-6 XLOT below 9 5/8”shoe.

FIP Fracture Initiating Pressure
FPP Fracture Propagation Pressure
FCP Fracture Closure Pressure
FRP Fracture Reopening Pressure

4.10 Formation temperature

The true formation temperature was estimated from the Horner plot method and experience from nearby wells. The Horner method is based on maximum recorded temperature and time since circulation stopped. The sea bed temperature was assumed to be 4ºC. The formation temperature at TD (1444.4m TVD) was estimated to be 46.9°C from Horner plot with local area correction applied. The formation temperature gradient from seabed is estimated to be 4.2º C / 100 m (TVD) in overburden and 3.76º C / 100 m (TVD) in reservoir section.

The Horner temperature correction method is based on maximum recorded temperature from different logging runs (listed in Table 4-7 Measured and Horner corrected temperatures). All temperature measurements and Horner corrected temperatures are plotted in Figure 4-7 Formation temperature plot
.

Table 4-7 Measured and Horner corrected temperatures

Tool Combination	Run	Thermometer depth (m TVD)	Max recorded temp. (^{0}C)	Time since circulation (hrs)	Horner Corrected temperature at measured depth (^{0}C)
PEX	1A	1417.0	39	09:20	45.86
MDT	1A	1401.0	38	15:50	45.26
VSP	1A	1374.0	38	20:15	44.24

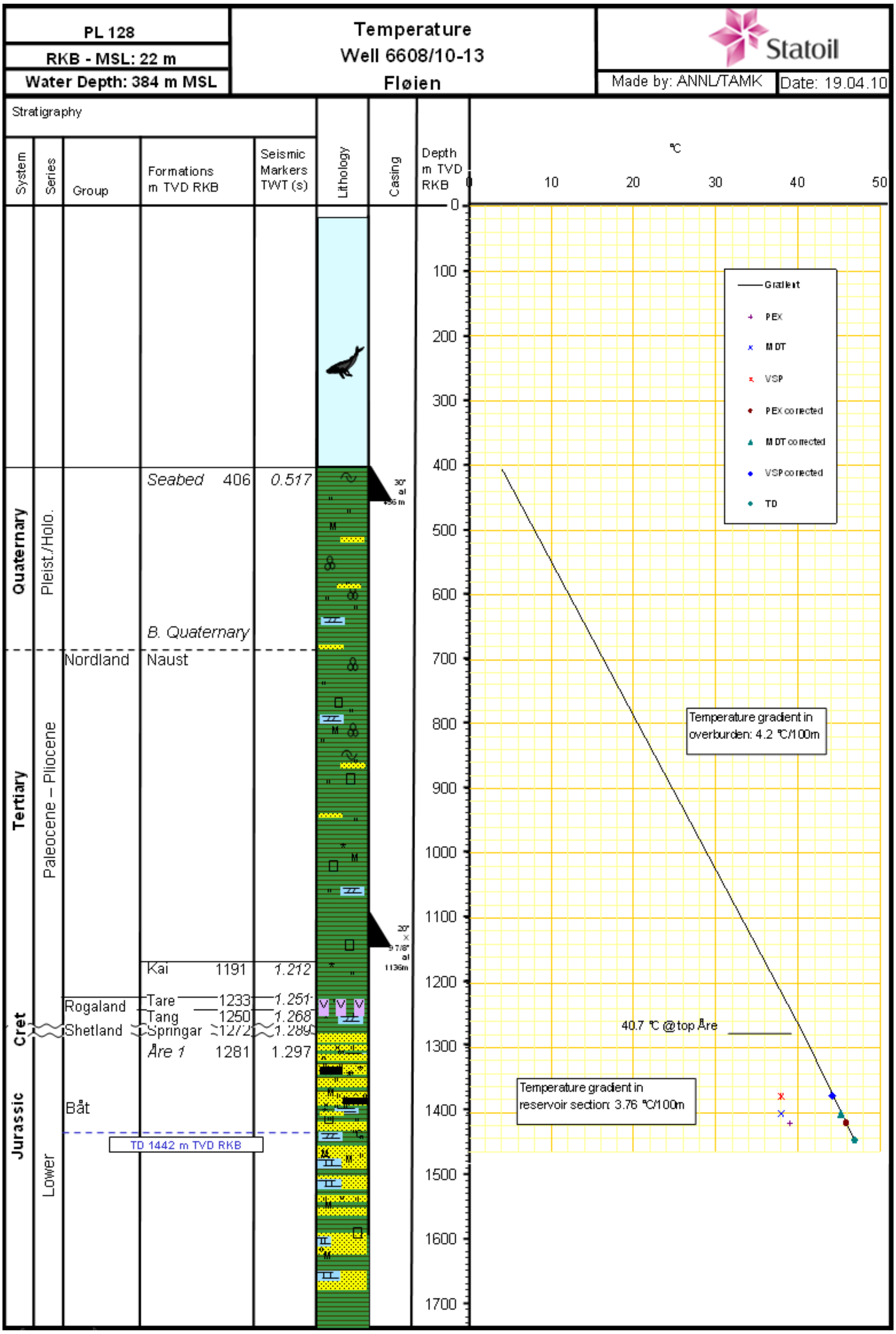


Figure 4-7 Formation temperature plot

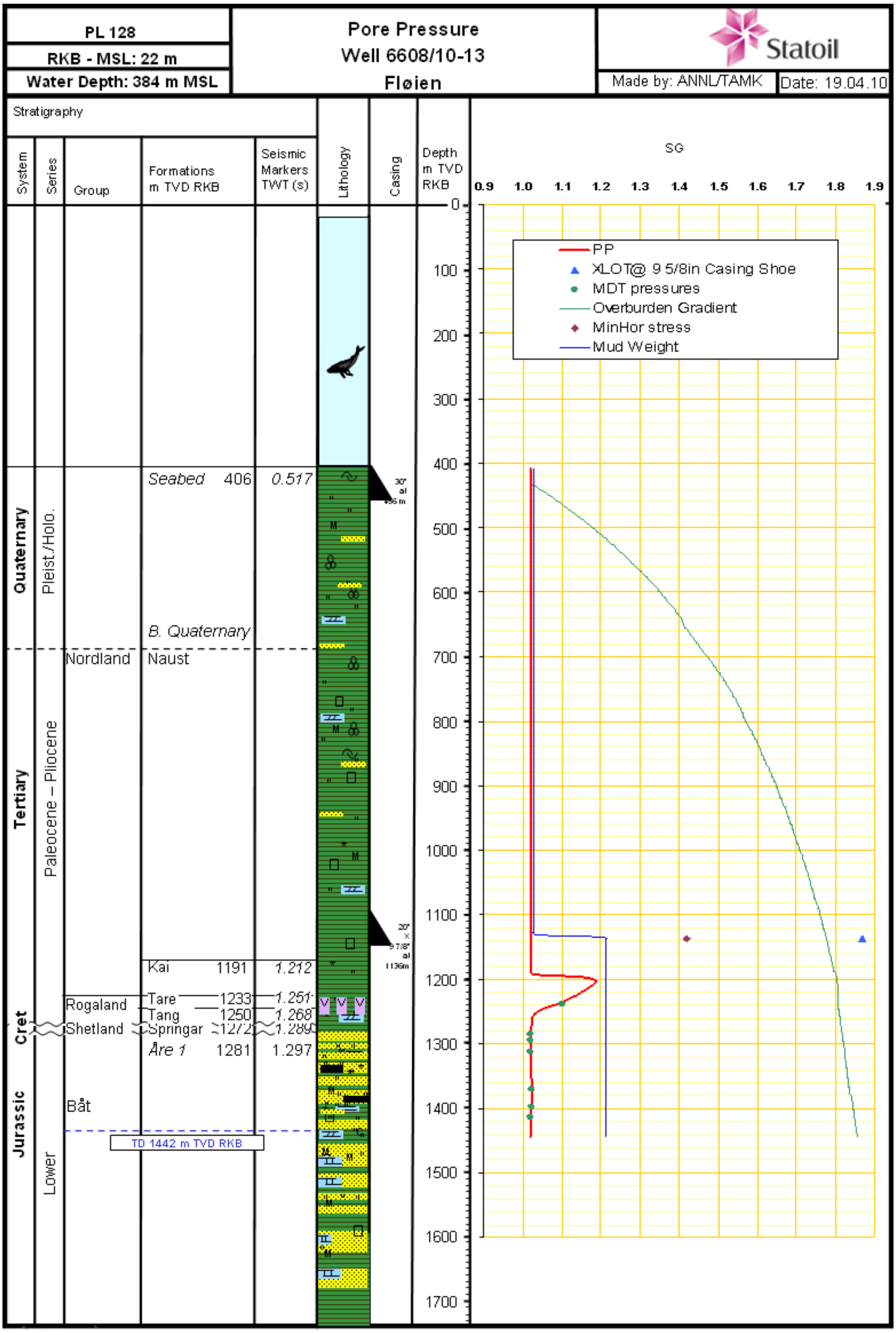


Figure 4-8 Formation pressures plot

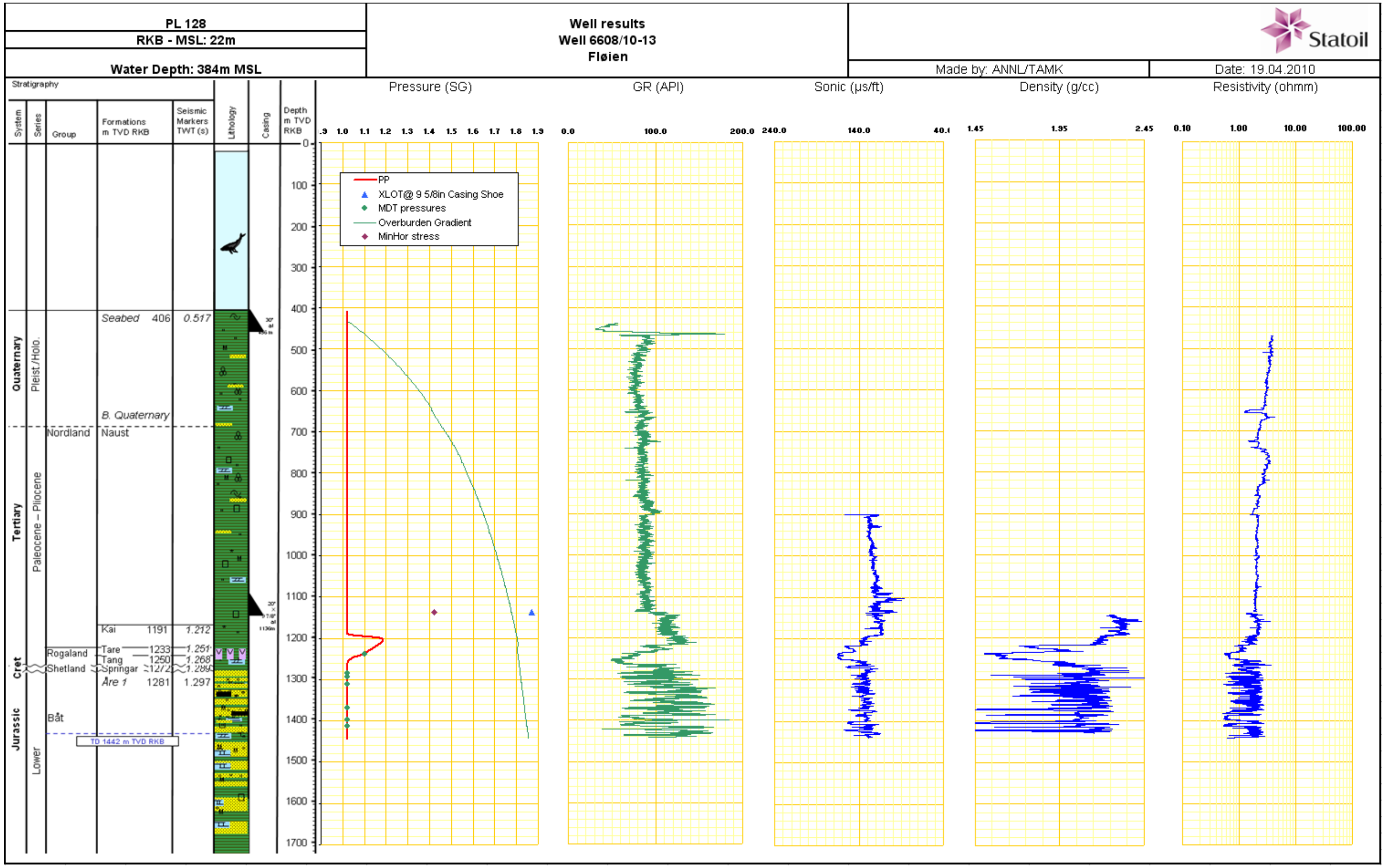


Figure 4-9 Composite plot: Wireline logging

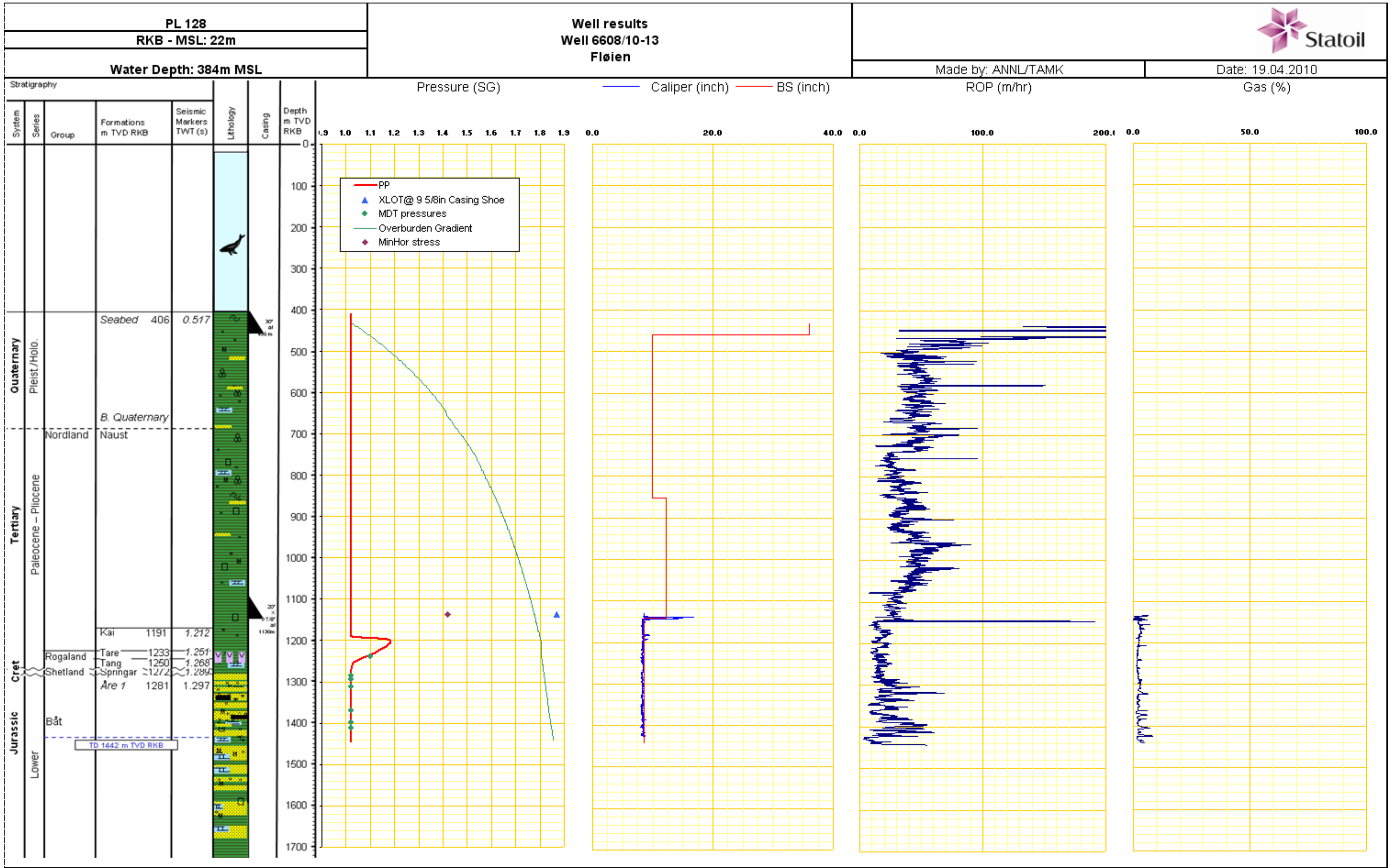


Figure 4-10 Composite plot: Drilling data

4.11 Experiences / recommendations

Cuttings: from 1296 m MD, spot and dry samples were taken every 6 m due to high ROP.

5 Drilling operations report

5.1 Rig move and anchor handling (12.Oct.2009 08:45)

START: 12.Oct.2009 08:45 0 mMD
END : 17.Oct.2009 12:00 408 mMD

OBJECTIVE
Move rig from Frigg Delta location towards Fløien location.

SUMMARY
Sailed from Frigg Delta location with Olympic Poseidon on tow bridle. Started sea voyage 12.10.09 at 08:45 and arrived at Fløien location at 15.10.09 at 06:00. Average speed 5.9 knt. Complete running anchors. Cross tension tested anchors, ballasted rig and performed anchors surveying.

5.1.1 Experiences / recommendations

The move and anchorhandling was done accoring to plan.

5.2 Comparison final / original well design

Refered to: Fig. 6.1 for final Well Schematic and to Figure 5-1 Well schematic, start of project.

5.3 PRESPUD (17.Oct.2009 12:00)

START: 17.Oct.2009 12:00 408 mMD
END : 18.Oct.2009 00:00 408 mMD

OBJECTIVE
Mu 36" BHA and RIH to seabed at 406 m. Position rig.

SUMMARY
PU 17 1/2" x 36" hole opener assmbly and ran in to seabed. Placed marker bouys with ROV and spudded the well.

5.4 Drilling top hole section : 36" (18.Oct.2009 00:00)

START: 18.Oct.2009 00:00 408 mMD

END : 21.Oct.2009 13:45 474 mMD

OBJECTIVE

Drill 36" hole to 474 m. Circulate hole clean and displace to 1.35 g/cm^3 mud. Run and cement 30" conductor.

SUMMARY

Verified position of rig, and placed marker buoys on seabed with ROV. Spudded well at 00:00 hours and drilled 17 1/2" x 36" hole from 408 m to 418 m with 1975 - 2820 lpm / 28 - 60 bar, 0 - 4 ton WOB, 30 - 70 RPM and 2 - 6 kNm torque. Hard formation. Drilled with restricted, i.e. low WOB to not build angle. Check survey at 417 m (survey depth 397 m) gave 0,11 degree inclination. Drilled 17 1/2" x 36" hole from 418 m to 452 m with 4830 lpm / 157 bar, 80 -100 RMP / 5 - 12 kNm torque and 4 - 10 ton WOB. Pumped 1,05 g/cm3 hi visc pills in the middle of each stand and prior to connections. Survey at survey depth 413 m and 420 m gave 0,96 and 0,93 degrees inclination. Drilled boulders from 437 m to 438 m with low WOB; 1 - 4 ton. However survey at 432 m showed 2,37 degree inclination. Decided to ream hole to reduce inclination. Reamed area from 420 m to 448 m with 4500 lpm / 140 bar, 110 - 130 RPM / 5 - 18 kNm torque after laying out one single. Inclination gradually got reduced from 2,37 to 2,01 to 1,69 degrees. Continued reaming area to get angle below 1,0 degree. Observed high vibration in drill string while reaming. Continued reaming area from 425 m to 452 m with 4400 lpm / 135 bar, 120 - 130 RPM / 4 - 18 kNm torque. Hole angle dropped to 1,38 degrees at 431 m.
Continued ream with parameters as above. Hole inclination at 433 m, taken at 03:09, showed 1,78 degrees. Drilled from 452 m to 454 m with 4450 lpm / 140 bar, 100 RPM / 3 - 5 kNm torque and 2 - 4 ton WOB. Drilled from 454 m to 464 m with 4500 lpm / 150 bar, 100 rpm / 7 - 8 kNm torque and 2-4 ton WOB. Reamed area from 464 m to 451 m to get angle down from 1.89 degrees. Obtained reduction in inclination from 1.88 degree to 1,33 degree at 445 m. Displaced hole to 1.45 g/cm3 mud , made wipertrip and displaced to 1.50 g/cm3. No drag or fill observed. Racked back the 36" BHA . Checked 36" hole opener assembly. Observed that one cutter on 36" upper hole opener was missing.

BIT RUNS

This section was drilled with 17 1/2" MX-ST31 115 Hughes Christensen bit SN 6027866 ; 3 x16/32" nozzles(TFA=1,54 in2).
3 stage 17 1/2" x 22" x36"x36” HO from Weatherford. Nozzles 3x10(36"), 3x12(26"), 3x12(22")

DRILLING FLUID

Section drilled with seawater and 10 m3 hi-visc sweeps for every 15 m drilled. At TD circulated BU and pumped 20 m3 sweeps out of hole. Displaced well to 1.50 g/cm3 mud prior to POOH and running conductor. Conductor landed and well displaced back to seawater prior to cement job.

CASING

Picked up 30" shoe joint and two intermediates after holding pre job meeting with involved crew. Filled casing while picking up. Observed excessive movement of elevators at rotary table, hence pulled back clear of splash zone. Rigged up rigs false rotary beam assembly. Lowered, landed and locked conductor housing in Permanent Guide Base (PGB). Removed PGB anchoring chains. Ran in with conductor and PGB on 5 1/2" DP to seabed. Checked bulls eyes with ROV (0 degrees on port side, and 0 - 0,5 degrees on starboard forward bulls eyes) Stabbed in 36" hole with 30" conductor while monitoring with ROV. Ran in hole with conductor on 5 1/2" DP to 450 m. Checked bulls eyes (port 1,5 degree and starboard forward and aft 1,25 degrees). ROV knocked two guidelines loose. Attempted to attach again, without success. Positioned PGB two meters above seabed. Checked orientation of PGB.

ROV unfolded umbilical which was tangled around PGB.

MWD/BHA

Powerpulse MWD tool included in the BHA for inclination purposes.

CEMENTING

Well displaced by circulating the string volume with seawater at 2500 lpm prior to cementing. Mixed and pumped 51 m3 of 1.54 g/cm3 Tuned Light XL cement slurry (300 % open hole excess), and displaced with 7,2 m3 of seawater to leave approximately 5 m cement inside conductor. Waited 6 hrs on cement to set before opening valve and ventilating pressure. Conductor was cemented successfully without problems. TOC inside conductor was later found to be at 316 m. Released 30" conductor running tool with 5 right-hand turns after checking bulls eyes with ROV. Prior to stab conductor bulls eye showed 0,5 degrees maximum. Prior to release of running tool bullseyes showed 1,5 degrees. POOH with landing string from 405 m.

LOGGING

No logging in this section.

5.5 Drilling intermediate sections

5.5.1 9 7/8" (21.Oct.2009 13:45)

START: 21.Oct.2009 13:45 474 mMD
END : 22.Oct.2009 11:30 857 mMD

OBJECTIVE

Drilled 9 7/8" pilot hole to 857 m MD

SUMMARY

MU and RIH with 9 7/8" pilot hole assembly. Drilled 9 7/8" pilot hole to 857 m, dynamically flow checked sand zones. Well stable. No gas observed with ROV sonar. Circulated well clean at TD. Performed 30 minutes flow check. Displaced hole to 1,35 g/cm^3 mud. Pulled out with 9 7/8" BHA from 857 m to 490 m. Relogg area from 469 - 490 m. Top up well with 1,35 g/cm^3 mud. POOH and rack 9 7/8" BHA.

BIT RUNS

This section was drilled with 9 7/8" MXC1 IADC 117 Hughes Christensen bit SN 5032868 ; 4 x20/32" nozzles(TFA=1,228 in2).

DRILLING FLUID

Section drilled with seawater and 10 m3 hi-visc sweeps for every 15 m drilled. At TD circulated BU and pumped 20 m3 sweeps out of hole. Displaced well to 1.35 g/cm^3 mud

LOGGING

The BHA consisted of 9 7/8" bit, bitsub, VisionRes LWD and Powerpuls

5.5.2 12 1/4" (22.Oct.2009 11:30)

START: 22.Oct.2009 11:30 474 mMD
END : 27.Oct.2009 21:15 1145 mMD

OBJECTIVE
Build 12 1/4" BHA. RIH and open up 9 7/8" hole to 12 1/4" hole from 466 m to 857 m. Drill 12 1/4" hole from 857 m to 1155 m.

SUMMARY
PU 12 1/4" bit. RIH with 12 1/4" BHA. Opened 9 7/8" hole to 12 1/4" hole from 466 m to 857 m with 3650 lpm / 130 - 140 bar, 115 - 135 RPM / 2 - 3 kNm torque and 0 -1 ton WOB. ECD varied between 1,040 and 1,060 g/cm3. Gross ROP: 30,3 m/h (including connections). Pumped 10 m3 1,05 g/cm3 hi-visc sweep 1- 2 times per stand.. Drilled 12 1/4" hole from 857 m to 1047 m with 3610 lpm / 140 bar, 120 RPM / 3 - 8 kNm torque and 3 - 14 ton WOB. ECD : 1,060 - 1,070 g/cm3. Gross ROP : 33 m/h. Pumped 1 - 2 hi-visc pills every stand. Drilled 12 1/4" hole from 1047 m to 1142 m with 3650 lpm / 145 bar, 115 RPM / 3 - 5 kNm torque and 3 - 10 ton WOB. ECD : 1,050 - 1,070 g/cm3. Gross ROP : 23,2 m/h. Pumped 1 - 2 hi-visc pills every stand. Displaced hole to 1,35 g/cm3 mud with 3600 lpm / 150 - 205 bar, 60 RPM and 2 kNm torque. Performed visual flow check with ROV at guide base. Well static. Pulled out of hole and rigged up to run casing.

BIT RUNS
This section was drilled with 12 1/4" EQHC1RC IADC 117W Seacurity bit SN 11378273 ; 2 x18/32"and 2 x16/32" nozzles(TFA=0.89 in2).

DRILLING FLUID
Section drilled with seawater and 10 m3 hi-visc sweeps for every 15 m drilled. At TD circulated BU and pumped 15 m3 sweeps out of hole. Displaced well to 1.35 g/cm^3 mud.

CASING
The casing consisted of 9-5/8" 53.5# P-110 Vam Top casing. The shoe track was approximately 37 meters. Centek bow type centralizers were on the frist 16 joints run starting with the shoe joint. The Dril-Quip 18 3/4" wellhead housing (WHH) included a 20" extension with a welded cross-over to 9 5/8" 53.5# P-110 Vam Top pup joint. The WHH Dril-Quip system with rigid lock down system.

MWD/BHA
BHA consisting of an ARCVision, a PowerPulse and a Milltooth bit. Drilled out the casing shoe and into new formation at a depth of 429 m. No high stickslip or shocks drilling out shoe track or during the run to TD. Surveys were taken on each connection. Inclination remained below 1 degree.

CEMENTING

Due to recommendation from earlier drilled wells in these are we used gas-tight cement for the surface casing. Wells in the area have experience gas bubbles when landing the well head housing even though nothing was seen when drilling.
Major focus on section was to ensure cement was to seabed. The 18 3/4" WHH had a 20" extension crossed over to 9 5/8"casing. For fatigue purposes it was important to get cement to seabed
Circulated seawater with 1950 - 2500 lpm / 20 - 30 bar. Pressure tested cement line to 300 bar / 5 minutes. Mixed and pumped 53 m3, 1,54 g/cm3 lead cement slurry with 565 lpm / 6 bar, followed by 8 m3 1,90 g/cm3 tail slurry pumped with 495 lpm / 7 bar. Released top dart. Displaced and sheared top wiper plug by pumping 7,0 m3 1,21 g/cm3 Performadrill mud with 1070 lpm / 12 bar, using cement pump.
Continued displacing with mud pumps with 2500 lpm / 59 - 75 bar. Slowed down pumping rate to 1200 lpm / 40 bar after pumping 3650 liters. Bumped plug with 110 bar pressure (70 bar above circulation pressure) with 97 % efficiency. Pressure leaked off to 20 bar. Lined well over to cement unit and pressured up well from 20 to 87 bar by pumping in 255 liters (could not get up to 120 bar). Bled back and checked for backflow. OK.

Ran BOP / riser

Ran BOP on 50 ft riser joints from 40 m to 193 m. Ran a total of 12 x riser joints. Pressure tested choke and kill lines to 20 / 345 bar for 5 / 10 minutes for every 5th joint. Did not get a good pressure test on choke line on last riser joint at 193 m. Hence pulled out with 3 riser elements from 193 m to 147 m. Rigged up testcap and performed a new pressure test. Still no good test on choke line. Pulled out one more riser element, i.e. pulled up to 132 m. Retested choke / kill line. Good test. One of the choke pin connector on last riser element was leaking. Landed BOP on 18 3/4" wellhead at 404 m while monitoring with ROV. Closed shear ram and lined up kill and choke line and pressure tested 9 5/8"casing and wellhead connector to 133 bar / 15 minutes by pumping in 300 litre. Bled back same

LOGGING
None planned or performed.

5.6 Drilling reservoir section 8 1/2" (27.Oct.2009 21:15)

START: 27.Oct.2009 21:15 1145 mMD
END : 29.Oct.2009 22:30 1442 mMD

OBJECTIVE
Drill out 9 5/8"shoe and 3 meters of new formation. Perform XLOT. Drill to potential reservoir coring points. Major emphasis is on balling the bit which has occurred in offsets. Keep high flowrate and RPM

to minimize balling. TD is prognosed at 1417m MD or maximum TD at 1537m MD: Criteria for TD is 50m into the Åre 1 formation. Any potential hydrocarbon zones will be cored.

SUMMARY

Drilled out 9 5/8”shoe 3 m new formation. Performed Extended Leak Off Test (XLOT) with two cycles against closed upper annular. Pumped in 1272 litres in first cycle and bled back in 32 minutes. Pumped in 1100 litres in second cycle and bled back in 10 minutes. Used different valve setting during bleed off in second cycle, after confirming with duty personnel. XLOT : 1,49 g/cm3. Drilled 8 1/2" hole from 1145 m to 1180 m with 2090 lpm / 120 bar, 75 - 115 RPM / 3 - 4 kNm torque and 3-4 ton WOB. ECD : 1,34 - 1,43 g/cm3. Added Barracarb LCM material to the drilling fluid (25 kg / m3) Continued drilling 8 1/2" hole from 1180 m to 1206 m with 2090 lpm / 120 bar, 75 - 115 RPM / 3 - 4 kNm torque and 3 - 4 ton WOB. ECD started at 1,41 g/cm3 and gradually increased to 1,47 g/cm3. Reamed stand for 20 minutes with 2090 lpm / 125 bar, 112 RPM / 2 - 3 kNm torque. ECD dropped down to 1,40 g/cm3.Drilled 8 1/2" hole from 1206 m to 1261 m with 2090 - 2120 lpm / 125 - 130 bar, 100 - 110 RPM / 2 - 4 kNm torque and 2 - 6 ton WOB. ECD started at 1,40 g/cm3 and dropped down and stabilized at 1,38 g/cm3. Drilled with restricted ROP (i.e. less than 20 m/h), due to evaluation of cuttings and 3 m samples, to evaluate core point. Continued drilling from 1322 m to 1329 m with parameters as above. ECD around 1,41 g/cm3. Observed losses over shaker. Drilled 8 1/2" hole from 1329 m to 1442 m with 2050 lpm / 125 bar, 100 RPM and 3- 6 kNm torque. WOB: 4 - 6 ton. ECD: 1,41 - 1,45 g/cm3.Used pump 1 on booster to aid in hole cleaning. Flow checked well at TD for 10 minutes. Well static. Pulled out with 8 1/2" BHA to surface after changing to 5" handling equipment. Racked BHA. Observed tight spot at 1288 m, i.e. took 22 ton overpull while pulling from 1307 - 1288 m. Wiped area. No overpull seen

BIT RUNS

This section was drilled with 8 1/2" MDI 616 IADC M222 Smith bit SN SY 9085 ; 5 x12/32"and 1 x11/32" nozzles(TFA=0,646 in2).

DRILLING FLUID

This section was drilled with Performadril WBM.
Drilled 9 5/8”shoe track and 3 m new formation while displaced to 1.21 g/cm3mud. Added Barracarb LCM material to the drilling fluid (25 kg / m^3) due to high overbalance in mud weight.
At TD circulated bottoms up two times with 2030 lpm / 125 bar, 100 RPM and 2 - 3 kNm torque, while slowly reciprocating string. Pumped 10 m3 barracarb pill after first circulation. Filtered out excess barracarb on shakers on second bottoms up.

CASING

None. The 7" contingency liner was not needed to drill to TD.

MWD/BHA

The BHA consisted of an ARCVision, a GeoVISION, a TeleScope and a PDC bit. Tools performed well throughout the run, giving good signal and data. No high stickslip or shocks observed during drilling the shoetrack or the run. Surveys were taken on each connection. ROP during most of the run varied between 20-30 m/hrs and inclination remained below 1 degree.

LOGGING

The Dry-Hole logging program consisted of three logging runs which took a total of 18 hours. Run # 1 consited of PEX / DSI DTC which is the density, neutron, gamma and sonic logs. Run # 2 MDT included 13 pressure points. Run # 3 was the ZOVSP seismic log.

5.7 P&A

START: 29.Oct.2009 22:30 1442 mMD
END : 02.Nov.2009 00:00

OBJECTIVE

Permanently plug back the well by setting 9 5/8 EZSV mechanical plug at 1080 m MD. Tag EZSV with 10 MT. Pressure test EZSV to 105 bar with 1.21 g/cm3 WBM which is 70 bar above prognoses formation strength (1.49 g/cm3) at 9 5/8 casing shoe. Set one standard 210 m balanced cement plug on top of 9 5/8 EZSV from 1080 m to 870 m. Set 2 x 210 m balanced cement plugs on top of bottom plug to get TOC @ 450 m , 50 m above the permeable zone behind 9 5/8 casing.

SUMMARY

A 9 5/8"EZSV bridge plug was set at 9 5/8"liner shoe at 1080 m, tagged and pressure tested as planned. The well displaced to seawater above the bridge plug. Set 3 cement plugs on top of EZSV and displaced same with 1 m^3 drill water using the cement unit.
The riser / BOP were pulled with no problems. Ran in with Weatherford's MOST casing cutting tool and pulled WHH/PGB.

5.7.1 Summary

The well was permanently plugged back according to Figure 5-2 Well schematic, end of project.

5.8 MOVE (02.Nov.2009 00:00)

START: 02.Nov.2009 00:00 408 mMD
END : 07.Nov.2009 11:30 0 mMD

OBJECTIVE
Deballast rig and pull anchors.

SUMMARY
Waiting on weather in 104 hrs before Deballasted the rig.
Deballasted the rig and pulled anchors.

6 Appendices

App A Operational listing

WELLBORE ID: NO 6608/10-13
INTERVAL: MOVE
START TIME: 12.Oct.2009 08:45
END TIME: 17.Oct.2009 12:00

Report date	Description
13.Oct.2009	Transit to well location 6608/10-13 (Fløien).
14.Oct.2009	Continued sailing towards Fløyen prospect (6608/10-13) with Olympic Poseidon on towing bridle. Sailed a total of 162 nm last 24 hours. Remaining 150 nm to new location.
15.Oct.2009	Sailed to Fløyen (6608/10-13) location. Positioned rig to commence anchor handling.
16.Oct.2009	Waiting on weather to set anchors at Fløyen location (6608/10-13). Meanwhile perfomed PM's and cleaning of rig.
17.Oct.2009	Set all anchors, except for anchor # 3.

INTERVAL: 36"
START TIME: 18.Oct.2009 00:00
END TIME: 21.Oct.2009 13:45

Report date	Description
18.Oct.2009	Restet anchor # 6. Ballasted rig down to drilling draft. Built 42 jts. 5 1/2" DP while cross tension testing anchors. Made up and racked cement stand. Picked up 36" hole opener assembly (17 1/2" x 22" x 26" x 36" x 36") and built 36" BHA by picking up components from deck. Ran in to seabed at 408 m on 5 1/2" DP. Verified rig position and placed marker bouys. Spudded well at 24:00 hours. Drilled 17 1/2" x 36" hole to 418 m.
19.Oct.2009	Drilled 17 1/2" x 36" hole from 418 m to 452 m. Had to ream hole since hole angle had built up to 2,37 degrees (at 432 m) after drilling through boulders at 437 - 438 m. Reamed hole for about 10 hours. Hole angle dropped to 1,38 degrees. Performed derrick inspection and PM's on DDM as drillstring had been subjected to excessive vibrations while reaming.
20.Oct.2009	Drilled 36" x 17 1/2" " hole from 452 m to 464 m. Displaced hole to 1.35 g/cm3made a wiper trip, displaced hole to 1.5 g/cm3mud and pulled out of the hole. L/D hole opener. Inspected derrick. Rigged up for and ran 51 m of 30" conductor. Landed 30" casing on 5 1/2" landing string @ 458 m.

Report date	Description
21.Oct.2009	Cemented 30" conductor. Held conductor until cement set up. Released RT and pulled out with innerstring. L/D same. MU and RIH with 17 1/2" clean out assy. Drilled out firm cement in shoe from 452 m to 456 m. Cleaned rathole. Drilled 4 m new formation from 462 m to 466 m. POOH and racked BHA (problems with drawwork; replaced control valve while pulling out.). Continued working on problem with drawwork clutch.

INTERVAL: 9 7/8"
START TIME: 21.Oct.2009 13:45
END TIME: 22.Oct.2009 11:30

Report date	Description
22.Oct.2009	Resolved low gear drawworks problem. L/D last part of 36" BHA. P/U 42 x 5 1/2" DP singles. MU and RIH with 9 7/8" pilot hole assembly. Drilled 9 7/8" pilot hole to 857 m, dynamically flow checked sand zones. Well stable. No gas observed with ROV sonar. Circulated well clean at TD. Performed 30 minutes flow check. Displaced hole to 1,35 g/cm3mud. Pulled out with 9 7/8" BHA from 857 m to 490 m. Started relogging.

INTERVAL: 12 1/4"
START TIME: 22.Oct.2009 11:30
END TIME: 27.Oct.2009 21:15

Report date	Description
23.Oct.2009	Relogg area from 480 - 490 m. POOH and racked 9 7/8" BHA. Dumped LWD memory data and initialized PowerPulse tool. Modified BHA by installing three 12 1/8" stabilizers. PU 12 1/4" bit. RIH with 12 1/4" BHA. Opened 9 7/8" hole to 12 1/4" hole from 466 m to 857 m. Drilled 12 1/4" hole from 857 m to 1047 m.
24.Oct.2009	Drilled 12 1/4" hole from 1047 m to 1142 m. Circulated hole clean. Displaced to 1,35 g/cm3mud. Pulled out and topped up hole at 440 m. Jetted PGB with bit. Pulled out and racked BHA (L/D LWD). Made up cement stand and 18 3/8" wellhead running tool and racked in derrick. Rigged up for running 9 5/8"casing. Picked up shoe, two intermediates and float collar and ran in with 9 5/8"casing to 717 m.
25.Oct.2009	Installed 18 3/4" wellhead with crossover to 9 5/8". MU innerstring with plugset. MU wellhead running tool to casing. RIH on 5 1/2" DP. Landed 9 5/8"casing. Circulated with seawater. Cemented 9 5/8"casing. Bumped plug, but plug leaking. Performed rigid lockdown on 18 3/4" wellhead. Release RT. POOH. L/D WH running tool. Rigged down casing handling equipment. Rigged up for running BOP. Ran BOP to 193 m.

Report date	Description
26.Oct.2009	Did not get pressure test on choke line at 193 m. Pulled out with 4 joints of 50 ft risers from 193 m to 132 m. Reinstalled riser element with leaking pin connector. RIH to 328 m (pressure tested every 6.th joint). Installed 25 / 10 ft pup joints before picking up slip-joint. Installed landing joint. Installed c/k lines to slip-joint in moonpool. Pressure tested same. Installed booster, load ring, saddles and pod hoses. Moved rig to well center and attached guide wires.
27.Oct.2009	Landed BOP and performed 25 ton overpull test. Tested 9 5/8"casing and connector to 133 bar against BSR. L/O landing joint and installed diverter. R/D BOP handling equipment. Broke down last part of 12 1/4" BHA and cement stand. Built 8 1/2" drilling BHA and ran in hole. Tagged cement at 1098 m. Function tested BOP and held choke drill. Pulled out 9 stands and built 250 m with 5 1/2" DP to reach max. TD. Drilled float.

INTERVAL: 8 1/2"
START TIME: 27.Oct.2009 21:15
END TIME: 29.Oct.2009 22:30

Report date	Description
28.Oct.2009	Drilled plugset in 9 5/8"casing at 1099 m with 8 1/2" BHA. Reamed down first 9 m of shoetrack (no cement). Drilled cement in remaining part of shoetrack from 1109 m to 1136 m (hard cement last 10 m). Drilled out 9 5/8"shoe at 1136 m. Cleaned out rathole. Drilled 3 m new formation from 1142 m to 1145 m. Flow checked well. Static. Pressure tested lineup. Performed XLOT to 1,49 g/cm^3. Built hangoff tool. Drilled 8 1/2" hole from 1145 m to 1261 m.
29.Oct.2009	Drilled 8 1/2" hole from 1261 m to 1442 m. No corepoint encountered. Had to circulate at 1322 to reduce ECD which peaked up to 1,50 g/cm^3, also changed screens at shakers due to overflow when drilling at 1329 m. Circ. 2 x BU at TD and circulated around Barracarb pill. POOH with 8 1/2" BHA. Took clustershot at 1179 m. POOH and R/B BHA, except for LWD which was laid out. R/U for wireline.

INTERVAL: PERM P&A
START TIME: 29.Oct.2009 22:30
END TIME: 02.Nov.2009 00:00

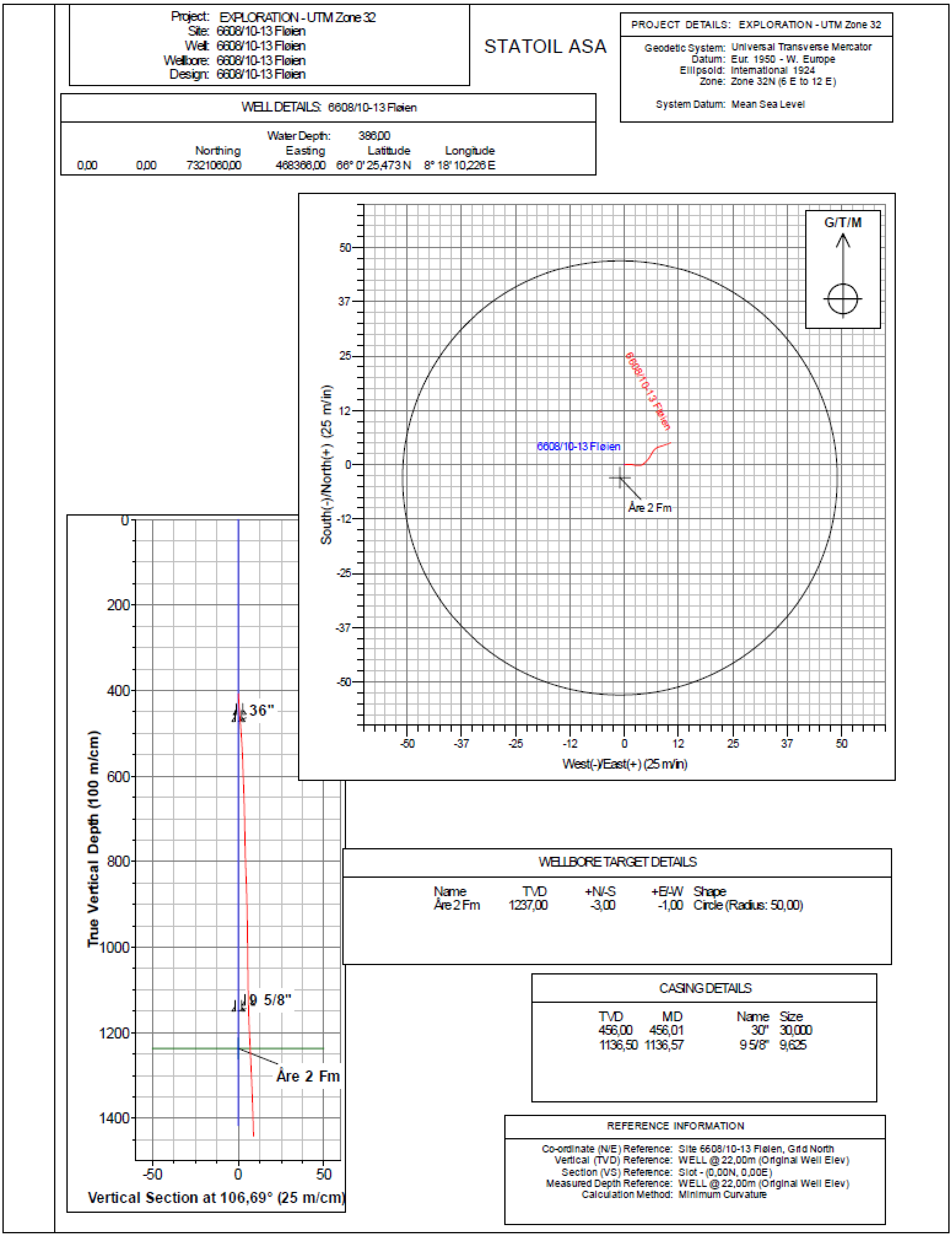
Report date	Description
30.Oct.2009	Performed 3 runs of formation evaluation logging on wireline (PEX-DSI, MDT & VSP). Laid down 3 stands of 6 1/2" DC and Jar. Picked up and racked cement head. Serviced rig while waiting on helicopter with EZSV running tool. Helicopter turned around at 05:00 due to poor visibility. Continued waiting while evaluating situation.
31.Oct.2009	Waited on EZSV running tool arriving with helicopter. Ran in and set EZSV retainer at 1080 m. Pressure tested same to 105 bar. Displaced well to seawater. Rigged up and set 3 cement plugs inside 9 5/8"casing from 1080 to 430 m. POOH with 5" cement stinger and washed wellhead and riser. Laid out cement stand. RIH and retrieved wearbushing.

Report date	Description
01.Nov.2009	Disconnected and pulled BOP's and riser

INTERVAL: MOVE
START TIME: 02.Nov.2009 00:00
END TIME: 07.Nov.2009 11:30

Report date	Description
02.Nov.2009	Pulled and secured BOP's and riser. Cut 20" and 30" casing and retrieved wellhead and PGB. Performed seabed survey and retrieved marker buoys. Waited on weather for pulling anchors.
03.Nov.2009	Waiting on weather.
04.Nov.2009	Waiting on weather
05.Nov.2009	Waiting on weather to pull anchors. Parallel activities: Perform PM's and other maintenance.
06.Nov.2009	Waiting on weather to pull anchors. Parallel activities: Perform PM's, general rig cleaning and other maintenance.
07.Nov.2009	Waiting on weather to pull anchors. Commence pulling anchors 08:00. Pull anchors and heave chain 2, 3, 4, 6, 7 & 8. Pull anchors 1 and 5 and commence heaving chain. Anchor handling operation delayed 3 hrs due to splitt grappling wire on anchor No.2 and 4.5 hrs so far at 06:00 due to twists in anchor chains 2, 4, 5, 6 and 8.

App B Directional data, survey listing



Schlumberger

Survey Report - Geographic

Company:	STATOIL NORWAY	**Local Co-ordinate Reference:**	Site 6608/10-13 Fløien
Project:	EXPLORATION - UTM Zone 32	**TVD Reference:**	WELL @ 22.00m (Original Well Elev)
Site:	6608/10-13 Fløien	**MD Reference:**	WELL @ 22.00m (Original Well Elev)
Well:	6608/10-13 Fløien	**North Reference:**	Grid
Wellbore:	6608/10-13 Fløien	**Survey Calculation Method:**	Minimum Curvature
Design:	6608/10-13 Fløien	**Database:**	edm_restore

Project	EXPLORATION - UTM Zone 32		
Map System:	Universal Transverse Mercator	**System Datum:**	Mean Sea Level
Geo Datum:	European 1950 - Mean		Using Well Reference Point
Map Zone:	Zone 32N (6 E to 12 E)		Using geodetic scale factor

Site	6608/10-13 Fløien				
Site Position:		**Northing:**	7,321,060.00 m	**Latitude:**	66° 0' 25.473 N
From:	Map	**Easting:**	468,366.00 m	**Longitude:**	8° 18' 10.226 E
Position Uncertainty:	0.00 m	**Slot Radius:**	in	**Grid Convergence:**	-0.64 °

Well	6608/10-13 Fløien					
Well Position	**+N/-S**	0.00 m	**Northing:**	7,321,060.00 m	**Latitude:**	66° 0' 25.473 N
	+E/-W	0.00 m	**Easting:**	468,366.00 m	**Longitude:**	8° 18' 10.226 E
Position Uncertainty		0.00 m	**Wellhead Depth:**	386.00 m	**Water Depth:**	386.00 m

Wellbore 6608/10-13 Fløien

Magnetics	**Model Name**	**Sample Date**	**Declination (°)**	**Dip Angle (°)**	**Field Strength (nT)**
	User Defined	20/09/2009	0.65	76.07	52,100

Design 6608/10-13 Fløien

Audit Notes:

Version: 1.0 **Phase:** ACTUAL **Tie On Depth:** 408.00

Vertical Section:	**Depth From (TVD) (m)**	**+N/-S (m)**	**+E/-W (m)**	**Direction (°)**
	408.00	0.00	0.00	106.69

Survey Program **Date** 23/11/2009

From (m)	**To (m)**	**Survey (Wellbore)**	**Tool Name**	**Description**
413.39	445.65	36in MWD Surveys - Incl Only (6608/10-13	Magnetic, other	Magnetic Tools (MWD, EMS)
477.14	843.51	9 7/8" Pilot Hole MWD Surveys (6608/10-1	Magnetic, std, non-mag	Magnetic Tools (MWD, EMS)
887.00	1,128.08	12 1/4" MWD (6608/10-13 Fløien)	Magnetic, std, non-mag	Magnetic Tools (MWD, EMS)
1,176.72	1,424.82	8 1/2" MWD (6608/10-13 Fløien)	Magnetic, std, non-mag	Magnetic Tools (MWD, EMS)

Schlumberger

Survey Report - Geographic

Company:	STATOIL NORWAY	**Local Co-ordinate Reference:**	Site 6608/10-13 Fløien
Project:	EXPLORATION - UTM Zone 32	**TVD Reference:**	WELL @ 22.00m (Original Well Elev)
Site:	6608/10-13 Fløien	**MD Reference:**	WELL @ 22.00m (Original Well Elev)
Well:	6608/10-13 Fløien	**North Reference:**	Grid
Wellbore:	6608/10-13 Fløien	**Survey Calculation Method:**	Minimum Curvature
Design:	6608/10-13 Fløien	**Database:**	edm_restore

Survey

Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Map Northing (m)	Map Easting (m)	Latitude	Longitude
408.00	0.00	0.00	408.00	0.00	0.00	7,321,060.00	468,366.00	66° 0' 25.473 N	8° 18' 10.226 E
413.39	0.96	109.86	413.39	-0.02	0.04	7,321,059.98	468,366.04	66° 0' 25.472 N	8° 18' 10.230 E
419.44	0.39	88.34	419.44	-0.03	0.11	7,321,059.97	468,366.11	66° 0' 25.472 N	8° 18' 10.235 E
429.11	1.02	71.96	429.11	0.00	0.23	7,321,060.00	468,366.23	66° 0' 25.472 N	8° 18' 10.244 E
445.65	1.44	94.22	445.64	0.03	0.57	7,321,060.03	468,366.57	66° 0' 25.474 N	8° 18' 10.272 E
456.01	1.35	91.41	456.00	0.01	0.82	7,321,060.01	468,366.82	66° 0' 25.473 N	8° 18' 10.292 E
30"									
477.14	1.17	84.35	477.13	0.03	1.29	7,321,060.03	468,367.29	66° 0' 25.474 N	8° 18' 10.328 E
505.38	0.86	103.22	505.36	0.01	1.78	7,321,060.01	468,367.78	66° 0' 25.473 N	8° 18' 10.368 E
534.12	0.74	112.71	534.10	-0.11	2.16	7,321,059.89	468,368.16	66° 0' 25.470 N	8° 18' 10.398 E
591.90	0.52	80.99	591.88	-0.22	2.76	7,321,059.78	468,368.76	66° 0' 25.467 N	8° 18' 10.446 E
620.55	0.70	90.73	620.52	-0.20	3.07	7,321,059.80	468,369.07	66° 0' 25.467 N	8° 18' 10.470 E
651.05	0.43	88.29	651.02	-0.20	3.37	7,321,059.80	468,369.37	66° 0' 25.467 N	8° 18' 10.494 E
680.44	0.40	103.54	680.41	-0.22	3.58	7,321,059.78	468,369.58	66° 0' 25.467 N	8° 18' 10.510 E
713.17	0.29	56.84	713.14	-0.20	3.76	7,321,059.80	468,369.76	66° 0' 25.467 N	8° 18' 10.525 E
738.99	0.33	71.76	738.96	-0.14	3.88	7,321,059.86	468,369.88	66° 0' 25.469 N	8° 18' 10.535 E
768.38	0.53	64.93	768.35	-0.06	4.09	7,321,059.94	468,370.09	66° 0' 25.472 N	8° 18' 10.551 E
826.32	1.15	44.66	826.28	0.47	4.74	7,321,060.47	468,370.74	66° 0' 25.489 N	8° 18' 10.602 E
843.51	1.06	46.79	843.47	0.70	4.98	7,321,060.70	468,370.97	66° 0' 25.497 N	8° 18' 10.620 E
887.00	1.04	37.47	886.95	1.29	5.51	7,321,061.29	468,371.51	66° 0' 25.516 N	8° 18' 10.662 E
972.26	0.73	25.40	972.20	2.40	6.21	7,321,062.40	468,372.21	66° 0' 25.552 N	8° 18' 10.717 E
1,001.18	0.76	34.88	1,001.12	2.72	6.40	7,321,062.72	468,372.40	66° 0' 25.563 N	8° 18' 10.732 E
1,060.82	0.55	36.63	1,060.76	3.27	6.80	7,321,063.27	468,372.80	66° 0' 25.581 N	8° 18' 10.763 E
1,089.76	0.51	42.01	1,089.70	3.48	6.97	7,321,063.48	468,372.97	66° 0' 25.587 N	8° 18' 10.776 E
1,118.31	0.41	45.23	1,118.24	3.65	7.13	7,321,063.65	468,373.12	66° 0' 25.593 N	8° 18' 10.788 E
1,128.08	0.45	39.20	1,128.01	3.70	7.18	7,321,063.70	468,373.17	66° 0' 25.595 N	8° 18' 10.792 E
1,136.57	0.51	48.16	1,136.50	3.75	7.22	7,321,063.75	468,373.22	66° 0' 25.596 N	8° 18' 10.796 E
9 5/8"									
1,176.72	0.86	70.84	1,176.65	3.97	7.64	7,321,063.97	468,373.64	66° 0' 25.603 N	8° 18' 10.829 E
1,204.12	0.73	78.34	1,204.05	4.07	8.01	7,321,064.07	468,374.00	66° 0' 25.607 N	8° 18' 10.858 E
1,232.99	0.80	68.20	1,232.92	4.19	8.37	7,321,064.18	468,374.37	66° 0' 25.611 N	8° 18' 10.887 E
1,263.14	0.79	69.22	1,263.06	4.34	8.76	7,321,064.34	468,374.76	66° 0' 25.616 N	8° 18' 10.917 E
1,292.14	0.80	72.08	1,292.06	4.47	9.14	7,321,064.47	468,375.14	66° 0' 25.620 N	8° 18' 10.947 E
1,321.50	0.80	71.03	1,321.42	4.60	9.53	7,321,064.60	468,375.53	66° 0' 25.624 N	8° 18' 10.978 E
1,350.68	0.72	75.72	1,350.60	4.71	9.90	7,321,064.71	468,375.90	66° 0' 25.628 N	8° 18' 11.007 E
1,407.89	0.46	59.14	1,407.80	4.92	10.45	7,321,064.92	468,376.44	66° 0' 25.635 N	8° 18' 11.051 E
1,424.82	0.60	61.37	1,424.73	5.00	10.58	7,321,064.99	468,376.58	66° 0' 25.638 N	8° 18' 11.061 E
1,442.00	0.60	61.37	1,441.91	5.08	10.74	7,321,065.08	468,376.74	66° 0' 25.640 N	8° 18' 11.074 E
Projection to TD									

23/11/2009 15:31:42 *Page 3* *COMPASS 2003.16 Build 65*

Schlumberger

Survey Report - Geographic

Company:	STATOIL NORWAY	**Local Co-ordinate Reference:**	Site 6608/10-13 Fløien
Project:	EXPLORATION - UTM Zone 32	**TVD Reference:**	WELL @ 22.00m (Original Well Elev)
Site:	6608/10-13 Fløien	**MD Reference:**	WELL @ 22.00m (Original Well Elev)
Well:	6608/10-13 Fløien	**North Reference:**	Grid
Wellbore:	6608/10-13 Fløien	**Survey Calculation Method:**	Minimum Curvature
Design:	6608/10-13 Fløien	**Database:**	edm_restore

Casing Points

Measured Depth (m)	Vertical Depth (m)	Name	Casing Diameter (in)	Hole Diameter (in)
456.01	456.00	30"	30.000	36.000
1,136.57	1,136.50	9 5/8"	9.625	12.250

Design Annotations

Measured Depth (m)	Vertical Depth (m)	Local Coordinates +N/-S (m)	Local Coordinates +E/-W (m)	Comment
1,442.00	1,441.91	5.08	10.74	Projection to TD

Checked By: ______________ Approved By: ______________ Date: ______________

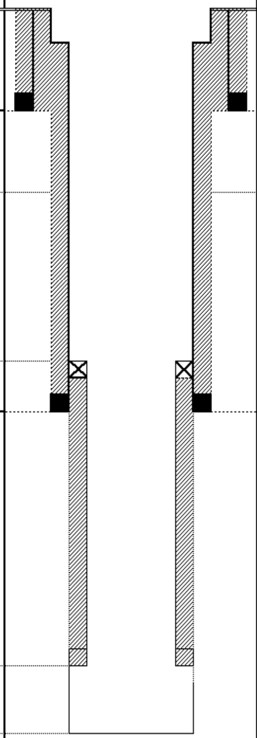
App C Figures and tables

Well: 6608/10-13
Field: Fløien
Rig: Ocean Vanguard

WELL SCHEMATIC

All depths refers to RKB.
RKB-MSL Ocean Vanguard:22 m

HOLE		CASING			LOT / FIT	TOC		CSG. SHOE		RKB		LWD LOGS	SURV CSG/ OH
SIZE	TVD MD	SIZE	TYPE / RAD. MARKERS	CENTRALIZERS	[SG]	TVD	MD	TVD	MD				
SB	408					408	408	408	408				
36"	466.5 466.5	30"	Type: 309.7 lb/ft, X-52, SL-60 Drift: 26,875"	None	N/A			456	456			N/A	MWD
12 1/4"	1145 1145	20" x 9 5/8"	18 3/4" WH housing with XO to 9 5/8" 53,5 lb/ft, P-110, Vam Top	1 x Bow type / 5 first joints	N/A			757 If Liner 1040 1136.5	757 If liner 1040 1136.5			Shallow gas warning Ref. Data Aquisition Program	MWD
8 1/2"	1442 1442		Contingency 7" liner* Interval: To be decided Type: 32 lbf/ft, P-110,Vam Top	As requested	XLOT			As req. 1442	As req. 1442			Ref. Data Aquisition Program	MWD



Comments: *) The 7" liner is a contingency which may be run in case of hole stability problems.

Figure 6-1 Wellbore schematic, well 6608/10-13

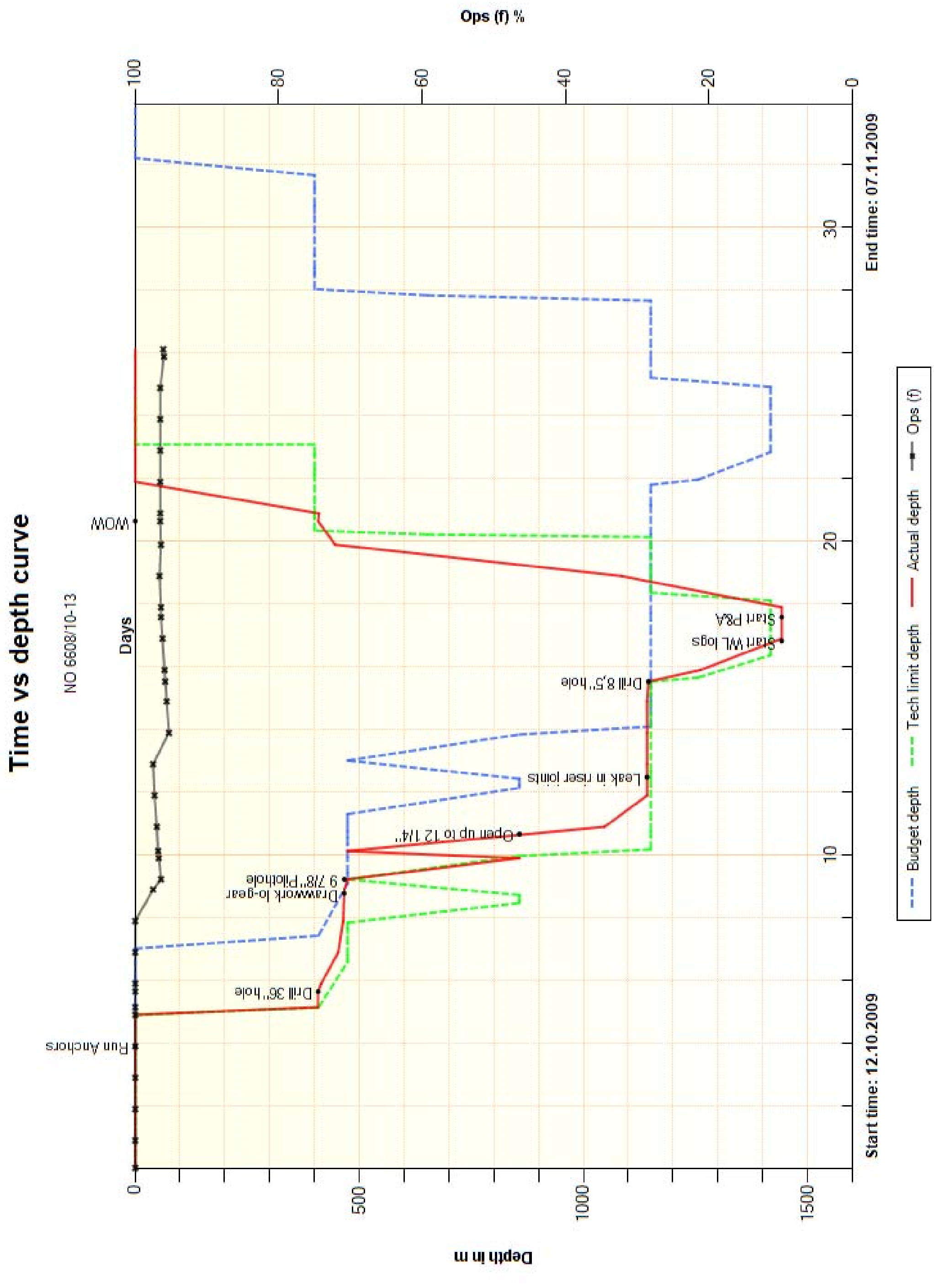


Figure 6-2 Time/depth curve

Table 6-1 Project planner (Project number T. O437A.AP.20100)

Start time	End time	Budget time hrs	Acc budget days	Tech limit hrs	Acc tech days	Plan time hrs	Act time hrs	Acc actual days	Down time	Description	Companies
12.Oct.2009 08:45	17.Oct.2009 12:00	146,0	6,1	102,0	4,2	121,0	123,3	5,1	0,0	MOVE [NO 6608/10-13]	
12.Oct.2009 08:45	15.Oct.2009 06:00	78,0	3,3	67,0	2,8	65,0	69,3	2,9	0,0	1 Transit from Frigg Delta to Fløien	DODI
15.Oct.2009 06:00	16.Oct.2009 06:00	30,0	4,5	0,0	2,8	30,0	24,0	3,9	0,0	2 WOW to commence anchor handling	DODI
16.Oct.2009 06:00	17.Oct.2009 12:00	38,0	6,1	35,0	4,3	26,0	30,0	5,1	0,0	3 AH / Cross tensioning and ballasting	Ocean,DODI
17.Oct.2009 12:00	18.Oct.2009 00:00	32,0	1,3	21,0	0,9	12,0	12,1	0,5	0,0	PRESPUD [NO 6608/10-13]	
17.Oct.2009 12:00	17.Oct.2009 15:15	22,0	7,0	15,0	4,9	4,0	3,3	5,3	0,0	4 P/U and rack back 400 m 5 1/2" DP (Mix Hi-visc, kill & disp.mud)	DODI,Hall,Ocean
17.Oct.2009 15:15	17.Oct.2009 16:15	0,0	7,0	0,0	4,9	3,0	1,0	5,3	0,0	4A MU conductor RT and cement stand and rack back	DODI,Hall,Ocean
17.Oct.2009 16:15	18.Oct.2009 00:00	10,0	7,4	6,0	5,1	5,0	7,8	5,6	0,0	5 MU 36" BHA and RIH to seabed at 409 m. Position rig.	DODI,Hall,Ocean,Schlum, Schlum
18.Oct.2009 00:00	21.Oct.2009 13:45	93,0	3,9	65,0	2,7	65,0	85,9	3,6	7,1	36" [NO 6608/10-13]	
18.Oct.2009 00:00	19.Oct.2009 13:30	40,0	9,1	35,0	6,6	12,0	37,5	7,2	0,0	6 Drill 36" hole from 409 to 474 m.	DODI,Hall,Ocean,Schlum, Schlum
19.Oct.2009 13:30	19.Oct.2009 15:00	3,0	9,2	2,0	6,7	2,0	1,5	7,3	0,0	7 Circulate hole clean and displace to 1,35 g/cm^{3}mud.	DODI,Hall,Ocean,Schlum, Schlum

Start time	End time	Budget time hrs	Acc budget days	Tech limit hrs	Acc tech days	Plan time hrs	Act time hrs	Acc actual days	Down time	Description	Companies
19.Oct.2009 15:00	19.Oct.2009 17:30	5,0	9,4	3,0	6,8	6,0	2,5	7,4	0,0	8 POOH and LD 36" BHA. Perform Safety Check / Derrick Inspection.	DODI,Hall,Ocean,Schlum, Schlum
19.Oct.2009 17:30	20.Oct.2009 05:30	12,0	9,9	9,0	7,2	8,0	12,0	7,9	0,0	9 RU and run 30" conductor w innersting.MU RT lamd in PGB.	DODI,Hall,DQ,Schlum,Ocean,Schlum
20.Oct.2009 05:30	20.Oct.2009 06:45	0,0	9,9	0,0	7,2	2,0	1,3	7,9	0,0	9A Land 30" conductor.	DODI,Hall,DQ,Ocean,Schlum,Schlum
20.Oct.2009 06:45	20.Oct.2009 14:30	10,0	10,3	8,0	7,5	3,0	7,8	8,2	0,0	10 Cement 30" conductor. WOC (approx 5 h).	DODI,DQ,Hall,Hall,Ocean, Schlum
20.Oct.2009 14:30	20.Oct.2009 15:30	5,0	10,5	3,0	7,6	3,0	1,0	8,3	0,0	11 Release RT. POOH.	DODI,DQ,Ocean,OWS,Schlum
20.Oct.2009 15:30	20.Oct.2009 21:30	0,0	10,5	0,0	7,6	8,0	6,0	8,5	0,0	11 A Make up 17 1/2" drillout BHA. RIH. Drill out cement and shoe and 2 m new fmn.	DODI,DQ,Ocean,Schlum
20.Oct.2009 21:30	21.Oct.2009 03:00	5,0	10,8	0,0	7,6	5,0	5,5	8,8	1,8	11 C POOH & L/D 17 1/2" BHA & 36" BHA.	DODI,Ocean
21.Oct.2009 03:00	21.Oct.2009 09:00	0,0	10,8	0,0	7,6	6,0	6,0	9,0	5,3	11 D Repair drawork	DODI
21.Oct.2009 09:00	21.Oct.2009 11:00	7,0	11,0	0,0	7,6	5,0	2,0	9,1	0,0	12A PU approx 600 m 5 1/2" DP	DODI
21.Oct.2009 11:00	21.Oct.2009 13:45	6,0	11,3	5,0	7,8	5,0	2,8	9,2	0,0	14 MU and RIH with 9 7/8" BHA to 469 m	DODI,Hall,Ocean,OWS,Schlum,Schlum
21.Oct.2009 13:45	22.Oct.2009 11:30	41,0	1,7	33,0	1,4	24,0	21,8	0,9	0,0	9 7/8" [NO 6608/10-13]	
21.Oct.2009 13:45	22.Oct.2009 03:15	20,0	12,1	15,0	8,5	12,0	13,5	9,8	0,0	16 Drill 9 7/8" pilot hole from 466 m to 857m (25 m/hr).	DODI,Ocean,Hall,Schlum, Schlum

Start time	End time	Budget time hrs	Acc budget days	Tech limit hrs	Acc tech days	Plan time hrs	Act time hrs	Acc actual days	Down time	Description	Companies
22.Oct.2009 03:15	22.Oct.2009 05:00	2,0	12,2	2,0	8,5	2,0	1,8	9,8	0,0	17 Circulate BU to evaluate for shallow gas. Flowcheck. Displace to 1.35 g/cm^3 mud.	DODI,Hall,Hall,Ocean,Schlum,Schlum
22.Oct.2009 05:00	22.Oct.2009 09:00	5,0	12,4	4,0	8,7	4,0	4,0	10,0	0,0	18 POOH & rack 9 7/8" BHA. L/D MWD. Verify marker buoy location. Perform Derrick Inspection.	DODI,Ocean,Hall,Hall,Schlum,Schlum
22.Oct.2009 09:00	22.Oct.2009 11:30	14,0	13,0	12,0	9,2	6,0	2,5	10,1	0,0	19 MU 12 1/4" BHA and RIH.	DODI,Schlum,Hall,Ocean, Schlum
22.Oct.2009 11:30	27.Oct.2009 21:15	211,0	8,8	151,5	6,3	118,5	129,9	5,4	7,5	12 1/4" [NO 6608/10-13]	
22.Oct.2009 11:30	23.Oct.2009 00:30	20,0	13,8	18,0	10,0	8,0	13,0	10,7	0,0	20 Open 9 7/8" pilot hole to 12 1/4" hole from 466 m to 857	DODI,Schlum,Hall,Ocean, Schlum
23.Oct.2009 00:29	23.Oct.2009 00:30	6,0	14,1	5,0	10,2	0,0	0,0	10,7	0,0	23 Wipertrip (optional)	DODI,Schlum,Hall,Ocean, Schlum
23.Oct.2009 00:30	23.Oct.2009 11:00	15,0	14,7	10,0	10,6	10,0	10,5	11.1	0,0	20 A Drill 12 1/4" hole from 857 m to 1142 m, 30 m/h	DODI,Hall,Ocean,Schlum, Schlum
23.Oct.2009 11:00	23.Oct.2009 13:00	5,0	14,9	4,0	10,7	4,0	2,0	11,2	0,0	21 Circulate hole clean and displace to 1,35 g/cm^3 mud.	DODI,Schlum,Hall,Ocean, Schlum
23.Oct.2009 13:00	23.Oct.2009 19:30	11,0	15,4	9,0	11.1	7,0	6,5	11,4	0,0	22 POOH, top up hole. Wash 30" con.housing. Rack BHA.	DODI,Schlum,Hall,Ocean, Schlum
23.Oct.2009 19:30	23.Oct.2009 21:00	6,0	15,6	0,0	11.1	5,0	1,5	11,5	0,0	23 A MU cement stand and 18 3/4" WH housing.	Hall,Ocean,Schlum,Schlum ,DODI
23.Oct.2009 21:00	24.Oct.2009 12:30	35,0	17,1	25,0	12,2	12,0	15,5	12,2	0,0	24 RU and run 9 5/8"c g/cm3./18 3/4" WH housing on 5 1/2" DP.	DODI,Schlum,DQ,Hall,Ocean,OWS
24.Oct.2009 12:30	24.Oct.2009 16:45	9,0	17,5	7,0	12,5	7,0	4,3	12,3	0,0	25 Circulate, pump and displace cement.	DODI,Schlum,Hall,Hall,Ocean,OWS

Start time	End time	Budget time hrs	Acc budget days	Tech limit hrs	Acc tech days	Plan time hrs	Act time hrs	Acc actual days	Down time	Description	Companies
24.Oct.2009 16:45	24.Oct.2009 20:00	6,0	17,7	5,0	12,7	5,0	3,3	12,5	0,0	26 Release RT and wash 18 3/4" WH. POOH.	DODI,Schlum,DQ,Hall,Ocean
24.Oct.2009 20:00	26.Oct.2009 12:00	60,0	20,2	40,0	14,3	30,0	40,0	14,1	7,5	27 Rig and run BOP	DODI,DQ,Hall,Ocean
26.Oct.2009 12:00	26.Oct.2009 15:00	0,0	20,2	0,0	14,3	3,0	3,0	14,3	0,0	27 a PU and Rack back hang off tool. LD excess DC`s	DODI,Hall,DQ,Ocean
26.Oct.2009 15:00	26.Oct.2009 23:00	22,0	21.1	18,0	15,1	14,0	8,0	14,6	0,0	28 MU/RIH with 8 1/2" BHA. P/U +/- 350 m 5" DP while RIH. (TD 1537m)	DODI,Schlum,Hall,Schlum, Schlum
26.Oct.2009 23:00	27.Oct.2009 00:00	1,0	21,2	0,5	15,1	0,5	1,0	14,6	0,0	29 Perform Kick Drill	DODI,Schlum,Hall,Schlum
27.Oct.2009 00:00	27.Oct.2009 01:45	0,0	21,2	0,0	15,1	3,0	1,8	14,7	0,0	29a Pick up 250 m with 5 1/2" DP. Rack back 9 stands. Build pipe running in.	DODI,Hall,Schlum,Schlum
27.Oct.2009 01:45	27.Oct.2009 16:15	10,0	21,6	6,0	15,4	6,0	14,5	15,3	0,0	30 Drill out shoe track and 3 m new formation while displacing to 1.25 g/cm^3 mud	DODI,Schlum,Hall,Schlum
27.Oct.2009 16:15	27.Oct.2009 21:15	5,0	21,8	4,0	15,5	4,0	5,0	15,5	0,0	31 Perform XLOT. Test surface lines.	DODI,Schlum,Hall,Schlum
27.Oct.2009 21:15	29.Oct.2009 22:30	75,0	3,1	62,0	2,6	39,0	49,4	2,1	0,0	8 1/2" [NO 6608/10-13]	
27.Oct.2009 21:15	28.Oct.2009 10:00	4,0	22,0	3,0	15,6	3,0	12,8	16,1	0,0	32 Drill 8 1/2" hole to core point.	DODI,Schlum,Hall,Schlum, Schlum
28.Oct.2009 09:59	28.Oct.2009 10:00	0,0	22,0	0,0	15,6	0,0	0,0	16,1	0,0	33 Option If HC. POOH and RIH with core assembly	DODI,Schlum,Hall,Schlum, Schlum
28.Oct.2009 09:59	28.Oct.2009 10:00	0,0	22,0	0,0	15,6	0,0	0,0	16,1	0,0	34 Option Cut core and POOH	DODI,Schlum,Hall,Schlum, Schlum

Start time	End time	Budget time hrs	Acc budget days	Tech limit hrs	Acc tech days	Plan time hrs	Act time hrs	Acc actual days	Down time	Description	Companies
28.Oct.2009 10:00	29.Oct.2009 04:15	21,0	22,8	17,0	16,4	14,0	18,3	16,8	0,0	34a Drill 8 1/2" hole to 1417 m. Dry case POOH, L/D BHA. RU wire line equipment	DODI,Hall,Schlum,Schlum, Schlum
29.Oct.2009 04:15	29.Oct.2009 10:15	12,0	23,3	10,0	16,8	8,0	6,0	17,1	0,0	35 WL run # 1 (PEXlite, DSI) Den, Neu, Sonic.	DODI,Schlum,Schlum,Hall, Schlum
29.Oct.2009 10:14	29.Oct.2009 10:15	12,0	23,8	10,0	17,2	0,0	0,0	17,1	0,0	36 Pressure test BOP (optional).	DODI,Schlum,Hall,Schlum
29.Oct.2009 10:15	29.Oct.2009 16:15	12,0	24,3	10,0	17,6	7,0	6,0	17,3	0,0	37 WL run # 2 (MDT).	DODI,Hall,Schlum,Schlum
29.Oct.2009 16:15	29.Oct.2009 22:30	14,0	24,9	12,0	18,1	7,0	6,3	17,6	0,0	38 WL run # 3 (VSP).	DODI,Hall,Schlum,Schlum
29.Oct.2009 22:30	02.Nov.2009 00:00	162,0	6,8	119,5	5,0	69,0	73,6	3,1	2,0	PERM P&A [NO 6608/10-13]	
29.Oct.2009 22:30	30.Oct.2009 18:30	7,0	25,2	6,0	18,4	6,0	20,0	18,4	0,0	45 RIH w/ 9 5/8”EZSV on DP. Set EZSV and pressure test .	DODI,Hall,Schlum
30.Oct.2009 18:29	30.Oct.2009 18:30	15,0	25,8	10,0	18,8	0,0	0,0	18,4	0,0	39 Cancelled: RIH w/ cmt stinger. PU 3 1/2" DP.Estimate 300 m.	DODI,Hall,Schlum
30.Oct.2009 18:29	30.Oct.2009 18:30	0,0	25,8	0,0	18,8	0,0	0,0	18,4	0,0	40 cancelled; Optional. Circulate out HC prior to plug back.	DODI,Hall,Hall,Schlum
30.Oct.2009 18:29	30.Oct.2009 18:30	9,0	26,2	6,0	19,0	0,0	0,0	18,4	0,0	41 Cancelled; Plug back OH and 100 m into 9 5/8”liner.	DODI,Hall,Schlum
30.Oct.2009 18:29	30.Oct.2009 18:30	24,0	27,2	18,0	19,8	0,0	0,0	18,4	0,0	42 Cancelled; LD exceesive DP while WOC	DODI
30.Oct.2009 18:29	30.Oct.2009 18:30	1,0	27,2	0,5	19,8	0,0	0,0	18,4	0,0	43 Cancelled; Pressure test cmt plug	DODI,Hall,Hall,Schlum

Start time	End time	Budget time hrs	Acc budget days	Tech limit hrs	Acc tech days	Plan time hrs	Act time hrs	Acc actual days	Down time	Description	Companies
30.Oct.2009 18:29	30.Oct.2009 18:30	10,0	27,7	8,0	20,1	0,0	0,0	18,4	0,0	44 Cancelled; Option Scraper run for EZSV.	DODI,Hall,Hall,Schlum
30.Oct.2009 18:30	31.Oct.2009 02:15	4,0	27,8	2,0	20,2	6,0	7,8	18,7	0,0	46 Set 3* 212 m cement plug on top of bridge plug.	DODI,Hall,Schlum
31.Oct.2009 02:15	31.Oct.2009 04:00	5,0	28,0	3,0	20,3	3,0	1,8	18,8	0,0	47 Pull out to TOC and displace well to SW. Wash wellhead and riser.	DODI,Hall,Hall,Schlum
31.Oct.2009 03:59	31.Oct.2009 04:00	11,0	28,5	9,0	20,7	0,0	0,0	18,8	0,0	48 POOH (LD excess DP will be minimised).	DODI
31.Oct.2009 04:00	31.Oct.2009 06:00	2,0	28,6	2,0	20,8	2,0	2,0	18,9	0,0	49 RIH and retrieve wear bushing. Lay out excess DP/BHA	DODI
31.Oct.2009 06:00	31.Oct.2009 11:00	3,0	28,7	2,0	20,9	2,0	5,0	19,1	0,0	50 RU riser handling gear.	DODI
31.Oct.2009 10:59	31.Oct.2009 11:00	10,0	29,1	8,0	21,2	2,0	0,0	19,1	0,0	55 Lay out tubulars from derrick. Backload equipment.	DODI
31.Oct.2009 11:00	01.Nov.2009 11:30	32,0	30,5	22,0	22,1	25,0	24,5	20,1	2,0	51 Pull RISER and BOP	DODI
01.Nov.2009 11:30	01.Nov.2009 16:00	6,0	30,7	4,0	22,3	3,0	4,5	20,3	0,0	52 MU and RIH w/MOST cut assy.	DODI
01.Nov.2009 16:00	01.Nov.2009 21:00	13,0	31,2	11,0	22,7	11,0	5,0	20,5	0,0	53 Cut and retrieve 20" x 30" c g/cm3and WH.	DODI,Hall
01.Nov.2009 21:00	02.Nov.2009 00:00	10,0	31,7	8,0	23,1	9,0	3,0	20,6	0,0	54 LD 18 3/4" WH. Remove PGB from moonpool area.	DODI
02.Nov.2009 00:00	07.Nov.2009 11:30	55,0	2,3	36,0	1,5	20,0	131,5	5,5	3,0	MOVE [NO 6608/10-13]	

Start time	End time	Budget time hrs	Acc budget days	Tech limit hrs	Acc tech days	Plan time hrs	Act time hrs	Acc actual days	Down time	Description	Companies
02.Nov.2009 00:00	06.Nov.2009 08:00	13,0	32,2	0,0	23,1	0,0	104,0	25,0	0,0	56 WOW (5.7 %). Inspect seabed with ROV and retrieve buoys.	DODI,Ocean
06.Nov.2009 08:00	07.Nov.2009 11:30	42,0	34,0	36,0	24,6	20,0	27,5	26,1	3,0	57 Deballast rig and pull anchors.	DODI

Bit Record

WELLBORE ID: NO 6608/10-13
INTERVAL: MOVE
START TIME: 12.Oct.2009 08:45
END TIME: 17.Oct.2009 12:00

Table 6-2 Bit record

Run no	Bit size	Bit no	BHA no	Bit type	IADC code	Bit manufacturer
1	36"	1	1	MX-ST31	115	Hughes Christensen
2	17 1/2"	2	2	XR+VEC	115	Smith Bits
3	9 7/8"	3	3	MXC1	117	Hughes Christensen
4	12 1/4"	4	4	EQHC1RC	117W	Security
5	8 1/2"	5	5	MDI 616		UNKNOWN

					Nozzles (n/32”)				
Run no	Bit size	Bit no	BHA no	Serial no	no x n	no x n	no x n	no x n	Flow area in2
1	36"	1	1	6027866	3 x 16	3 x 12	3 x 13	3 x 10	1,540
2	17 1/2"	2	2	PM5384	4 x 14	x	x	x	0,602
3	9 7/8"	3	3	5032868	4 x 20	x	x	x	1,228
4	12 1/4"	4	4	11378273	2 x 18	2 x 16	x	x	0,890
5	8 1/2"	5	5	SY 9085	5 x 12	1 x 11	x	x	0,646

Run no	Bit size	Pump rate l/min	Pump press bar	Depth in mMD	Depth out mMD	Form drld m	Total drld m	Drld hrs	Circ hrs	ROP m/hr
1	36"	4500	155	408,0	452,0	44,0	44,0	17,1	22,5	2,6
2	17 1/2"	3600	200	452,0	466,0	14,0	5,0	0,5	1,8	28,0
3	9 7/8"	3770	135	466,0	857,0	391,0	391,0	10,8	14,3	36,2
4	12 1/4"			466,0	1142,0	676,0	676,0	18,4	22,3	36,7
5	8 1/2"			1142,0	1442,0	300,0	337,0	19,0	36,9	15,8

Run no	Bit size	Min WOB tonne	Max WOB tonne	Min RPM	Max RPM	Torque Min Nm	Torque Max Nm	Con drag Min 1000 1000 daN	Con drag Max 1000 1000 daN
1	36"	4	10	80	110	4	10	1	1

Run no	Bit size	Min WOB tonne	Max WOB tonne	Min RPM	Max RPM	Torque Min Nm	Torque Max Nm	Con drag Min 1000 1000 daN	Con drag Max 1000 1000 daN
2	17 1/2"	4	6	60	60	4	6		
3	9 7/8"	5	8	80	85	2	4		
4	12 1/4"	3	14	115	120	3	8	1	1
5	8 1/2"	2	6	110	115	2	6		

Run no	Bit size	I	O	DC	L	B	G	OC	RP
1	36"	3	3	WT	A	E	I	NO	TD
2	17 1/2"	1	1	NO	A	E	I	NO	TD
3	9 7/8"	2	2	WT	N	E	I	BT	TD
4	12 1/4"	2	2	WT	S	E	I	ER	TD
5	8 1/2"	1	2	BT	S	X	I	PN	TD

Run no	Bit size	Remarks
1	36"	float/3 stage HO,36"HO,bitsub 9 1/2" pony, 17 1/2" BIT MX-ST36
2	17 1/2"	
3	9 7/8"	
4	12 1/4"	TBR: 141 krevs
5	8 1/2"	159 krevs. Smith, 6 bladed PDC. Bit data at 04:00 One partially plugged waterway

BHA List

WELLBORE: NO 6608/10-13
BHA NO: 1
BHA KIND:
DESCRIPTION: 36" hole
BHA NAME: 1

String component	OD in	ID in	Length m	Acc length m
MX-ST31~17 1/2"	36,000		12,02	12,02
POWERPULSE	9,125		8,82	20,84
NM DRILL COLLAR	9,125	3,000	9,09	29,93
NM DRILL COLLAR	9,125	3,000	9,18	39,11

String component	OD in	ID in	Length m	Acc length m
DRILL COLLAR	9,500	3,000	9,44	48,55
DRILL COLLAR	9,500	3,000	8,71	57,26
XO	9,500	3,000	1.15	58,41
DRILL COL	8,000	3,000	9,42	67,83
DRILL COL	8,000	3,000	9,41	77,24
DRILL COL	8,000	3,000	9,43	86,67
DRILL COL	8,000	3,000	9,17	95,84
DRILL COL	8,000	3,000	8,88	104,72
DRILL COL	8,000	3,000	9,36	114,08
DRILL COL	8,000	3,000	9,05	123,13
XO			1,01	124,14
HW DRILL PIPE, 5½"			114,22	238,36

BHA NO: 2
BHA KIND:
DESCRIPTION: 17 1/2" drillout assembly.
BHA NAME: 2

String component	OD in	ID in	Length m	Acc length m
XR+VEC~26"	17,500		0,45	0,45
BIT SUB	9,500	3,000	0,91	1,36
9 1/2" DRILL COLLAR	9,500	3,000	9,44	10,80
9 1/2" DRILL COLLAR	9,375	3,000	8,71	19,51
XO	9,500	3,000	1.15	20,66
DRILL COLLAR	8,000	3,000	37,43	58,09
JAR	8,250	3,000	9,39	67,48
DRILL COL	8,000	3,000	27,29	94,77
XO	8,000	3,000	1,01	95,78
HW DRILL PIPE, 5½"	7,250	3,000	114,22	210,00

BHA NO: 3
BHA KIND:
DESCRIPTION: 9 7/8" pilot
BHA NAME: 3

String component	OD in	ID in	Length m	Acc length m
MXC1~12 1/4"	9,875		0,26	0,26
BIT SUB	8,000		1,22	1,48
VISION RES 8W / APWD	8,250		6,15	7,63
POWERPULSE	8,250		8,36	15,99

String component	OD in	ID in	Length m	Acc length m
NM DRILL COLLAR	8,250	2,750	9,25	25,24
NM DRILL COLLAR	8,188	2,750	9,08	34,32
DRILL COL	8,000	3,000	46,87	81.19
JAR	8,250	3,000	9,36	90,55
DRILL COL	8,000	3,000	54,18	144,73
XO	7,250	3,000	1,01	145,74
HW DRILL PIPE, 5½"	5,500		114,22	259,96

BHA NO: 4
BHA KIND:
DESCRIPTION: 12 1/4" hole
BHA NAME: 4

String component	OD in	ID in	Length m	Acc length m
EQHC1RC~NA	12,250		0,33	0,33
NB STAB W/FLOAT	12,125		1,71	2,04
VISION RES 8W / APWD	8,250		6,15	8,19
STABILIZER, NM	12,125		1,53	9,72
POWERPULSE	8,250		8,36	18,08
STABILIZER, NM	12,125		1,79	19,87
NM DRILL COLLAR	8,250	2,750	9,25	29,12
NM DRILL COLLAR	8,188	2,750	9,08	38,20
DRILL COL	8,000	3,000	46,87	85,07
JAR	8,250	3,000	9,36	94,43
DRILL COL	8,000	3,000	54,63	149,06
XO	7,250	3,000	1,01	150,07
HW DRILL PIPE, 5½"	5,500		114,22	264,29

BHA NO: 5
BHA KIND:
DESCRIPTION: 8 1/2" hole
BHA NAME: 5

String component	OD in	ID in	Length m	Acc length m
MDI 616~8 1/2"	8,500	2,250	0,26	0,26
GVR 675	6,750	4,880	3,88	4,14
VISION RES 6 W/ APWD	6,875	2,810	5,90	10,04
STABILIZER, NM	6,750	2,810	1,52	11,56
POWERPULSE	6,750	5,110	8,63	20,19
STABILIZER, NM	6,875	2,810	2,16	22,35

String component	OD in	ID in	Length m	Acc length m
NM DRILL COLLAR	6,875	3,000	9,39	31,74
DRILL COLLAR	6,500	2,875	54,89	86,63
JAR	6,500		9,90	96,53
DRILL COLLAR	6,500	2,875	27,23	123,76
HW DRILL PIPE, 5"	5,000	3,000	111,35	235,11
DP 5"	5,500	4,670	173,83	408,94
XO	5,500	2,750	1.13	410,07
DP 5 1/2"	5,500	4,670		410,07

Drilling Fluids

Well: 6608/10-13
Field: Fløien
Rig: Ocean Vanguard

DRILLING FLUIDS PROGRAMME

All depths refers to RKB.
RKB-MSL Ocean Vanguard: 22 m

HOLE SIZE	HOLE TVD MD	CASING SIZE	CASING TVD MD	MUD TYPE	MW [SG]	LGS [kg/m3]	10 sec. [Pa]	10 min. [Pa]	Funnel Visc. Sec	Fann 3 rpm	Fann 100 rpm	PV [mPa]	API FL [ml]	HTHP FL [ml]	MBT [KG/m3]	pH	KCl [KG/m3]	Glyc. [%]	Ca ++ [mg/l]	Sulphat [mg/l]	Usage Volume [m3]
36"	466.5 466.5	30"	456 456	Sweep Spec 2a Dispacement Spec 2b	1.03-1.06 1.03 - 1.04 1.35 1.35 - 1.60				>100 105 - 116				<10 3.0 - 3.2			>9 9.0-9.5 >9 9.3 - 9.9					100 128
				Drill with seawater at the highest possible rate and flush the hole with 10 m3 high viscosity sweep mud for every 15 m drilled. PAC RE will be used for viscosifying seawater. Have at least 60 m3 of 1.60 SG bentonite mud in a pit as kill mud before drilling this section. Pump 2 X 20 m3 of sweep mud at TD and displace the hole 1½ times the open hole volume to 1.35 SG displacement mud before pulling out to run 30" casing. If hole has to be reamed, RIH to TD, sweep and displace as above. PAC RE is a sacked viscosifying alternative for seawater. During the drilling phase the formation was seen to be very hard. The hole was drilled with low WOB to reduce build angle. Check survey carried out 417m, 0.11 degree inclination. A 10m3 Hi-vis sweep was pumped at the first connection (4m drilled) and generally 5m3 pumped every mid stand and again prior to connections. Boulders where drilled from 437-438m with low weight on bit. Survey's where taken at 432m = 2.37 degrees inclination. The string was rack back one single and commenced reaming area between 420-448m. Reaming operations where continued from 425m to 452m. Drilling operations commenced again from 452m to 364m. Operations required flushing and cleaning the hole, bottoms up was carried out followed by pumping a 20m3 Hi-vis sweep. The hole was then displaced with 1.35sg mud. A wiper tyrip was made and no fill was observed. The well was then re-displaced to 1.50sg mud at 4000 litres per minute. POOH. (One cone m HO). The conductor was landed conductor 455m. The conductor required to be held with tension on during cement job with approx 1m stick up.																	
12 1/4"	1145 1145	20" x 9 5/8"	1136.5 1136.5	Sweep Spec 2a Dispacement Spec 2b	1.03-1.06 1.03 1.35 1.35 - 1.60				>100 109 - 118 >100				<10 3.0 - 9.6			>8 9.0 - 9.3 >8 9.0 - 9.6					643 356
				Pilot Hole: Drill with seawater at the highest possible rate and flush the hole with 5 m3 high viscosity sweep mud for every 15 m drilled. Have at least 60 m3 of 1.60 SG bentonite mud in a pit as kill mud before drilling this section. Pump 2 X 20 m3 of sweep mud at TD and displace the hole 1½ times the open hole volume to 1.35 SG displacement mud before pulling out to open the hole. PAC RE will be used for viscosifying seawater. The operations required to Pump Hi-vis sweeps, 5m3 mid stand and 5m3 prior to connection. Boulders where encountered at 772m, drilled through and ahead to the section total depth of 857m. Through out the drilling of the section the well was dynamically flow checked at all potential gas zones, no gas observed. The hole was circulated clean by pumping 2 x 10m3 hi-vis sweeps, before displacing to 1.35sg mud (50 m3 pumped). **If no shallow gas:** Drill with seawater at the highest possible rate and flush the hole with 5 - 8 m3 high viscosity sweep mud for every 15 m drilled. Have at least 60 m3 of 1.60 SG bentonite mud in a pit as kill mud before drilling this section. Pump 2 X 20 m3 of sweep mud at TD and displace the hole 1½ times the open hole volume to 1.35 SG displacement mud before pulling out to run 9 5/8" casing. If hole has to be reamed, RIH to TD, sweep and displace as above. PAC RE will b Opened up hole with typically 3650 lpm sea water and 5-10m3 Hi-vis sweeps at mid stand and prior to connections, or as hole dictated. From 857m drilling proceeded with similar parameters and ROP until 1047m MD where gross 23m/hr and continued to T.D. at 1144m. Pumped a 15m3 Hi-vis sweep and circulated hole clean. Repeated above before displacing the wellbore to 1.35sg Spec 2a mud and POOH with BHA. When inside the 30" conductor topped up 1.35sg mud and L/D BHA. Rigged up and RIH 95/8" casing without incident or hole problems. Cemented the casing.																	
8 1/2"	1442 1442	7"		PERFORMADR IL Hi Perform WBM	1.19 - 1.25 1.20 - 1.22	<170 3.2 - 57		<20 2 - 2.4		5 - 12 1 - 3	<30 11 - 23		<4 2.6 - 3.0		<50 0 - 28		50 - 150 Contract Use high side of spec 138 - 154	3 - 5 3.5 - 5	<800 100 - 200		186
				Drill out the shoe track with the spud mud from the previous section. Displace the hole with 1.21 SG PERFORMADRIL mud and perform the EXLOT. Maintain the properties with premix, or direct additions on the rig. Maintain the density as hole conditions dictate. To help alleviate differential sticking, treat the active system with 25 kg/m3 of BARACARB 50 and add BARACARB 150 & BARACARB 600 at about 2 sacks per hour each while drilling. Once the drilling is finished stop the mixing of the coarse BARACARBS and screen out the coarser materials. All available solids control equipment must be utilised to its maximum efficiency to control the low gravity drilled solids (LGS) content.It is important to keep the MBT level and the Calcium level as low as possible at all times. If poor hole cleaning is experienced, start pumping HiVis sweeps to clean the hole. Baroid's DFG Drill Ahead hydraulics will be run as required to ensure proper hole cleaning and minimum ECDs are obtained. Commenced BARACARB 50 additions up to total concentration of 25 kg/m3 and observed ECD increase from 1.34 to 1.43 sg. Further ECD increase to 1.47 sg required reaming of the stand between 1180-1206m. Continued drilling with ECD in range of 1.38-1.40 sg at controlled ROP to allow evaluation of cuttings to pick core point. Drilled through Åre 2 sand without HC shows, thus no coring. Drilling progressed at controlled ROP due to ECD limitation of 1.45 sg. Low end rheology was increased to improve hole cleaning in attempt to reduce ECD but whole mud losses were experienced at the shakers. The fine API 140 (200 mesh equivalent) screens were reduced to coarse API 70 (84 mesh equivalent) on two shakers to allow circulation of cutting out of the hole, thereafter one shaker was fined up again and one shaker left with coarse screens to allow drilling to progress at an acceptable rate. Tried boosting riser to aid hole cleaning but ECD remained around 1.45 sg until section T.D at 1442m MD. Circulated 1 x B.U prior to pumping 10m3 BARACARB pill and circulating second B.U. to clean hole (2030 lpm / 125 bar, 100 rpm / 2-3 kNm TRQ). POOH with 10-15 tons overpull on first three stands, wiped back through area until no overpull observed. One further tight spot at 1288m, which again was wiped through. L/D drilling assembly and rigged up for wireline logging.																	

Cementing operation

Casing/Liner cementing

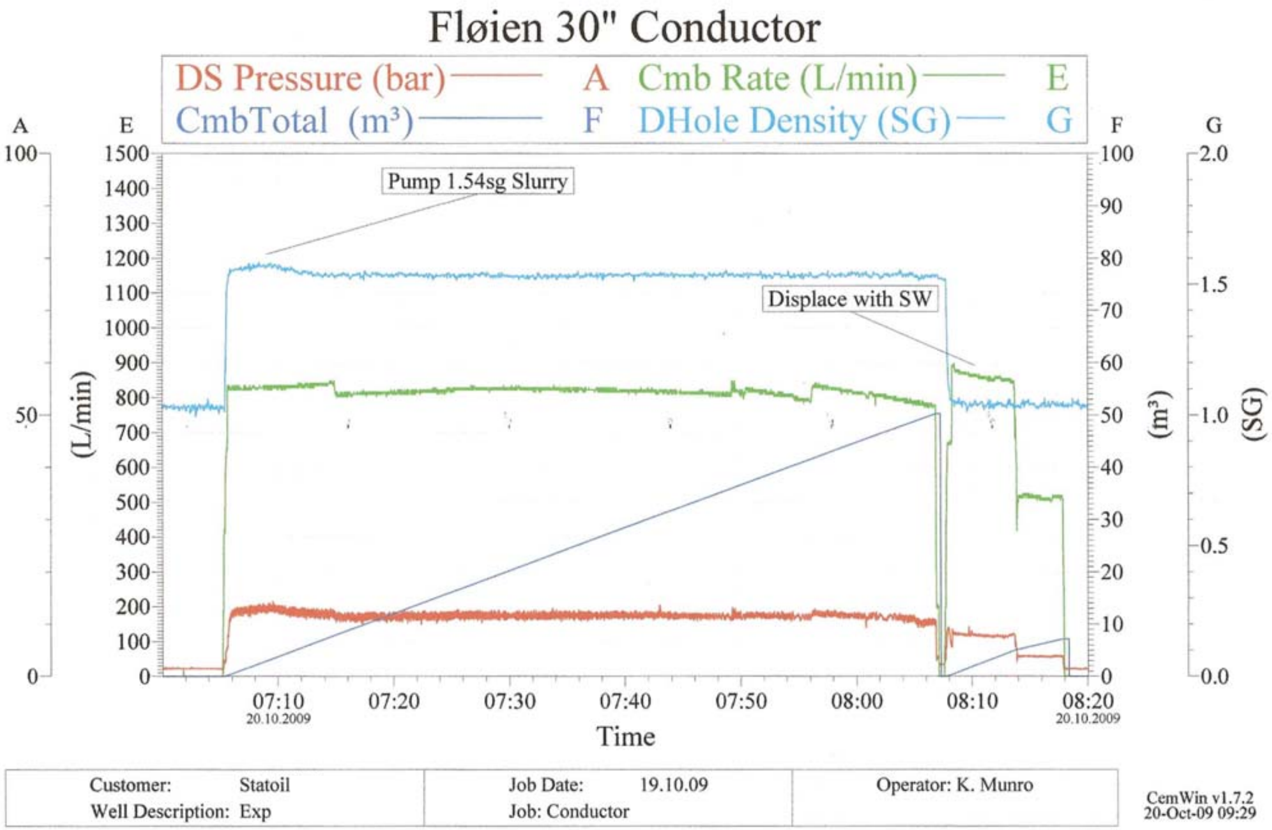
WELLBORE ID: NO 6608/10-13
CASING SIZE: 30"
STAGE CEMENTING: No
REPORT DATE: 21.Oct.2009
THEORETICAL TOC: 409 mMD
EVALUATED TOC: 0 mMD
EVALUATED BY BOND LOG:
LINER ROTATION PLANNED:
LINER ROTATION ACHIEVED:
REMARKS: Tuned Light cement, slurry 6 - DWLSP.

Objective:
Objective: Provide structural support for the conductor and bring TOC up to sea bed without losses

Execution:
Mixing and pumping according to plan. No operational problems.

Fluids pumped	Type	Density g/cm3	Volume m3	Pump Rate l/min	Pump Press bar	Return	Component	Quantity	Unit	Density g/cm3	Premixed in mixwater
Tail	Cement Slurry DWSLP	1,54	51,0	850	14	F	CALCIUM CHLORIDE LIQUID	4,50	l/100kg	1,32	
							NF-6, DEFOAMER	0,14	l/100kg	0,94	
							Seawater	59,75	l/100kg	1,03	

Fluids pumped	Type	Density g/cm3	Volume m3	Pump Rate l/min	Pump Press bar	Return	Component	Quantity	Unit	Density g/cm3	Premixed in mixwater
							Tuned Light cement	100,00	kg/100kg		
Displacement	Sea Water	1,03	7,2	900	8	F					



Evaluation:
Drill ahead ok.

CASING SIZE: 9 5/8"
STAGE CEMENTING: No
REPORT DATE: 25.Oct.2009
THEORETHICAL TOC: 405 mMD
EVALUATED TOC: 405 mMD
EVALUATED BY BOND LOG:
LINER ROTATION PLANNED:
LINER ROTATION ACHIEVED: Y
REMARKS: GasStop (PO-slurry) and GasStop (13-GTTLT)

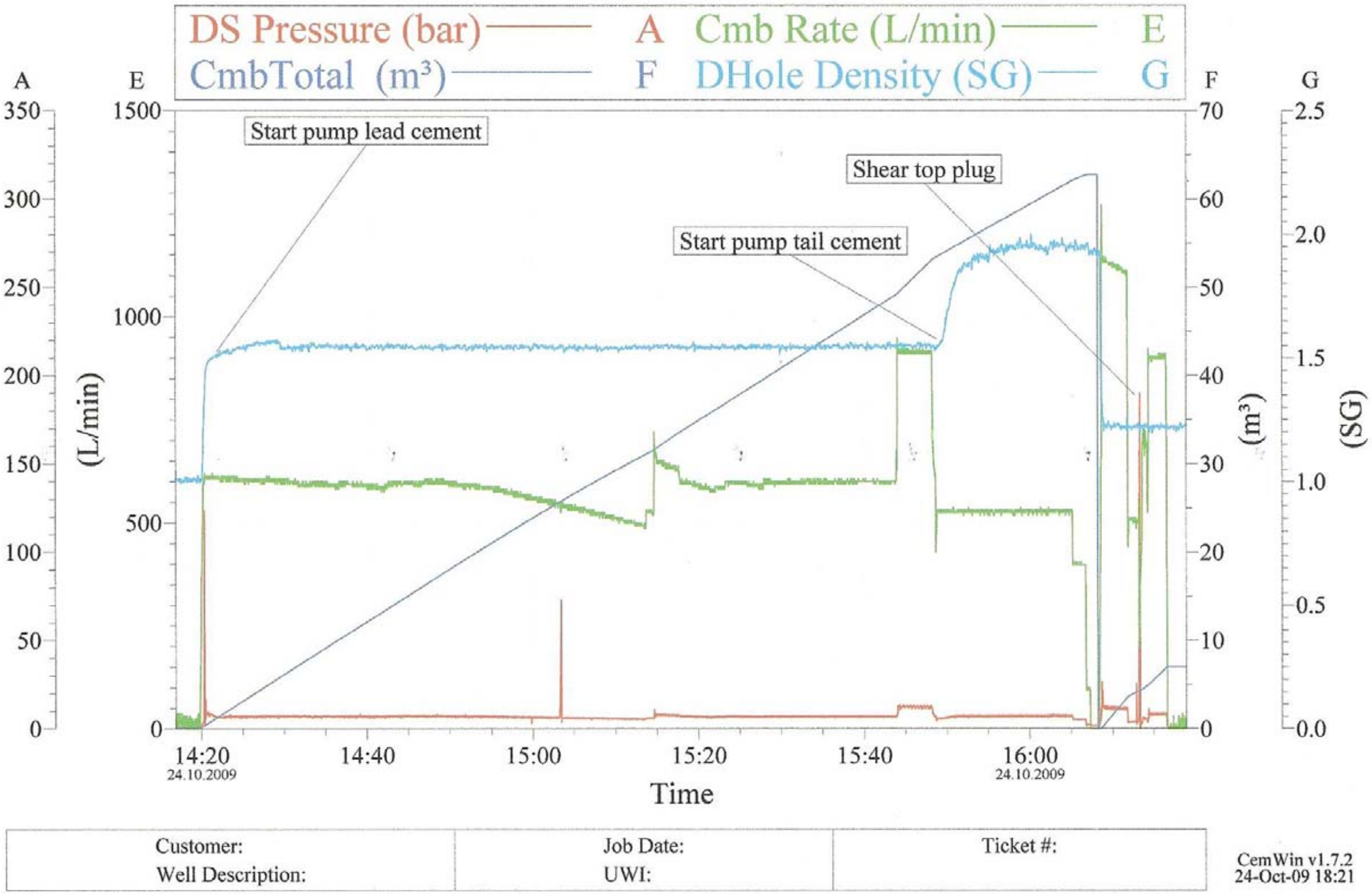
Objective:
Minimum cement requirement to 50 m above shallowest permable zone (507 m) -required top of cement 457 m

Execution:
No problems during mixing and displacing of the cement. The top plug bumped, but could not get a good pressure test on liner. Had to postpone the pressure test until cement was set up

Fluids pumped	Type	Density g/cm3	Volume m3	Pump Rate l/min	Pump Press bar	Return	Component	Quantity	Unit	Density g/cm3	Premixed in mixwater
Lead	Cement Slurry GTLLT	1,65	54,0	600	6	F	CFR-8L	0,00	l/100kg		
							Freshwater	74,37	l/100kg	1,00	
							HALAD 400L	8,00	l/100kg		
							HALAD-99LE+	4,00	l/100kg	1.10	
							Microsilica Liquid	40,00	l/100kg	1,39	
							NF-6, DEFOAMER	0,10	l/100kg	0,94	
							Norcem "G"	635,00	kg/m3	3,22	
Tail	Cement Slurry GTTLT	1,90	8,0	600	7	F	CFR-8L	1,50	l/100kg		

Fluids pumped	Type	Density g/cm3	Volume m3	Pump Rate l/min	Pump Press bar	Return	Component	Quantity	Unit	Density g/cm3	Premixed in mixwater
							HALAD 400L	2,50	l/100kg		
							HALAD-99LE+	0,00	l/100kg	1,02	
							HR-4L	0,80	l/100kg	1.18	
							Microsilica Liquid	12,00	l/100kg	1,39	
							NF-6, DEFOAMER	0,10	l/100kg	0,94	
							Norcem "G"	1215,00	kg/m3	3,22	
							Seawater	34,28	l/100kg	1,03	
Displacement	Water based mud	1,21	7,0	1068	12	F	CFR-8L	1,50	l/100kg		

Table 6-3 Cement details 9 5/8"casing


DS Pressure (bar) A
Cmb Rate (L/min) E
CmbTotal (m³) F
DHole Density (SG) G
A
E
F
G
Start pump lead cement
Shear top plug
Start pump tail cement
(L/min)
(m³)
(SG)
350
300
250
200
150
100
50
0
1500
1000
500
0
70
60
50
40
30
20
10
0
2.5
2.0
1.5
1.0
0.5
0.0
14:20
24.10.2009
14:40
15:00
15:20
15:40
16:00
24.10.2009
Time
Customer:
Well Description:
Job Date:
UWI:
Ticket #:
CemWin v1.7.2
24-Oct-09 18:21

Evaluation:
Liner pressure testd to 133 bar afterlanding of BOP. No cement in first part of the shoe, hard cement the last 10 m. XLOT performed to 1,49 g/cm^3 EMW.

Plugging

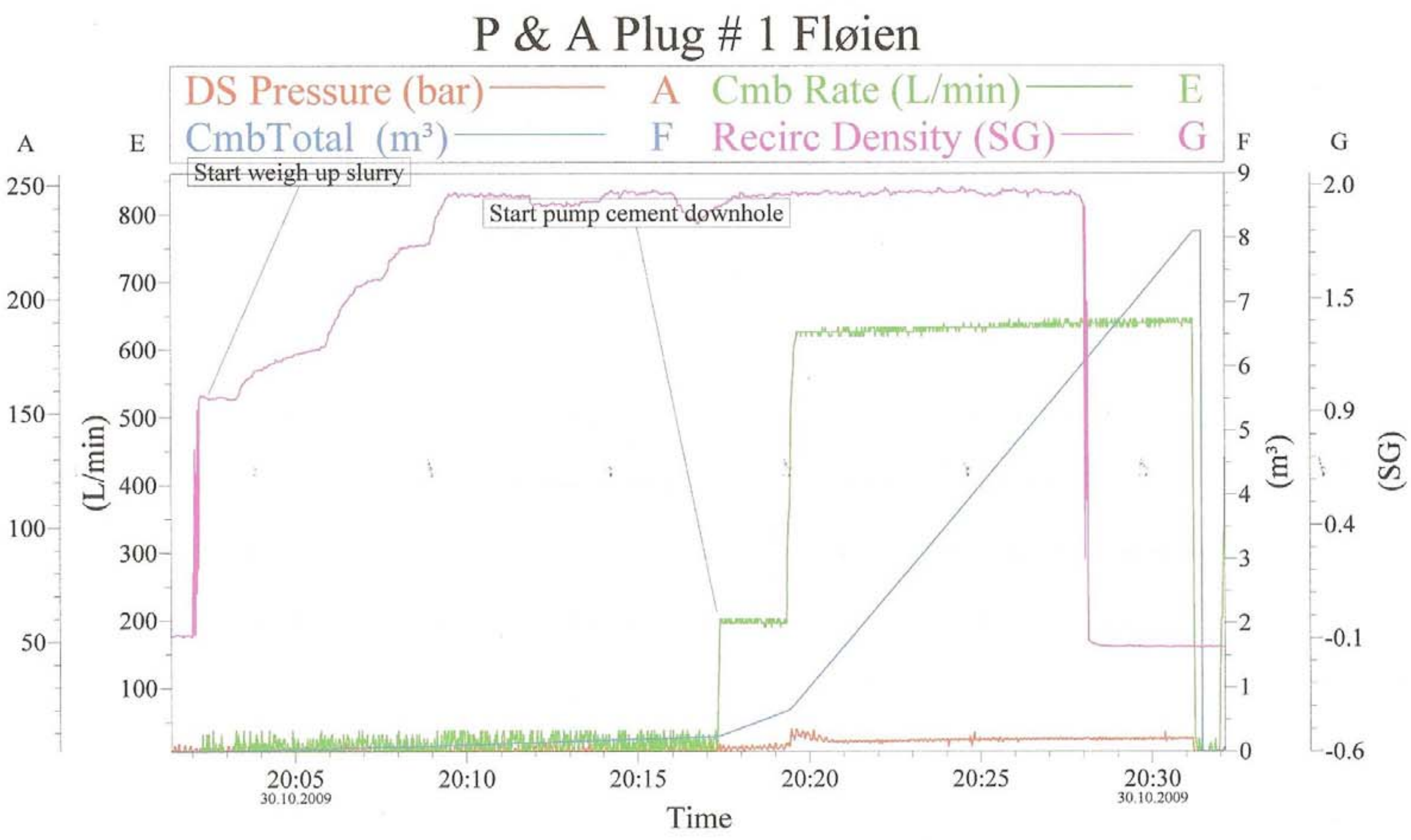
WELLBORE ID: NO 6608/10-13
REPORT DATE: 31.Oct.2009
PLUG NO: 1
TOP DEPTH: 868
BOTTOM DEPTH: 1077
CASING SIZE: 9 5/8"
HOLE DIAMETER: NA
CEMENTING TYPE: CASING
REMARKS:

Objective:
Objective: Permanently plug and abandon the well. Due to dry well, open hole plugs were cancelled and 3 plugs set in 9 5/8”liner on top of 9 5/8”EZSV

Execution:
No problems during mixing and pumping.

Fluids pumped	Type	Density g/cm3	Volume m3	Pump Rate l/min	Pump Press bar	Return	Component	Quantity	Unit	Density g/cm3	Premixed in mixwater
Slurry	Cement Slurry GTT90	1,92	7,9	636	10		CFR-8L	2,50	l/100kg		
							Freshwater	33,11	l/100kg	1,00	
							HALAD 400L	2,00	l/100kg		
							HR-4L	1,00	l/100kg	1.18	
							Microsilica Liquid	10,00	l/100kg	1,39	
							NF-6, DEFOAMER	0,10	l/100kg	0,94	
							Norcem "G"	1253,00	kg/m3	3,22	

Table 6-4 Cement details P&A, plug 1



Customer: Well Description:	Job Date: UWI:	Ticket #:

CemWin v1.7.2
30-Oct-09 21:10

Evaluation:

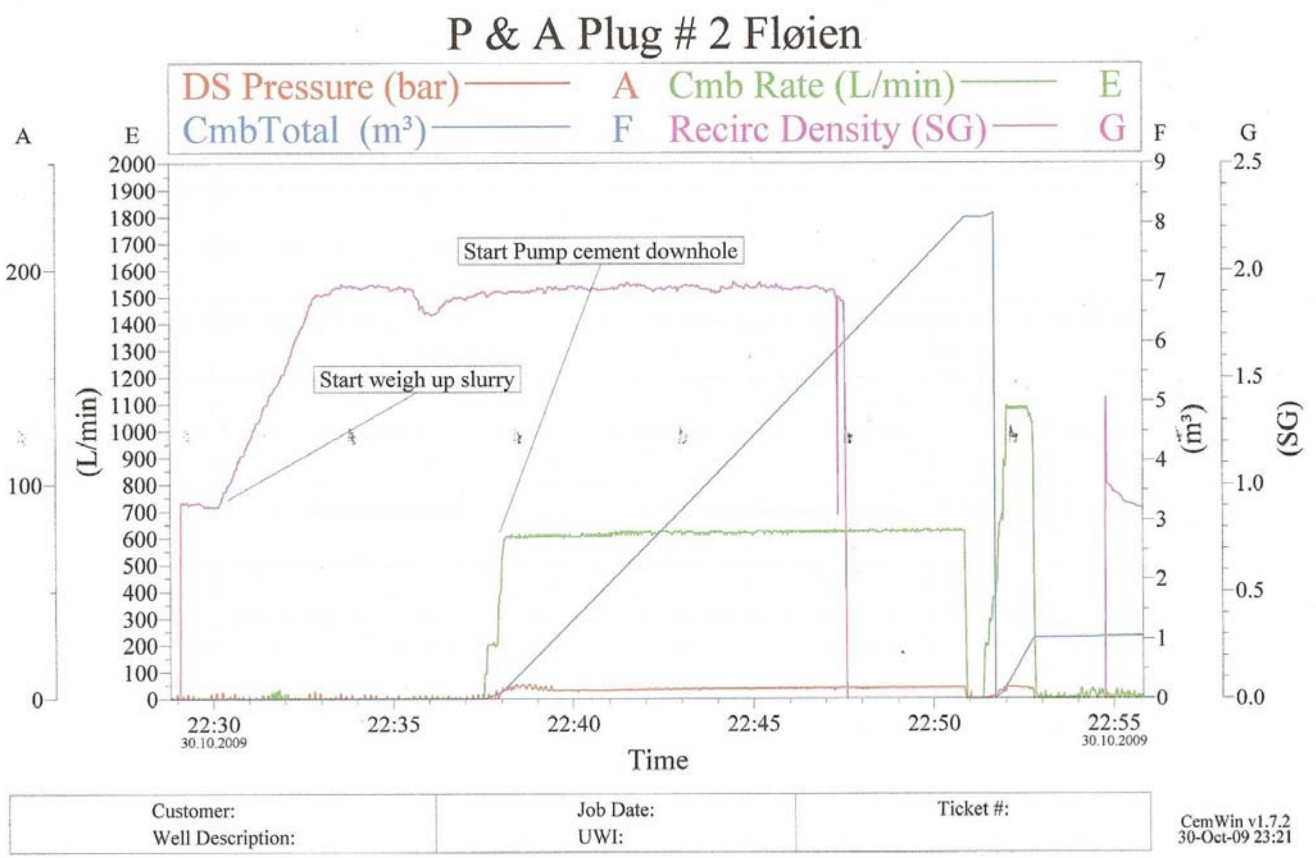
PLUG NO: 2
TOP DEPTH: 658
BOTTOM DEPTH: 868
CASING SIZE: 9 5/8"
HOLE DIAMETER:
CEMENTING TYPE: CASING
REMARKS: Plug 2a

Objective:
Objective: Permanently plug and abandon the well. Due to dry well, open hole plugs were cancelled and 3 plugs set in 9 5/8"liner on top of 9 5/8"EZSV

Execution:
Job went according to plan.

Fluids pumped	Type	Density g/cm3	Volume m3	Pump Rate l/min	Pump Press bar	Return	Component	Quantity	Unit	Density g/cm3	Premixed in mixwater
Slurry	Cement Slurry STTNT	1,92	7,9	636	10		Freshwater	43,53	l/100kg	1,00	
							NF-6, DEFOAMER	0,10	l/100kg	0,94	
							Norcem "G"	1338,00	kg/m3	3,22	

Table 6-5 Cement details P&A, plug 2


P & A Plug # 2 Fløien
DS Pressure (bar) A
Cmb Rate (L/min) E
CmbTotal (m³) F
Recirc Density (SG) G
A
E
F
G
200
100
0
2000
1900
1800
1700
1600
1500
1400
1300
1200
1100
1000
900
800
700
600
500
400
300
200
100
0
(L/min)
9
8
7
6
5
4
3
2
1
0
(m³)
2.5
2.0
1.5
1.0
0.5
0.0
(SG)
Start Pump cement downhole
Start weigh up slurry
22:30
30.10.2009
22:35
22:40
22:45
22:50
22:55
30.10.2009
Time
Customer:
Well Description:
Job Date:
UWI:
Ticket #:
CemWin v1.7.2
30-Oct-09 23:21

Evaluation:

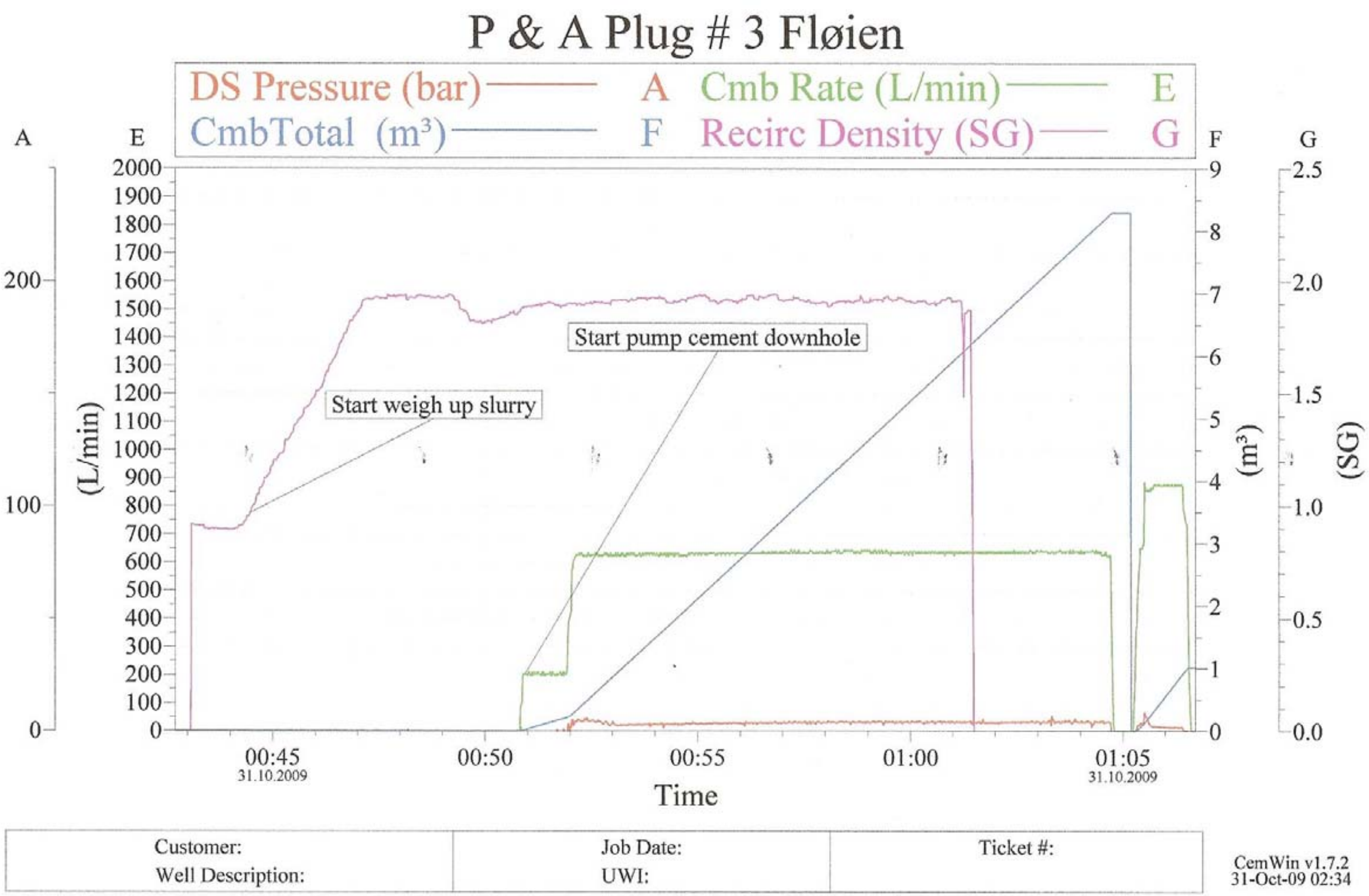
PLUG NO: 3
TOP DEPTH: 446
BOTTOM DEPTH: 656
CASING SIZE: 9 5/8"
HOLE DIAMETER:
CEMENTING TYPE: CASING
REMARKS:

Objective:
Objective: Permanently plug and abandon the well. Due to dry well, open hole plugs were cancelled and 3 plugs set in 9 5/8"liner on top of 9 5/8"EZSV

Execution:
No problems during the job. All according to plan.

Fluids pumped	Type	Density g/cm3	Volume m3	Pump Rate l/min	Pump Press bar	Return	Component	Quantity	Unit	Density g/cm3	Premixed in mixwater
Slurry	Cement Slurry STTNT	1,92	7,9				NF-6, DEFOAMER	0,10	l/100kg	0,94	
							Norcem "G"	1338,00	kg/m3	3,22	
							Seawater	43,53	l/100kg	1,03	

Table 6-6 Cement details P&A, plug 3


P & A Plug # 3 Fløien
DS Pressure (bar) A
Cmb Rate (L/min) E
CmbTotal (m³) F
Recirc Density (SG) G
Start weigh up slurry
Start pump cement downhole
(L/min)
(m³)
(SG)
Time
00:45 31.10.2009
00:50
00:55
01:00
01:05 31.10.2009
Customer:
Well Description:
Job Date:
UWI:
Ticket #:
CemWin v1.7.2
31-Oct-09 02:34

Evaluation:
No problems during the job. All according to plan

Mechanical plug

WELLBORE ID: NO 6608/10-13

Report date	Depth mMD	Diam. in	Plug type	Tagged depth mMD	Pressure test bar	Setting time	Pulling time	Remarks
31.oct.2009	1080,0	9 5/8"	EZSV	1080,0	105,0	30.okt.2009		Run as retainer

Table 6-7 Mechanical plug

Well schematic – wellbore NO 6608/10-13

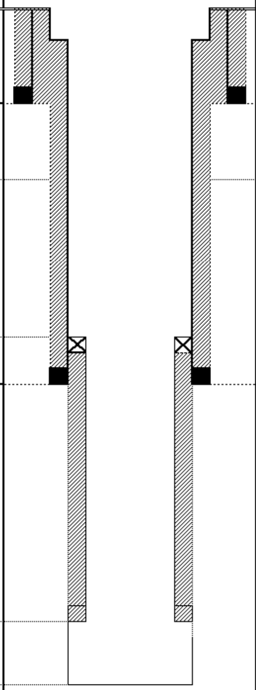
Start of project (12.Oct.2009)

WELL SCHEMATIC

Well: **6608/10-13**
Field: **Føien**
Rig: **Ocean Vanguard**

All depths refers to RKB.
RKB-MSL Ocean Vanguard:22 m

HOLE		CASING			LOT / FIT	TOC		CSG. SHOE		RKB		LWD LOGS	SURV CSG/ OH
SIZE	TVD MD	SIZE	TYPE / RAD. MARKERS	CENTRALIZERS	[SG]	TVD	MD	TVD	MD				
SB	408					408	408	408	408				
36"	474 474	30"	Type: 309.7 lb/ft, X-52, SL-60 Drift: 26,875"	None	N/A			472	472			N/A	MWD
12 1/4"	1144 1144	20" x 9 5/8"	18 3/4" WH housing with XO to 9 5/8" 53,5 lb/ft, P-110, Vam Top	1 x Bow type / 5 first joints	N/A			757 If Liner 1040 1140	757 If liner 1040 1140			Shallow gas warning Ref. Data Aquisition Program	MWD
8 1/2"	1417 1417		Contingency 7" liner* Interval: To be decided Type: 32 lbf/ft, P-110,Vam Top	As requested	XLOT			As req. 1417	As req. 1417			Ref. Data Aquisition Program	MWD



Comments: *) The 7" liner is a contingency which may be run in case of hole stability problems.

Figure 6-3 Well schematic, start of project

End of project (07.nov.2009)

Field	Well	Total Depth (MD)	Datum	Water Depth	RKB-WH	Schematic Date	Print Date
EXPLORATION	NO 6608/10-13	1 442,0m	RKB 22,0m	384,0m	0,0m	07.11.2009	14.12.2009 14:39

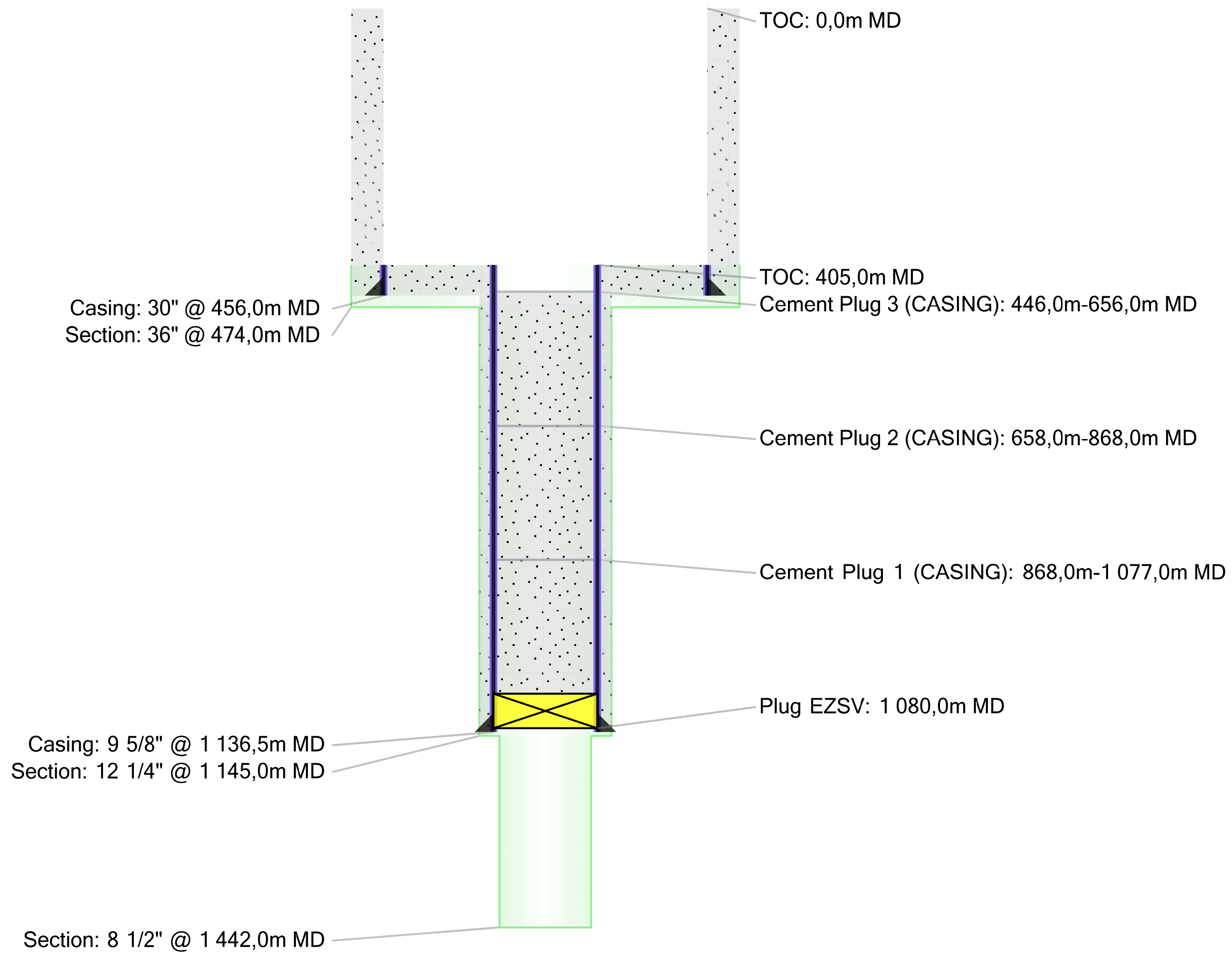


Figure 6-4 Well schematic, end of project

App D Contractors list

Service	Contract no	Contractor
Cementing	SAP4600007720	Halliburton
Mud Logging	SAP4600010265	Schlumberger
Electric Wireline Logging	SAP4600007180	Schlumberger
Drilling Fluids	SAP4600007720	Halliburton
Subsea WH/X-mas tree	SAP4600005076	Drill Quip
Directional Drilling	SAP4600010265	Schlumberger
Measurement While Drilling	SAP4600010265	Schlumberger
ROV Systems	SAP4600006702	Oceaneering
Casing Crew Service	SAP4600009488	Odfjell Well Services
Rig Operations	SAP4600007841	Diamond Offshore Netherlands BV

App E Wellsite sample description

WELLSITE CUTTINGS SAMPLE DESCRIPTION				Page 1 of 6
Country: Norway	Area: Norwegian Sea Nordland II		Field: Fløien	
Well no: 6608/10-13	Company: Statoil			
RKB: 22 metres	Geologist: L. Rasmussen, M.Gohil			
Hole size: 8 ½”	Cut solvent: Iso Propanol Alcohol		Date: 27 – 28.10.2009	
Depth (m RKB)	Lithology (%)	Lithological Description Rock name, mod.lith, colour, grain size, sorting, roundness, matrix, cementation, hardness, sed.structures, accessories, fossils, porosity, contamination		Remarks Shows, cavings, mud additives, etc.

			8 ½” started 27.10.09 at 21:00 hrs LOT= 1.49 sg at 1136,5m	
1150	90	Cmt	Cement + mud add. (baracarb, CaCO3)	Cont. mud add.
	10	Clst	m gry, grn gry, sft, amor, slty	baracarb, CaCO3
	Tr	Sst	clr- trnsl Qtz Gr, m-crs, sbang – sbrndd, pr-mod srt, lse	
1160	80	Cmt	Cement + mud add.	Cont. mud add
	20	Clst	m-dk gy, grn gry – dk grn gry, sol, sft, stky, amor – sbblky, slily slty, slily calc, slily micromic	
1170	70	Cmt	Cement + mud add.	a.a.
	30	Clst	a.a.	
	Tr	Sst	a.a.	
1180	50	Cmt	Cement + mud add.	a.a.
	50	Clst	a.a.	
	Tr	Sst	a.a.	
	Tr	Ls	v lt gry – yel gry, frm – hd, occ arg, occ slty, microxln	
1190	80	Clst	a.a.	Cont. mud add
	20	Sst	a.a. Tr Rk frag, dk grn – blk, v hd	
	Tr	Ls	a.a.	
1200	90	Clst	a.a.	Cont. mud add
	10	Sd	a.a.	
	Tr	Ls	a.a.	
1203	90	Clst	a..a	Cont. mud add
	10	Sd		
	Tr	Ls		
1206	90	Clst	a.a.	Cont. mud add
	10	Sd		
	Tr	Ls		
				Cont. mud add
1209	90	Clst	a.a.	
	10	Sd		
	Tr	Ls		
1212	80	Clst	a.a.	Cont. mud add
	20	Sd	clr- trnsl Qtz Gr, m-crs, sbang – sbrndd, pr-mod srt, lse	
	Tr	Ls	v lt gry – yel gry, frm – hd, occ arg, occ slty, microxln	
1215	90	Clst	grn gry – dk grn gry, occ olv gry, sol, sft, stky, amor – sbblky, slily slty, slily calc, slily micromic, Tr Glauc	Cont. mud add
	10	Sd	a.a.	
	Tr	Ls	a.a.	

WELLSITE CUTTINGS SAMPLE DESCRIPTION				Page 2 of 6
Country: Norway	Area: Norwegian Sea Nordland II		Field: Fløien	
Well no: 6608/10-13	Company: Statoil			
RKB: 22 metres	Geologist: L. Rasmussen, M.Gohil			
Hole size: 8 ½"	Cut solvent: Iso Propanol Alcohol		Date: 27 – 28.10.2009	
Depth (m RKB)	Lithology (%)	Lithological Description Rock name, mod.lith, colour, grain size, sorting, roundness, matrix, cementation, hardness, sed.structures, accessories, fossils, porosity, contamination		Remarks Shows, cavings, mud additives, etc.

1218	90	Clst	brn gry – brn blk, occ dk grn gry, amor – sbblky, slty, slily calc, slily micromic, slily micropyr, Tr Glauc, Tr Pyr	
	10	Sd	f- crs, else a.a.	
	Tr	Ls	a.a.	
1221	90	Clst	brn gry – brn blk, occ dk grn gry, amor – sbblky, slty, slily calc, slily micromic, slily micropyr, Tr Glauc, Tr Pyr	
	10	Sd	clr- trnsl Qtz Gr, f-crs, sbang – sbrndd, pr-mod srt, lse	
	Tr	Ls	v lt gry – yel gry, frm – hd, occ arg, occ slty, microxln	
1224	90	Clst	a.a.	
	10	Sd	a.a.	
	Tr	Ls	a.a.	
1227	90	Clst	a.a.	
	10	Sd	a.a.	
	Tr	Ls	a.a.	
1230	90	Clst	varicol, m-dk gry, dk grn gry, brn gry, else a.a.	
	10	Sd	a.a.	
	Tr	Ls	a.a.	
1233	90	Clst	lt –m blsh gry, occ brn gry, m-dk gry, blky – plty, sft – mod hd, non – slily calc, blk min spkld, tf	
	10	Sd	a.a.	
	Tr	Ls	a.a.	
1236	100	Clst	dk gry – olv gry, occ lt-m blsh gry, sbblky – blky, sft-mod hd, non – slily calc, slily slty, slily micromic	
	Tr	Sd	a.a.	
	Tr	Ls	a.a.	
1239	100	Clst	a.a.	
	Tr	Sd	a.a.	
	Tr	Ls	a.a.	
1242	100	Clst	olv gry- dk gry, occ brn gry, sbblky-blky, sft-mod hd, slty, non –slily calc, slily micromic, Tr Glauc	
	Tr	Sd	a.a.	
1245	100	Clst	a.a.	
	Tr	Sd	a.a.	
1248	100	Clst	a.a.	
	Tr	Sd	a.a.	

WELLSITE CUTTINGS SAMPLE DESCRIPTION			
Country: Norway	Area: Norwegian Sea Nordland II		Field: Fløien
Well no: 6608/10-13	Company: Statoil		
RKB: 22 metres	Geologist: L. Rasmussen, M.Gohil		
Hole size: 8 ½”	Cut solvent: Iso Propanol Alcohol		Date: 27 – 28.10.2009

Depth (m RKB)	Lithology (%)	Lithological Description Rock name, mod.lith, colour, grain size, sorting, roundness, matrix, cementation, hardness, sed.structures, accessories, fossils, porosity, contamination	Remarks Shows, cavings, mud additives, etc.

1251	100	Clst	olv gry, dk gry, occ m blsh gry, else a.a	
1254	100	Clst	m-dk gry, m blsh gry, sbblky-blky, sft-mod hd, slty, non –slily calc, , blk min spkld, tf, Tr Glauc	
	Tr	Sd	a.a.	
	Tr	Ls	a.a.	
1257	100	Clst	a.a.	
	Tr	Sd	a.a.	
	Tr	Ls		
1260	100	Clst	a.a.	Baracarb
	Tr	Sd	a.a.	
	Tr	Ls	a.a.	
1263	100	Clst	a.a.	a.a.
	Tr	Ls	a.a.	
1266	100	Clst	also dsky yelsh gry, else a.a.	a.a.
	Tr	Ls	v lt gry, else a.a.	
1269	100	Clst	m dk gry, m blsh gry, dsky yelsh gn, tr pa rdsh brn, sbblky-blky, sft-mod hd, slty, non –slily calc, , blk min spkld, tf, Tr Glauc	Baracarb
	Tr	Ls	v lt gry, frm – hd, occ arg, occ slty, microxln	
1272	100	Clst	both m dk gry, m blsh gry and pa rdsh brn, else a.a.	a.a.
	Tr	Ls	v lt gry-yel gry, frm-hd, occ slty-sdy, microxln	
1275	95	Clst	a.a.	a.a.
	5	Ls	a.a.	
1278	100	Clst	a.a.	a.a.
	Tr	Ls	a.a.	
1281	100	Clst	micropyr, else a.a.	a.a.
	Tr	Ls	a.a.	
	Tr	Sd?	clr- trnsl Qtz Gr, f-crs, sbang – sbrndd, pr-mod srt, lse	
1284	100	Clst	a.a.	a.a.
	Tr	Ls	a.a.	
	Tr	Sd?	a.a.	
1287	100	Clst	blk – brnsh blk, pa yelsh brn, m dk gry, m blsh gry, sbblky – blky, frm, slty, non calc, slily micromic, slily micropyr, occ microglauc	a.a.
	Tr	Ls	a.a.	
1290	100	Clst	a.a.	a.a.

WELLSITE CUTTINGS SAMPLE DESCRIPTION				Page 4 of 6
Country: Norway	Area: Norwegian Sea Nordland II			Field: Fløien
Well no: 6608/10-13	Company: Statoil			
RKB: 22 metres	Geologist: L. Rasmussen, M.Gohil			
Hole size: 8 ½"	Cut solvent: Iso Propanol Alcohol			Date: 27 – 28.10.2009
Depth (m RKB)	Lithology (%)	Lithological Description Rock name, mod.lith, colour, grain size, sorting, roundness, matrix, cementation, hardness, sed.structures, accessories, fossils, porosity, contamination		Remarks Shows, cavings, mud additives, etc.

	Tr	Ls	a.a.	
	Tr	Sst	Agg of clr – trnsl Qtz, m – crs, mod srtd, sbang – sbrnd, silic cmtd, no vis por	No shows
1293	100	Clst	a.a.	a.a.
	Tr	Ls	a.a.	
	Tr	Sst	a.a.	
1296	95	Clst	pred blk – brnsh blk, pa yelsh brn, but also m dk gry and m blsh gry, sbblky – blky, frm, slty, non – slily calc, micromic, occ micropyr, occ microglauc	a.a.
	5	Sst	agg of clr – trnsl Qtz Grns, also lse, f – crs, pr – mod srtd, sbang – sbrnd, silic cmtd?, no vis por	
1299	95	Clst	pred blk – brnsh blk, also pa yelsh brn, else a.a.	a.a.
	5	Sst	a.a.	
1305	90	Clst	pyr, else a.a.	a.a.
	10	Sst	Both as Agg and as lse Grns, else a.a.	
1308	40	Clst	a.a.	a.a.
	60	Sst	pred lse, f – crs, also as Agg, else a.a.	
1311	40	Clst	a.a.	a.a.
	60	Sst	a.a.	
1317	80	Clst	pred blk – brnsh blk, also pa yelsh brn, sbblky – blky, frm, slty, non – slily calc, micromic, micropyr, occ microglauc	Baracarb
	20	Sst	pred lse clr – trnsl Qtz Grns, also as Agg, f – crs, pr – mod srtd, sbang – sbrnd, silic cmtd?	No shows
	Tr	Ls	yel gry – lt gry, frm – hd, occ slty – sdy, microxln	
	Tr	Pyr	as nodules	
1323	90	Clst	blk – brnsh blk, brnsh gry, occ pa yelsh brn, else a.a.	a.a.
	10	Sst	mainly as agg, else a.a.	
1329	90	Clst	blk – brnsh blk, brnsh gry, else a.a.	a.a.
	10	Sst	mainly as agg, else a.a.	
1335	90	Clst	a.a.	a.a.
	10	Sst	Both lse Qtz Grns and as Agg, spl a.a.	
1341	10	Clst	a.a.	a.a.
	90	Sst	pred lse, else a.a.	
1347	20	Clst	a.a.	Lss Baracarb
	80	Sst	a.a.	No shows

WELLSITE CUTTINGS SAMPLE DESCRIPTION				Page 5 of 6
Country: Norway	Area: Norwegian Sea Nordland II		Field: Fløien	
Well no: 6608/10-13	Company: Statoil			
RKB: 22 metres	Geologist: L. Rasmussen, M.Gohil			
Hole size: 8 ½”	Cut solvent: Iso Propanol Alcohol		Date: 27 – 28.10.2009	
Depth (m RKB)	Lithology (%)	Lithological Description Rock name, mod.lith, colour, grain size, sorting, roundness, matrix, cementation, hardness, sed.structures, accessories, fossils, porosity, contamination		Remarks Shows, cavings, mud additives, etc.

1353	60	Clst	blk – brnsh blk, brnsh gry – olv gry, else a.a.	a.a.
	40	Sst	a.a.	
1359	80	Clst	a.a.	a.a.
	20	Sst	a.a.	
1365	80	Clst	a.a.	a.a.
	20	Sst	a.a.	
1371	100	Clst	blk – brnsh blk, occ brnsh gry, else a.a.	a.a.
	Tr	Sst	pred lse, else a.a.	
1377	90	Clst	brnsh gry – lt brnsh gry, occ blk - brnsh blk, else a.a.	a.a.
	10	Sst	both lse Qtz Grns and as Agg, else a.a.	
1383	50	Clst	a.a.	a.a.
	50	Sst	pred lse Qtz, else a.a.	
1389	30	Clst	a.a.	a.a.
	70	Sst	a.a.	
1395	10	Clst	a.a.	a.a.
	90	Sst	a.a.	
1401	60	Clst	a.a.	a.a.
	40	Sst	a.a.	
1407	50	Clst	brnsh gry – lt brnsh gry, blk – brnsh blk, sft – frm, slty, non – slily calc, micromic, occ micropyr	Baracarb
	50	Sst	pred as lse Qtz Grns, also as agg, f – crs, pr – mod srtd, sbang – sbrnd, occ Qtz cmtd	No shows
1413	50	Clst	a.a.	a.a.
	50	Sst		
1419	40	Clst	a.a., Pyr nod	a.a.
	60	Sst	pred as Agg, else a.a.	
1425	40	Clst	a.a	a.a.
	60	Sst	a.a.	
1431	80	Clst	a.a.	
	20	Sst	pred lse Qtz Gr	a.a.
	gd Tr	Coal	brn blk – blk, frm – hd, brit, ang, occ arg	
1437	90	Clst	a.a.	
	10	Sst	a.a.	a.a.

WELLSITE CUTTINGS SAMPLE DESCRIPTION

Country:	Norway	Area:	Norwegian Sea Nordland II	Field:	Fløien
Well no:	6608/10-13	Company:	Statoil		
RKB:	22 metres	Geologist:	L. Rasmussen, M.Gohil		
Hole size:	8 ½”	Cut solvent:	Iso Propanol Alcohol	Date:	27 – 28.10.2009

Depth (m RKB)	Lithology (%)		Lithological Description Rock name, mod.lith, colour, grain size, sorting, roundness, matrix, cementation, hardness, sed.structures, accessories, fossils, porosity, contamination	Remarks Shows, cavings, mud additives, etc.
	Tr	Coal	a.a.	
1440	80	Clst	a.a.	
	20	Sst	a.a.	a.a.
	r	Coal	a.a.	
1442	50	Clst	a.a., Pyr Nod	
	50	Sst	pred lse Qtz Gr	a.a.
	r	Coal	a.a.	
			TD 1442m MD at 20:05 hrs 28.10.2009	

Other reports

Nr	Title	Company	Date	Location	Index	Doc ID
001	Drilling Program, Fløien-6608/10-13	StatoilHydro	30.09.09	ST-HAR	Digital	EPDS-6608/10-13-001
002	Site survey Fløien.	Fugro	17.06.09	ST-FH	Digital	EPDS-6608/10-13-002
003	Borestartsnotat letebrønn 6608/10-13	StatoilHydro	29.09.09	ST-HAR	Digital	EPDS-6608/10-13-003
004	Fløien, 6608/10-13 wireline-Run 1A	Schlumberger	29.10.09	ST-HAR	CD/DVD nr.1407	EPDS-6608/10-13-004
005	PL437 Boresluttsnotat letebrønn 6608/10-13	Statoil ASA	17.11.09	ST-HAR	Digital	EPDS-6608/10-13-005
006	Final Well Report Fløien	Statoil ASA	08.05.10	ST-HAR	Digital	EPDS-6608/10-13-006

NPD standart for reporting shallow gas

WELL DATA: 6608/10-13, Fløien

1.	**Distance from drill floor to sea level:**	22 m
2.	**Water depth:**	384 m MSL
3a.	**Setting depth for conductor:**	456 m RKB
3b.	**Leak Off/Formation Integrity Test:**	N/A
4a	**Setting depth for casing on which BOP is installed:**	1136.5mMD
4b.	**Leak Off/Formation Integrity test:**	1.49 sg

5. Depth (mTVD RKB) and TWT to formation/section/layer tops:

Seabed	406.0 mTVD RKB / 507 ms
Top Naust Formation	691.0 mTVD RKB / 758 ms

6. Depth interval (mTVD RKB and TWT) and age of sand layers shallower than 1000 m below seabed. State which layers, if any, containing gas.

No returns of cuttings or gas to rig surface down to 858mMD RKB.
A preliminary interpretation of LWD logs indicates sand layers at:

509 – 513 mMD RKB, Pleistocene, sandy claystone, water filled
530 – 533 mMD RKB, Pleistocene, sandy claystone, water filled
652 – 654 mMD RKB, Pleistocene, sandstone, water filled
738 – 740 mMD RKB, Late Pliocene, sandy claystone, water filled

No gas observed with ROV either.

7. How was presence of gas proven?

No shallow gas was encountered in the well. No gas observed with ROV either.

8. Composition and origin of gas:

N/A

9. Describe all measurements performed in gas bearing layers:

N/A

10. Indicate the depths (mTVD RKB and TWT) of unconformities in the well bore:

An unconformity is possible at 740.0mMD RKB.

11. Indicate depth and extension of sand layers (communication, continuity, truncation, etc.):

The sand layer at 650 – 652mMD RKB could possibly be the continuation of the sand layer from 643 – 645.5mMD RKB in well 6608/11-1.

12. Indicate depth and extension of any gas blanking, seismic anomalies, etc:

N/A

13. State possible seismic indications that the gas originates from deeper levels. Description if gas originates from deeper levels:

N/A

14. How does the interpretation of the site survey correspond with well data with respect to:

Shallow gas:

From the site survey there was a possibility for shallow gas Class I at this location prognosed at 757mMD RKB. No shallow gas was observed, neither by ROV nor on the MWD log. Not even an indication of sand was seen in the well bore. The amplitude anomalies close by the well location could therefore either by due to locally distributed sand lenses or maybe more likely limestone lenses?

It seems that there is no correspondence between the Class I shallow gas warning and the observations in the well bore.

Sand layers:

From site survey seven intervals were prognosed with sand layers.

474mMD RKB –> not observed
502mMD RKB –> 509 – 513m sandy claystone
523mMD RKB –> 530 – 533m sandy claystone
664mMD RKB –> 652 – 654m sandstone
676mMD RKB –> not observed
741mMD RKB –> 738 – 740.0m sandy claystone
757mMD RKB –> not observed

Unconformities:

None interpreted but possible unconformity at 740.0mMD RKB.

Correlation with adjacent wells:

The nearest well is 6608/11-1, which is located 3.2 km to the northeast. The sand layer encountered in this well 6608/10-13 Fløien pilot hole at 652.0mMD RKB may correlate to sand layer seen in above mentioned offset well.

Enclosures

Composite log

Formation evaluation log (Geoservices)

Pressure evaluation log (Geoservices)

FINAL LOG 23.04.2010

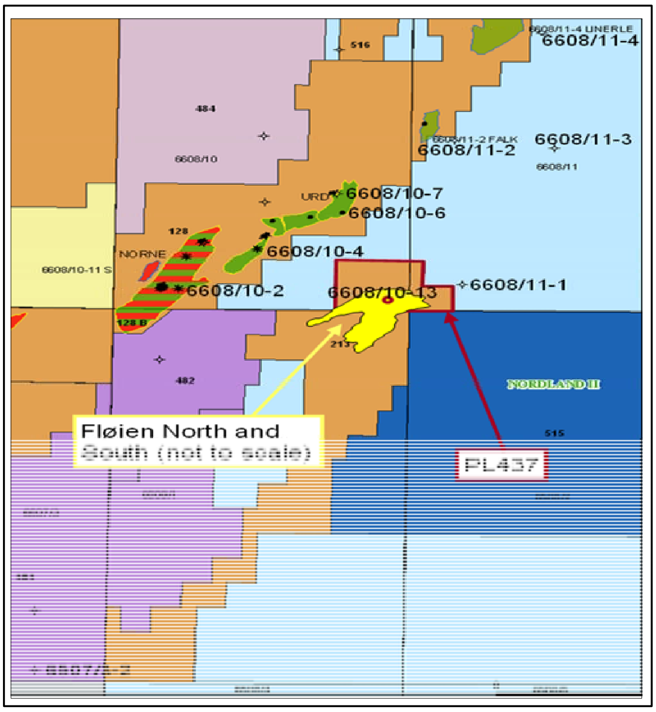
COMPLETION LOG

Fløien prospect

Scale: 1 / 500

Well: 6608/10-13

Location Map



KB Elevation	22 m	**Country**	Norway
Water Depth	384 m MSL	**Licence**	PL 437
Depth Reference	RKB		
		Owners	StatoilHydro 100%
Total Depth (Driller)	1442 m MD RKB 1442 m TVD RKB		
Total Depth (Logger)	1444.4 m MD RKB 1444.4 m TVD RKB		
		Field	Fløien Prospect
Formation at Total Depth	Åre Formation	**Rig**	Ocean Vanguard
Date Spudded	18.10.2009		
Date Reached TD	28.10.2009	**Drilling Contractor**	Diamond Offshore
Date Abandoned	07.11.2009	**Mudlogging Company**	Geoservices
Well Status	P&A	**Logging Company**	Schlumberger
		MWD Company	Schlumberger
Well Classification	Exploration Well	**Geologists**	L.Rasmussen, M.Gohil, E.Skottlien
Prepared by	J.Hopkins		
Controlled by	A. Larsen		
Date	21.04.2010		
Revision	0		

Well Coordinates	**Geographical (datum ED50)**	66° 00' 25.5" North	08° 18' 10.2" East
	UTM Zone 32 N	7 321 060 m N	468 366 m E

Casing Records

Diameter	Shoe depth m MD RKB	Shoe depth m TVD RKB
30" Casing shoe	456	456.0
9 7/8" Casing shoe	1136.5	1136.4

LOT / FIT

Type	Result g/cc	Depth m MD RKB	Depth m TVD RKB
ELOT	1.490	1136.5	1136.4

Comments

- A Class 1 shallow gas warning was given at 757 m RKB from site survey. Therefore a 9 7/8" pilothole was drilled from 466 m to 857 m for shallow gas evalution. No shallow gas was encountered in the well. The pilothole was opened up to 12 1/4" hole.

Logs

Run no.	Log type	Logged interval
1	DIR (Powerpulse) 36 " hole	408 - 466 m
2	GR-RES-ECD-DIR (ARCVRES) 9 7/8" pilothole	466 - 857 m
3	GR-RES-ECD-DIR (ARCVRES) 12 1/4" hole	466 - 1142 m
4	GR-RES-ECD-DIR (GVR-ARCRES) 8 1/2" hole	1142 - 1442 m

Cores

Core no.	Cored interval m MD RKB	Recovered interval m MD RKB	Rec. mMD	Rec. %
No cores				

Pressure points ()

Test no.	Depth m MD RKB	Depth m TVD RKB	Pressure bar	Test no.	Depth m MD RKB	Depth m TVD RKB	Pressure bar
No formation pressure tests							

Lithology Legend

CONGLOMERATE
SAND/SANDSTONE
SILT/SILTSTONE
CLAY/CLAYSTONE
SHALE
COAL
LIMESTONE
DOLOMITE
DOL. LIMESTONE
CALC. DOLOMITE
CHALK
MARL
ANHYDRITE
SALT
GYPSUM
TUFF
INTR. IGNEOUS
EXTR. IGNEOUS
PYROCLASTIC
BRECCIA
METAMORPHIC
SANDY, VERY
SANDY
SANDY, SLIGHTLY
SILTY, VERY
SILTY
SILTY, SLIGHTLY
ARGILLACEOUS
CALCAREOUS
DOLOMITIC
CHALKY
OOZE
CLASTS
OOLITE
PSEUDO OOLITE
PELLETS
C CARBONACEOUS MAT.
CHERT
GLAUCONITE
M MICA
PYRITE
v TUFFACEOUS
B BITUMINOUS
ANHYDRITIC
Chl CHLORITE
MACROFOSSILS
MACROFOSSILS FRAG
MICROFOSSILS
PLANT REMAINS
COAL FRAG
Q CEMENT: QUARTZ
CEMENT: CARBONATE
K CEMENT: KAOLINITIC

Symbol legend

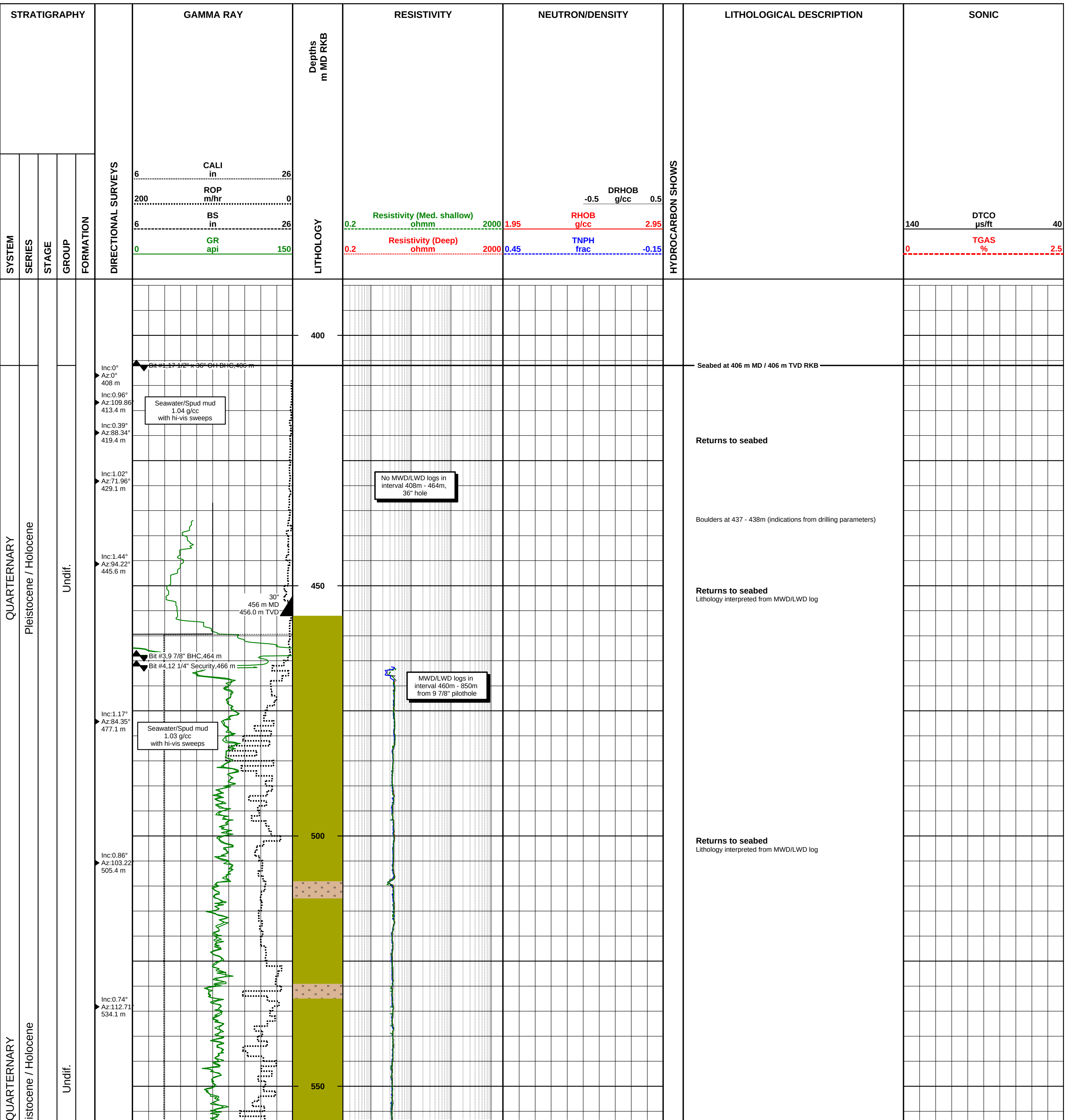
Casing shoe
Liner hanger
Liner shoe
Milled window
Deviation survey - MWD
Deviation survey - Other
Core interval
Sidewall core
(No recovery)

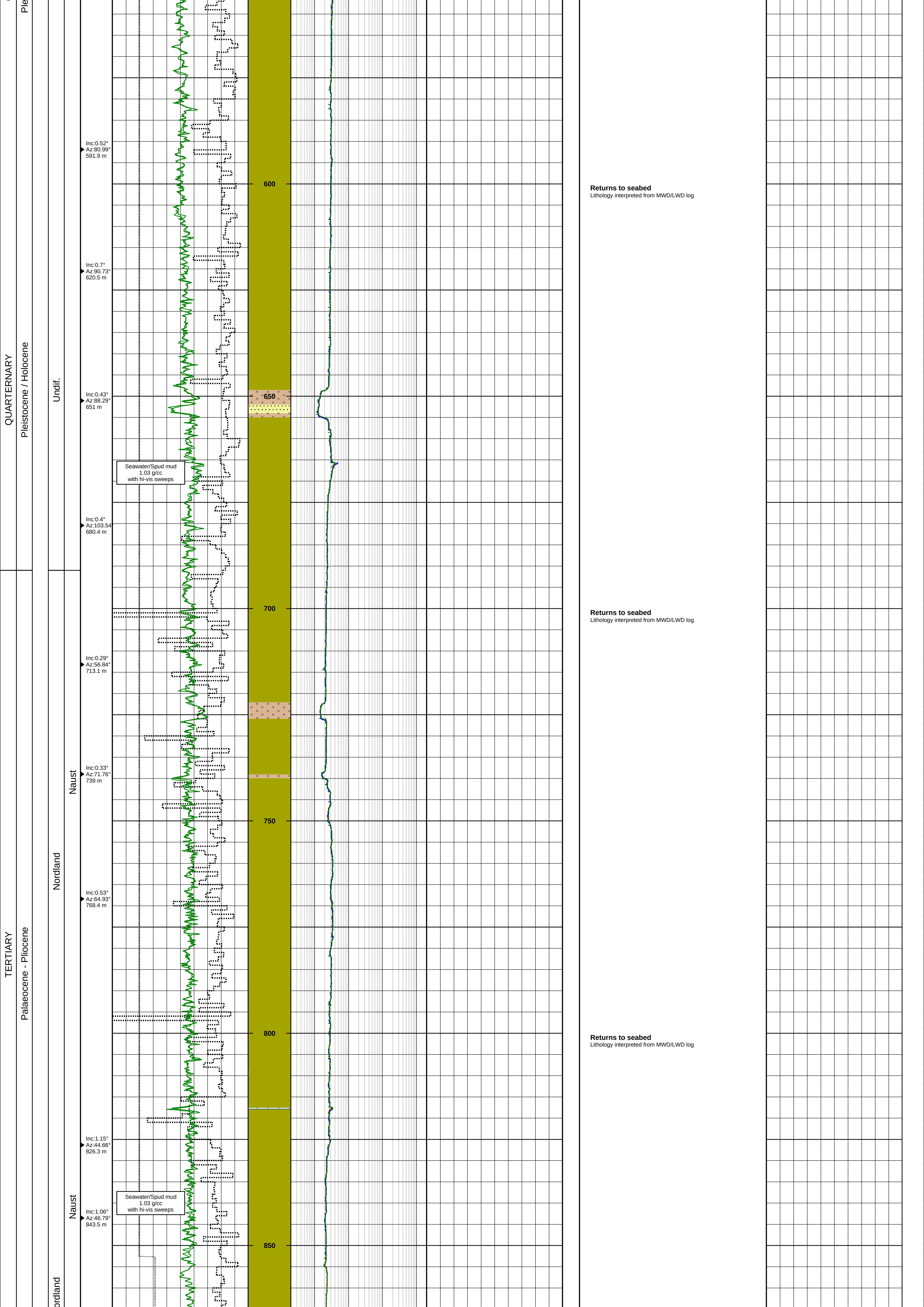
DST

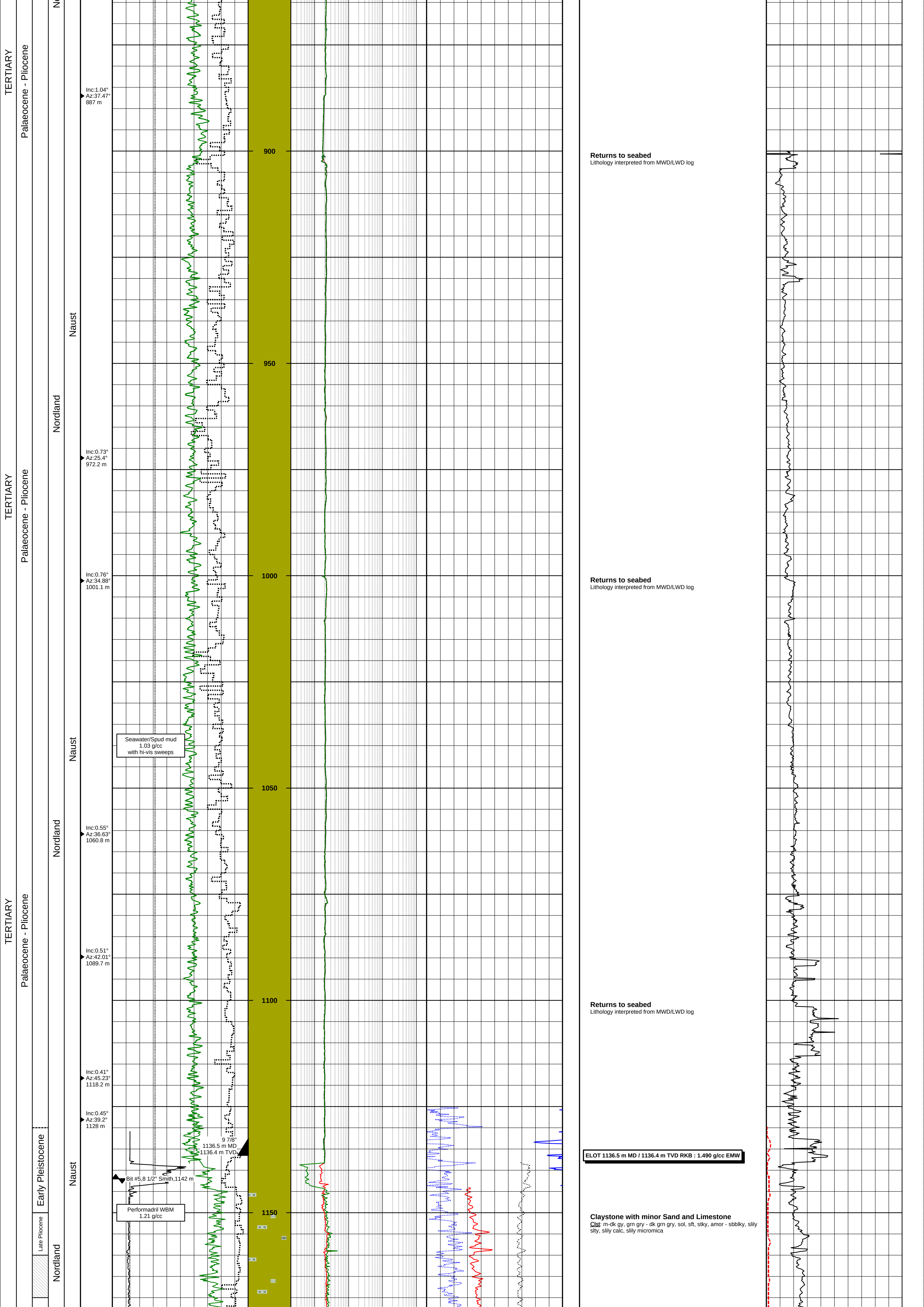
Open hole
Open hole
(rerun)
Through casing
Through casing
(rerun)
Perforation interval
Formation
pressure test

SHOWS

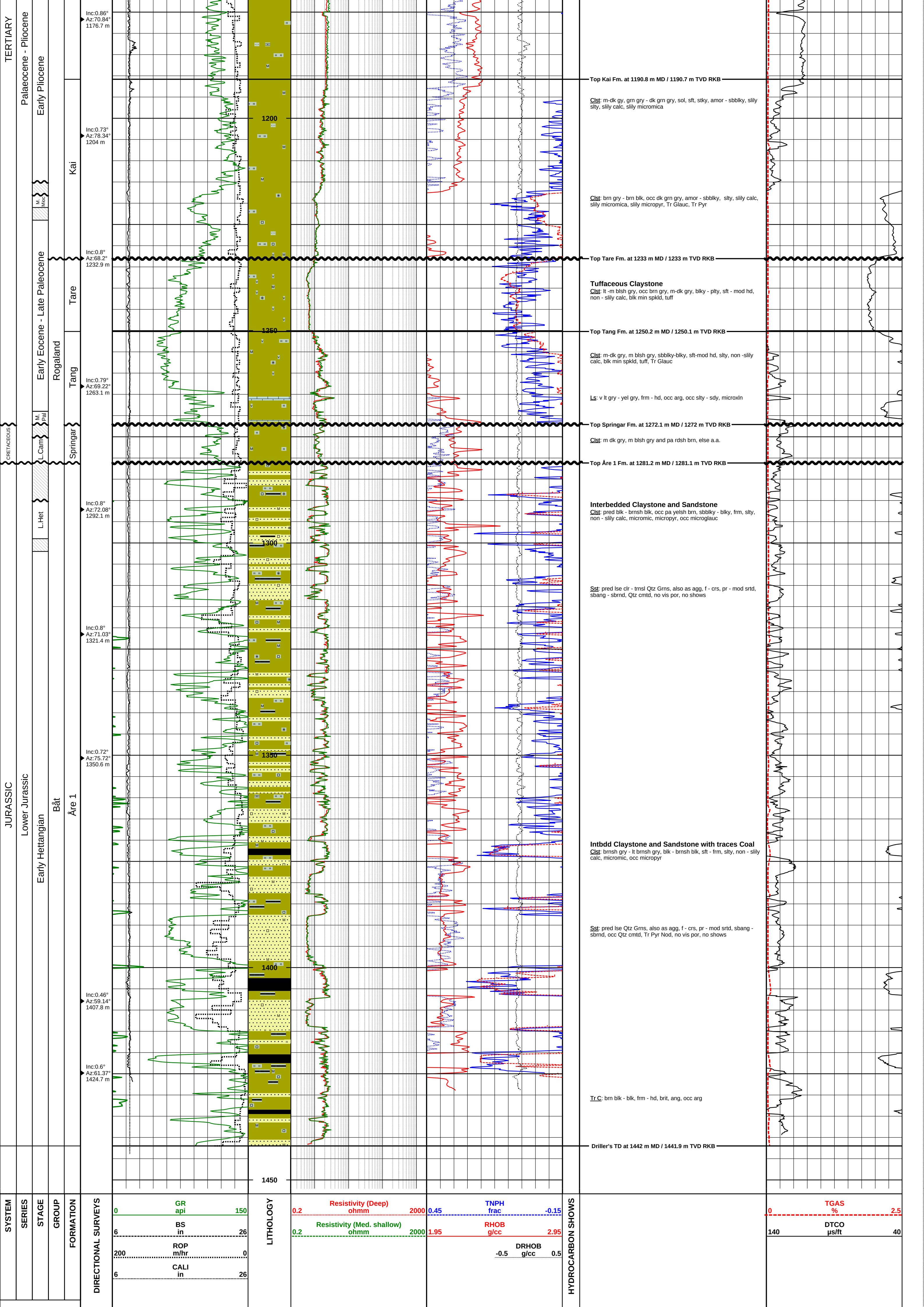
Oil stain
Fluorescence
Visible cut
Fluorescence cut
Gas shows
Dipmeter
Azi / Inc
Mud gain
Mud loss




QUARTERNARY
Pleistocene / Holocene
Undif.
TERTIARY
Palaeocene - Pliocene
Nordland
Naust
Inc:0.52°
Az:80.99°
591.9 m
Inc:0.7°
Az:90.73°
620.5 m
Inc:0.43°
Az:88.29°
651 m
Seawater/Spud mud
1.03 g/cc
with hi-vis sweeps
Inc:0.4°
Az:103.54°
680.4 m
Inc:0.29°
Az:56.84°
713.1 m
Inc:0.33°
Az:71.76°
739 m
Inc:0.53°
Az:64.93°
768.4 m
Inc:1.15°
Az:44.66°
826.3 m
Seawater/Spud mud
1.03 g/cc
with hi-vis sweeps
Inc:1.06°
Az:46.79°
843.5 m
600
650
700
750
800
850
Returns to seabed
Lithology interpreted from MWD/LWD log
Returns to seabed
Lithology interpreted from MWD/LWD log
Returns to seabed
Lithology interpreted from MWD/LWD log



TERTIARY
Palaeocene - Pliocene
Nordland
Naust
Early Pleistocene
Late Pliocene
Inc:1.04°
Az:37.47°
887 m
Inc:0.73°
Az:25.4°
972.2 m
Inc:0.76°
Az:34.88°
1001.1 m
Inc:0.55°
Az:36.63°
1060.8 m
Inc:0.51°
Az:42.01°
1089.7 m
Inc:0.41°
Az:45.23°
1118.2 m
Inc:0.45°
Az:39.2°
1128 m
900
950
1000
1050
1100
1150
Seawater/Spud mud
1.03 g/cc
with hi-vis sweeps
9 7/8"
1136.5 m MD
1136.4 m TVD
Bit #5,8 1/2" Smith,1142 m
Performadril WBM
1.21 g/cc
Returns to seabed
Lithology interpreted from MWD/LWD log
Returns to seabed
Lithology interpreted from MWD/LWD log
Returns to seabed
Lithology interpreted from MWD/LWD log
ELOT 1136.5 m MD / 1136.4 m TVD RKB : 1.490 g/cc EMW
Claystone with minor Sand and Limestone
Clst: m-dk gy, grn gry - dk grn gry, sol, sft, stky, amor - sbblky, slily slty, slily calc, slily micromica


TERTIARY
Palaeocene - Pliocene
Early Pliocene
M. Mioc.
Early Eocene - Late Paleocene
M. Pal
Rogaland
Kai
Tare
Tang
Springar
CRETACEOUS
L.Camp
L.Het
JURASSIC
Lower Jurassic
Early Hettangian
Båt
Åre 1
Inc:0.86° Az:70.84° 1176.7 m
Inc:0.73° Az:78.34° 1204 m
Inc:0.8° Az:68.2° 1232.9 m
Inc:0.79° Az:69.22° 1263.1 m
Inc:0.8° Az:72.08° 1292.1 m
Inc:0.8° Az:71.03° 1321.4 m
Inc:0.72° Az:75.72° 1350.6 m
Inc:0.46° Az:59.14° 1407.8 m
Inc:0.6° Az:61.37° 1424.7 m
1200
1250
1300
1350
1400
1450
Top Kai Fm. at 1190.8 m MD / 1190.7 m TVD RKB
Clst: m-dk gy, grn gry - dk grn gry, sol, sft, stky, amor - sbblky, slily slty, slily calc, slily micromica
Clst: brn gry - brn blk, occ dk grn gry, amor - sbblky, slty, slily calc, slily micromica, slily micropyr, Tr Glauc, Tr Pyr
Top Tare Fm. at 1233 m MD / 1233 m TVD RKB
Tuffaceous Claystone
Clst: lt -m blsh gry, occ brn gry, m-dk gry, blky - plty, sft - mod hd, non - slily calc, blk min spkld, tuff
Top Tang Fm. at 1250.2 m MD / 1250.1 m TVD RKB
Clst: m-dk gry, m blsh gry, sbblky-blky, sft-mod hd, slty, non -slily calc, blk min spkld, tuff, Tr Glauc
Ls: v lt gry - yel gry, frm - hd, occ arg, occ slty - sdy, microxln
Top Springar Fm. at 1272.1 m MD / 1272 m TVD RKB
Clst: m dk gry, m blsh gry and pa rdsh brn, else a.a.
Top Åre 1 Fm. at 1281.2 m MD / 1281.1 m TVD RKB
Interbedded Claystone and Sandstone
Clst: pred blk - brnsh blk, occ pa yelsh brn, sbblky - blky, frm, slty, non - slily calc, micromic, micropyr, occ microglauc
Sst: pred lse clr - trnsl Qtz Grns, also as agg, f - crs, pr - mod srtd, sbang - sbrnd, Qtz cmtd, no vis por, no shows
Intbdd Claystone and Sandstone with traces Coal
Clst: brnsh gry - lt brnsh gry, blk - brnsh blk, sft - frm, slty, non - slily calc, micromic, occ micropyr
Sst: pred lse Qtz Grns, also as agg, f - crs, pr - mod srtd, sbang - sbrnd, occ Qtz cmtd, Tr Pyr Nod, no vis por, no shows
Tr C: brn blk - blk, frm - hd, brit, ang, occ arg
Driller's TD at 1442 m MD / 1441.9 m TVD RKB
SYSTEM
SERIES
STAGE
GROUP
FORMATION
DIRECTIONAL SURVEYS
GR api 0 150
BS in 6 26
ROP m/hr 200 0
CALI in 6 26
LITHOLOGY
Resistivity (Deep) ohmm 0.2 2000
Resistivity (Med. shallow) ohmm 0.2 2000
TNPH frac 0.45 -0.15
RHOB g/cc 1.95 2.95
DRHOB g/cc -0.5 0.5
HYDROCARBON SHOWS
TGAS % 0 2.5
DTCO μs/ft 140 40

STRATIGRAPHY		GAMMA RAY		RESISTIVITY	NEUTRON/DENSITY		LITHOLOGICAL DESCRIPTION	SONIC

Output log file created : 23.04.2010 11:14:40
Winlog well ID : 6608_10_13 Floien
StatoilHydro (BORE) well ID : 99999
Template version : v3.10 [20070731] - AJC
Project file version : v3.11
Designed for Winlog Version 3 (TVD)

StatoilHydro

FORMATION EVALUATION LOG

MA32_N1Q.PR1

Well name : 6608/10-13

Client name : Statoil

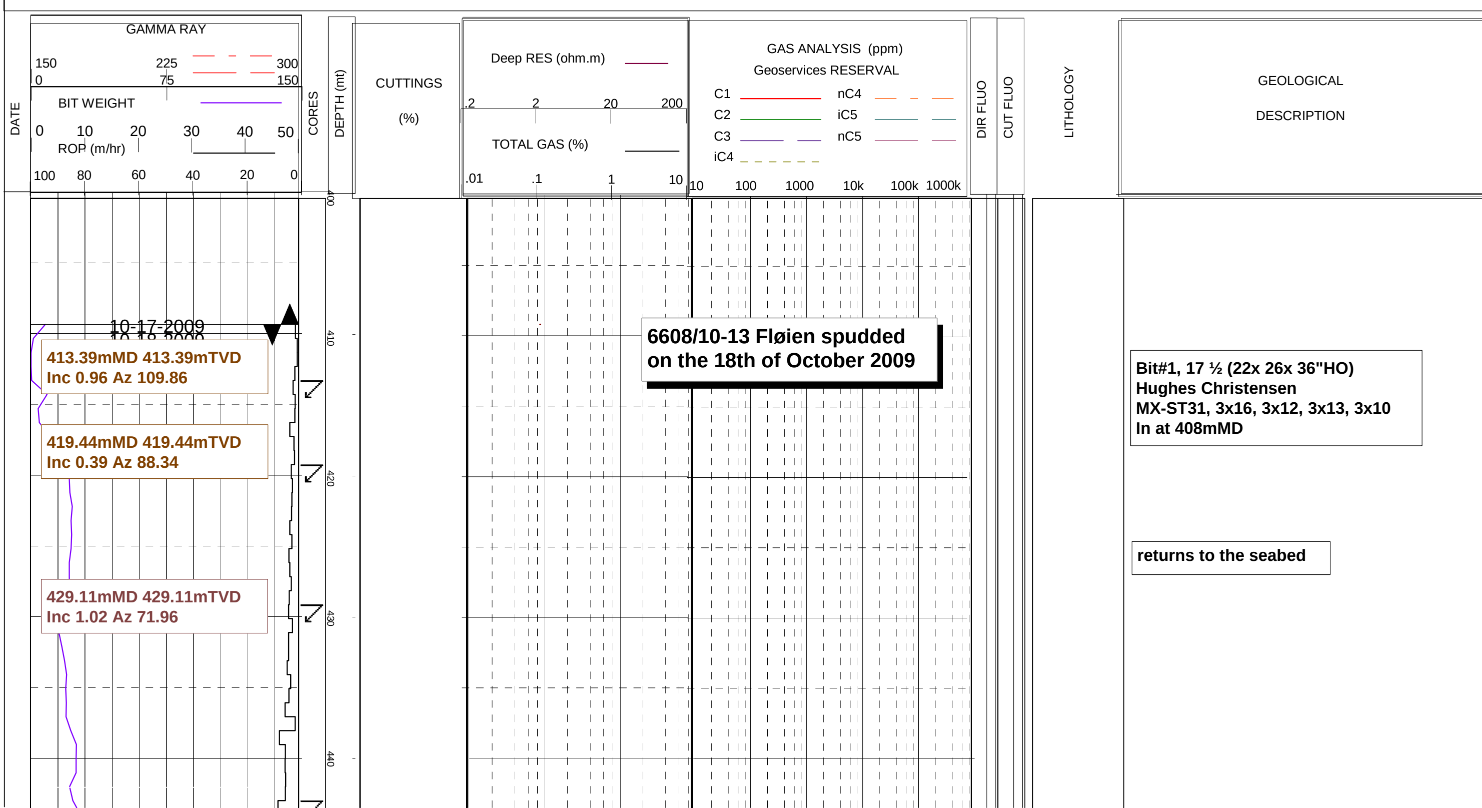
Rig Name	: OCEAN VANGUARD	Field	: Floien	Spud Date	: 18th October 2009
Rig Type	: Semisubmersible	Country	: NORWAY	TD Reached	: 28th October 2009
		Area	: North Sea	Total Drill Days	: 10
				Total Depth (m)	: 1442.00
				TVD (m)	: 1441.90
Location long	: 08deg 18' 10.2" E	UTM loc [E] (m)	: 468 366	RT - MSL (m)	: 22
Location lat	: 66deg 00' 25.5" N	UTM loc [N] (m)	: 7 321 060	MSL - Seabed (m)	: 386
				Depth Reference	: ROTARY TABLE
		Plot Scale	: 1/ 500		

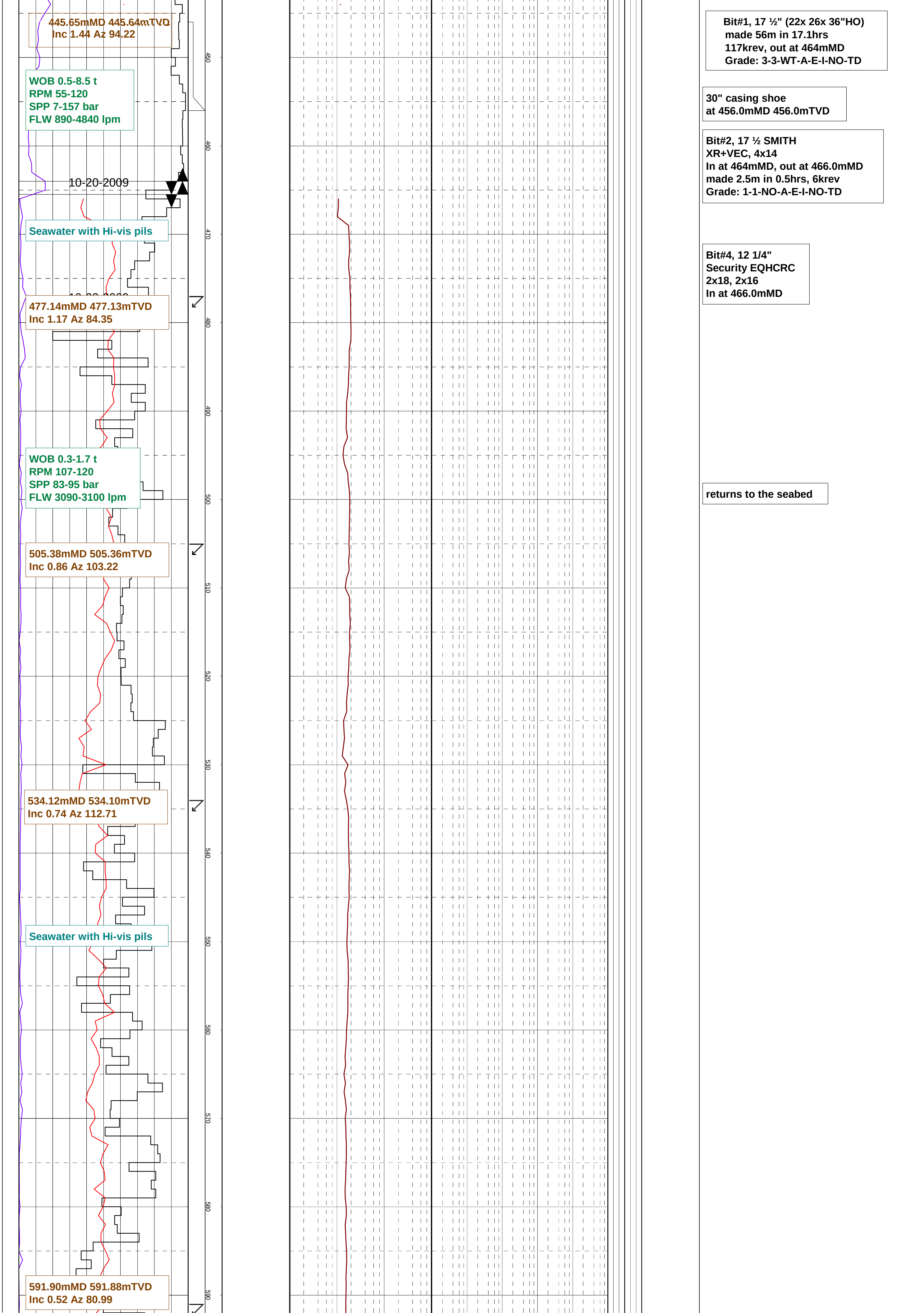
RT - MSL (m)

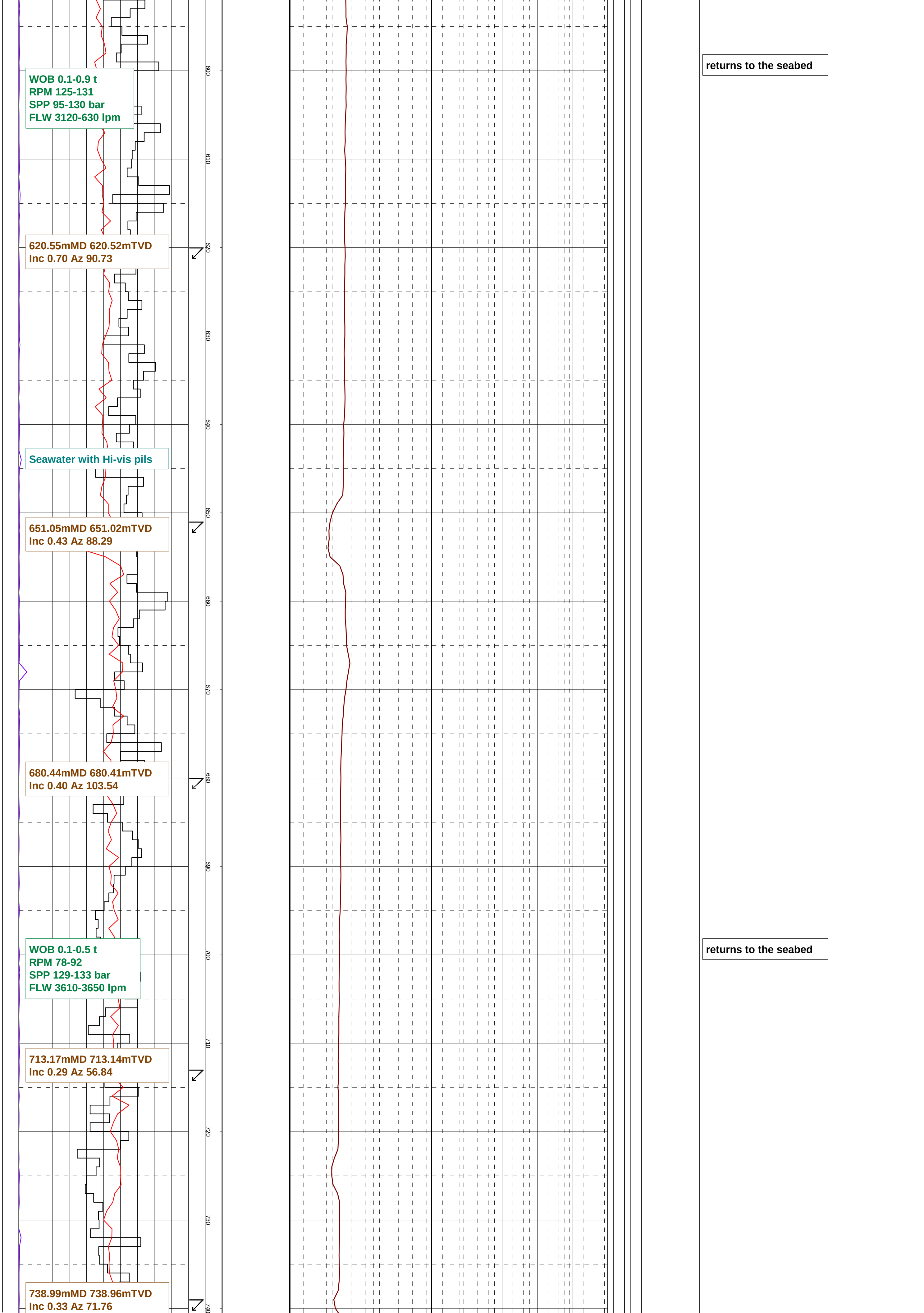
Depth Reference

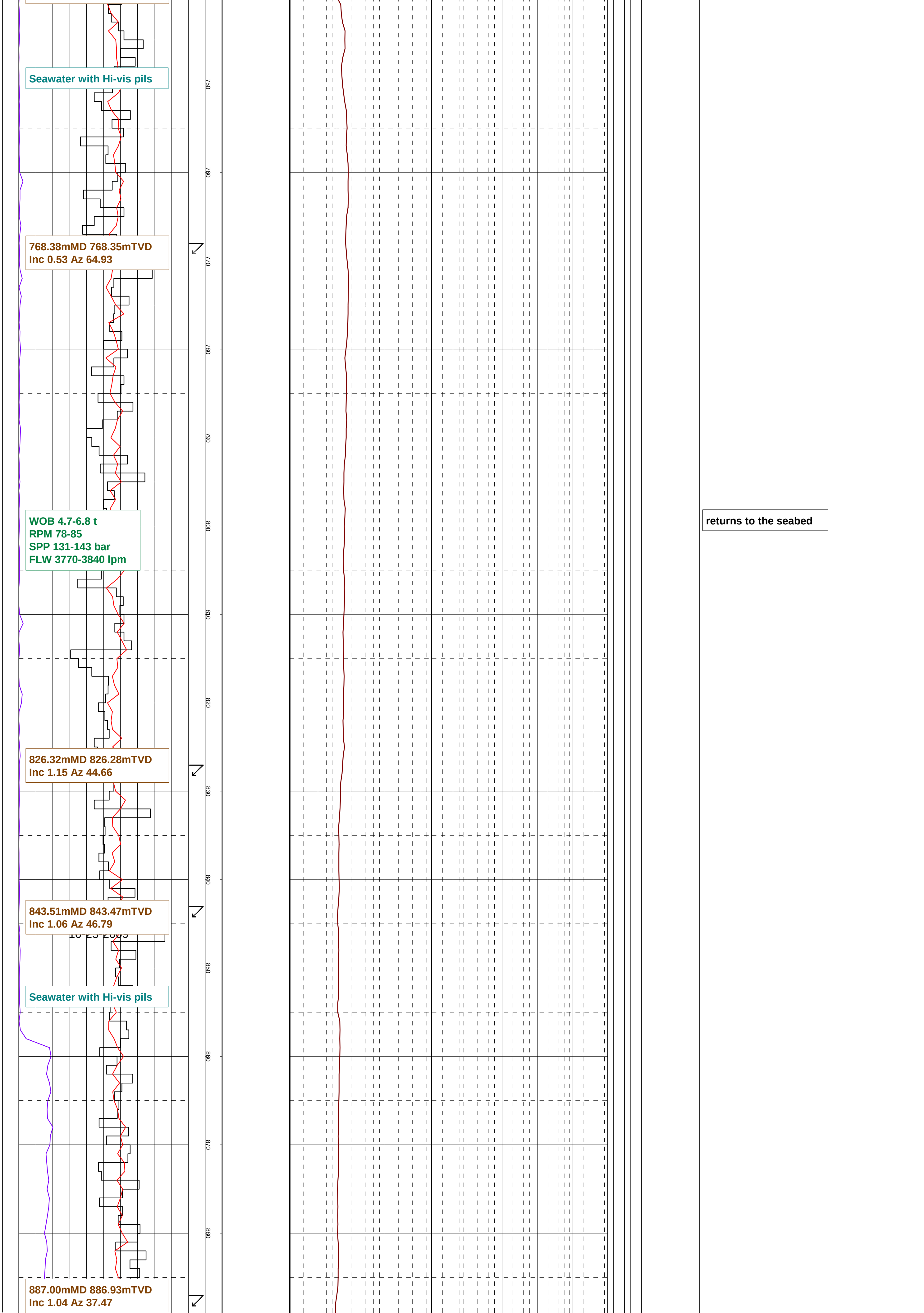
Generated by ALX Package

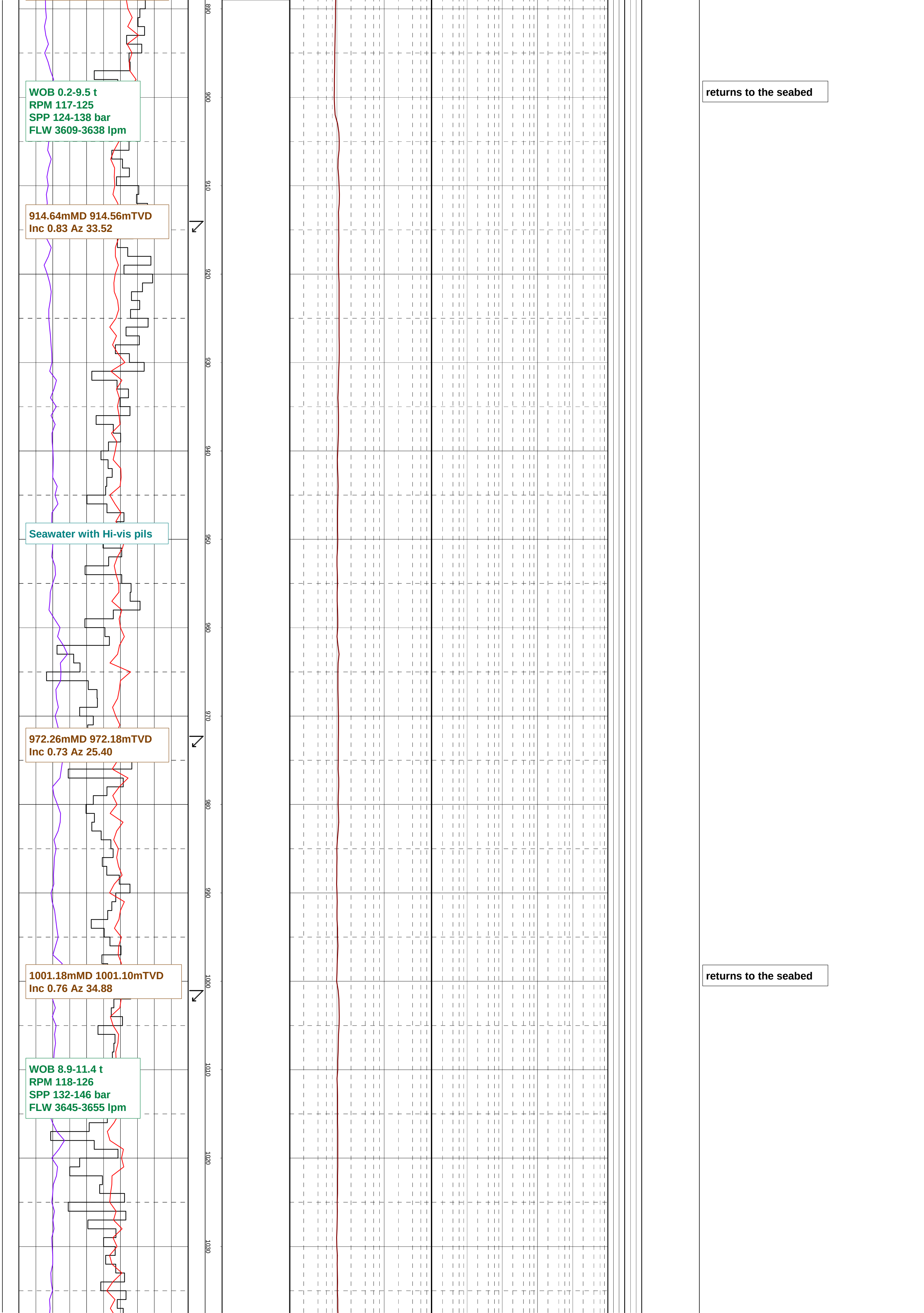
Clay-shale	Calcareous clay-shal	Dolomitic clay-shale	Silty clay-shale	Marl	Dolomitic marl	Silty marl
Dolomite	Calcareous dolomite	Argillaceous dolomit	Sandy dolomite	Argill and calc dolo	Argill and sandy dol	Argill and sandy dol
Limestone	Dolomitic limestone	Argillaceous limesto	Sandy limestone	Argill and dol lime	Argill and sandy lim	Sandy and dol limest
Silt	Fine sand	Medium sand	Coarse sand	Silstone	Fine sandstone	Medium sandstone
Coarse sandstone	Argillaceous siltsto	Argillaceous sandsto	Calcareous sandstone	Dolomitic sandstone	Calc and dol sandsto	Chert
Coal-lignite	Halite	Anhydrite	Gypsum	Volcanics	Granit	
Survey	FIT	Casing Top	Casing shoe	Mud gain	Mud loss	Core
Silty and dol marl	Silicified claystone	Anhydritic clay-shal	Saliferous clay-shal	Gypsiferous clay-sha	Bituminous clay-shal	Organic clay-shale

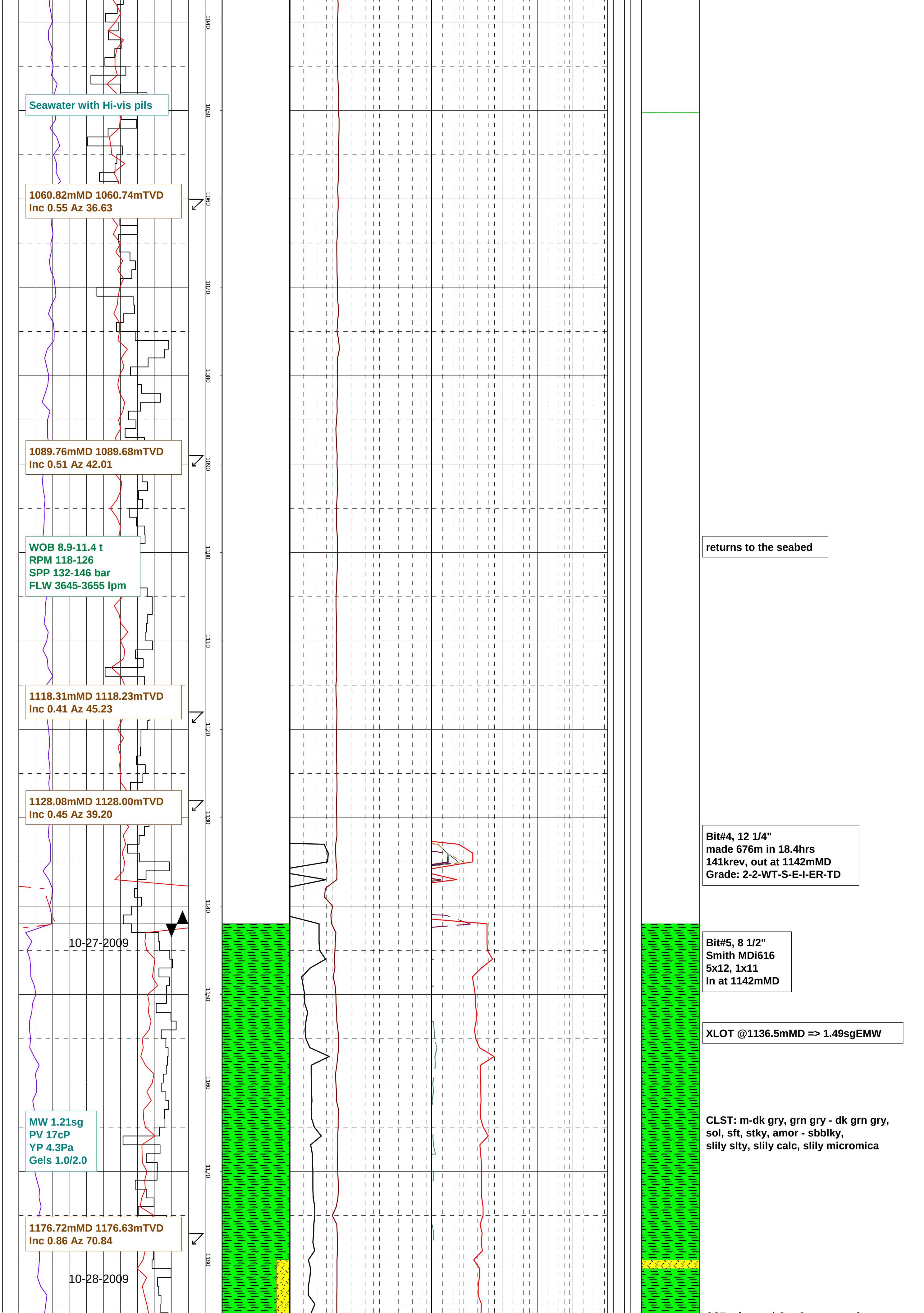



445.65mMD 445.64mTVD
Inc 1.44 Az 94.22
WOB 0.5-8.5 t
RPM 55-120
SPP 7-157 bar
FLW 890-4840 lpm
10-20-2009
Seawater with Hi-vis pils
477.14mMD 477.13mTVD
Inc 1.17 Az 84.35
WOB 0.3-1.7 t
RPM 107-120
SPP 83-95 bar
FLW 3090-3100 lpm
505.38mMD 505.36mTVD
Inc 0.86 Az 103.22
534.12mMD 534.10mTVD
Inc 0.74 Az 112.71
Seawater with Hi-vis pils
591.90mMD 591.88mTVD
Inc 0.52 Az 80.99
450
460
470
480
490
500
510
520
530
540
550
560
570
580
590
Bit#1, 17 ½" (22x 26x 36"HO)
made 56m in 17.1hrs
117krev, out at 464mMD
Grade: 3-3-WT-A-E-I-NO-TD
30" casing shoe
at 456.0mMD 456.0mTVD
Bit#2, 17 ½ SMITH
XR+VEC, 4x14
In at 464mMD, out at 466.0mMD
made 2.5m in 0.5hrs, 6krev
Grade: 1-1-NO-A-E-I-NO-TD
Bit#4, 12 1/4"
Security EQHCRC
2x18, 2x16
In at 466.0mMD
returns to the seabed

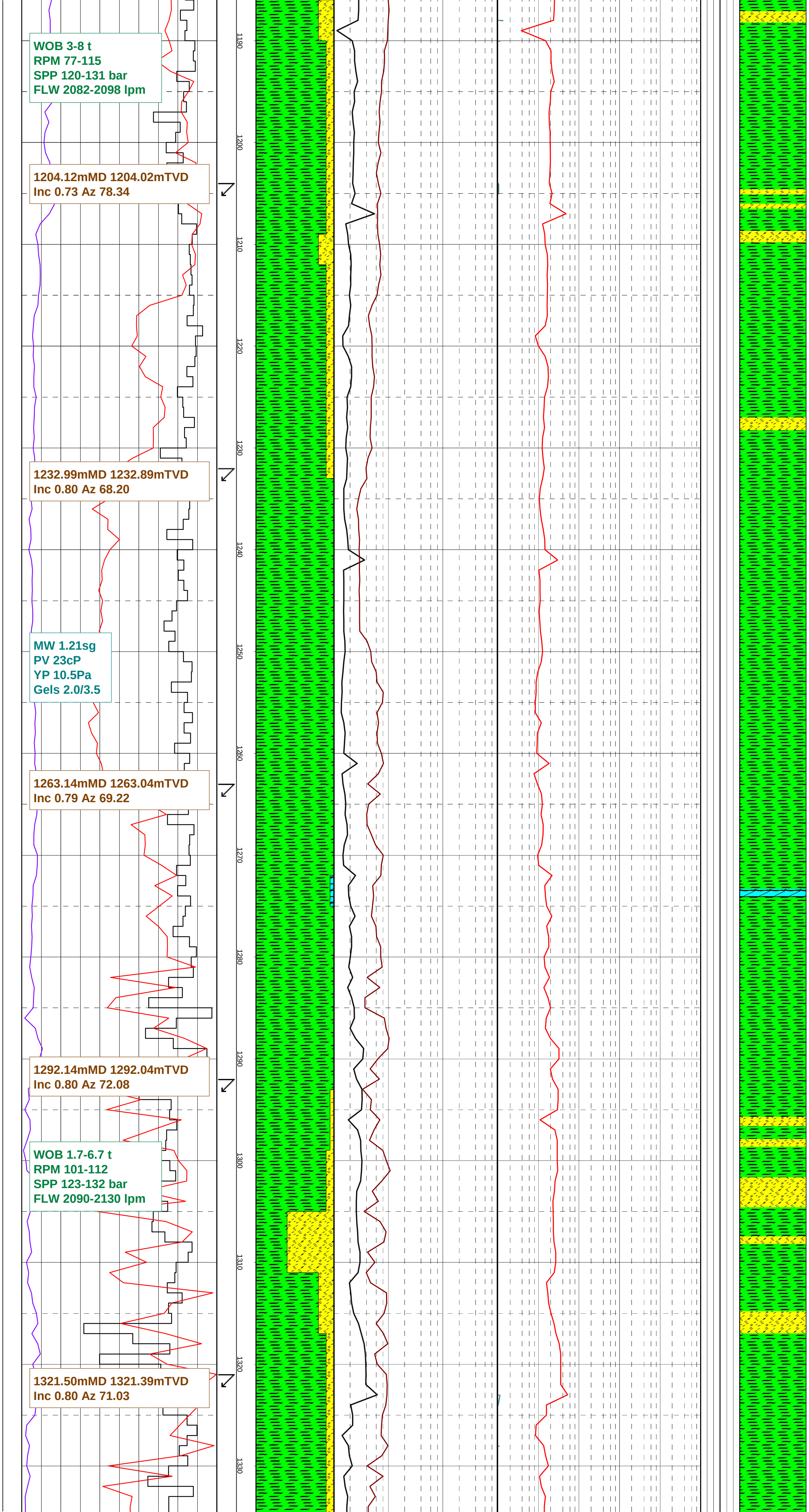

WOB 0.1-0.9 t
RPM 125-131
SPP 95-130 bar
FLW 3120-630 lpm
returns to the seabed
620.55mMD 620.52mTVD
Inc 0.70 Az 90.73
Seawater with Hi-vis pils
651.05mMD 651.02mTVD
Inc 0.43 Az 88.29
680.44mMD 680.41mTVD
Inc 0.40 Az 103.54
WOB 0.1-0.5 t
RPM 78-92
SPP 129-133 bar
FLW 3610-3650 lpm
returns to the seabed
713.17mMD 713.14mTVD
Inc 0.29 Az 56.84
738.99mMD 738.96mTVD
Inc 0.33 Az 71.76
600
610
620
630
640
650
660
670
680
690
700
710
720
730
740


Seawater with Hi-vis pils
768.38mMD 768.35mTVD
Inc 0.53 Az 64.93
WOB 4.7-6.8 t
RPM 78-85
SPP 131-143 bar
FLW 3770-3840 lpm
returns to the seabed
826.32mMD 826.28mTVD
Inc 1.15 Az 44.66
843.51mMD 843.47mTVD
Inc 1.06 Az 46.79
Seawater with Hi-vis pils
887.00mMD 886.93mTVD
Inc 1.04 Az 37.47
750
760
770
780
790
800
810
820
830
840
850
860
870
880


WOB 0.2-9.5 t
RPM 117-125
SPP 124-138 bar
FLW 3609-3638 lpm
914.64mMD 914.56mTVD
Inc 0.83 Az 33.52
Seawater with Hi-vis pils
972.26mMD 972.18mTVD
Inc 0.73 Az 25.40
1001.18mMD 1001.10mTVD
Inc 0.76 Az 34.88
WOB 8.9-11.4 t
RPM 118-126
SPP 132-146 bar
FLW 3645-3655 lpm
returns to the seabed
returns to the seabed
890
900
910
920
930
940
950
960
970
980
990
1000
1010
1020
1030



Seawater with Hi-vis pils
1060.82mMD 1060.74mTVD
Inc 0.55 Az 36.63
1089.76mMD 1089.68mTVD
Inc 0.51 Az 42.01
WOB 8.9-11.4 t
RPM 118-126
SPP 132-146 bar
FLW 3645-3655 lpm
1118.31mMD 1118.23mTVD
Inc 0.41 Az 45.23
1128.08mMD 1128.00mTVD
Inc 0.45 Az 39.20
10-27-2009
MW 1.21sg
PV 17cP
YP 4.3Pa
Gels 1.0/2.0
1176.72mMD 1176.63mTVD
Inc 0.86 Az 70.84
10-28-2009
1040
1050
1060
1070
1080
1090
1100
1110
1120
1130
1140
1150
1160
1170
1180
returns to the seabed
Bit#4, 12 1/4"
made 676m in 18.4hrs
141krev, out at 1142mMD
Grade: 2-2-WT-S-E-I-ER-TD
Bit#5, 8 1/2"
Smith MDi616
5x12, 1x11
In at 1142mMD
XLOT @1136.5mMD => 1.49sgEMW
CLST: m-dk gry, grn gry - dk grn gry,
sol, sft, stky, amor - sbblky,
slily slty, slily calc, slily micromica



SST: clr- trnsl Qtz Gr, m-crs, sbang- sbrndd, pr- mod srt, lse

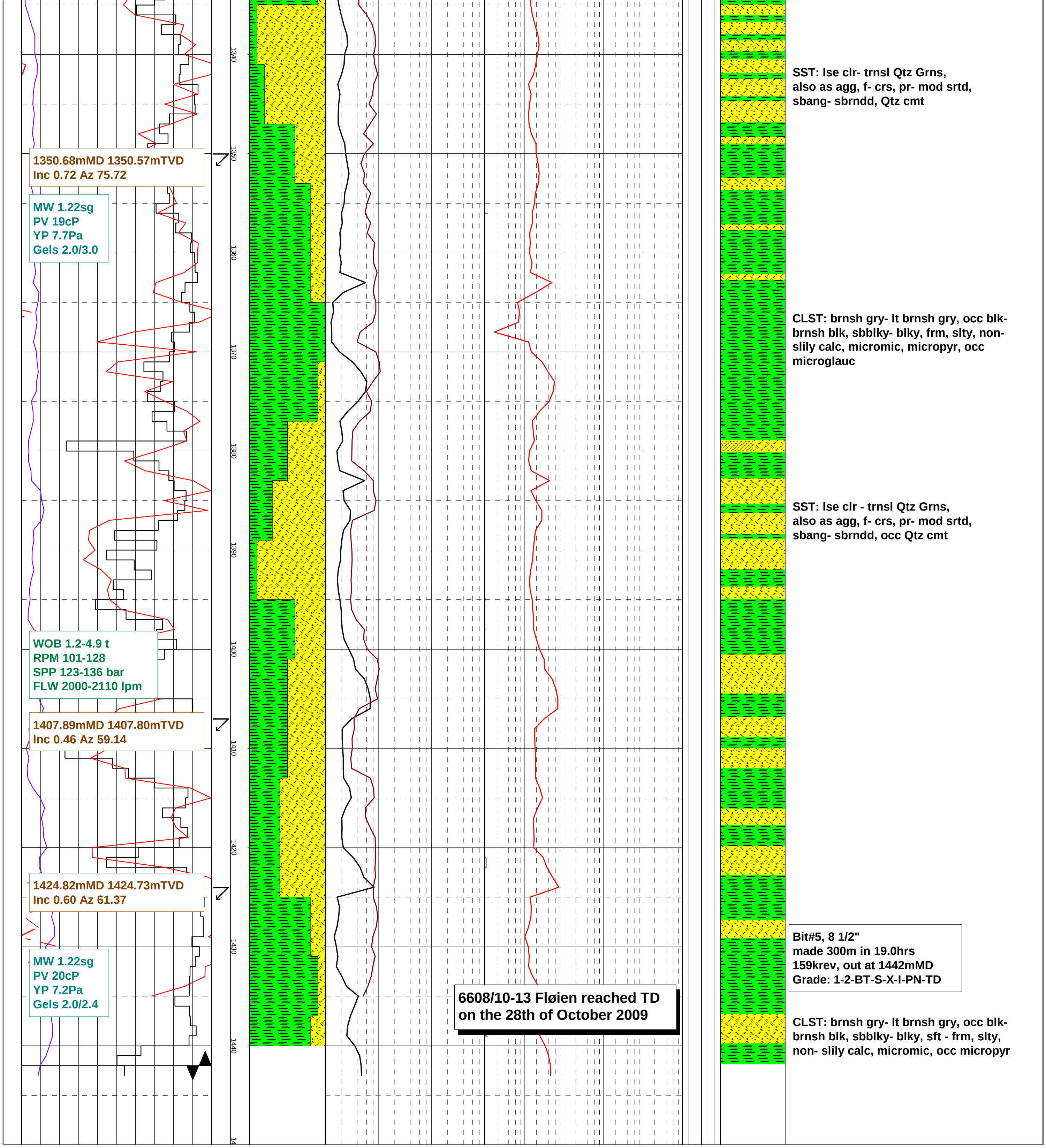
CLST: brn gry - brn blk, grn gry - dk gry, grn, sol, sft, stky, amor - sbblky, slty, slily calc, slily micropyr, Tr Glauc, Tr Pyr

CLST: brn gry - brn blk, grn gry - dk grn gry, sol, sft, stky, amor - sbblky, slty, slily calc, slily micropyr, Tr Glauc, Tr Pyr

LS: v lt gry, frm- hd, occ arg, occ slty, microxln

SST: lse clr - trnsl Qtz Grns, also as agg, f- crs, pr- mod srtd, sbang- sbrndd, Qtz cmt

CLST: blk- brnsh blk, also pa yelsh brn, sbblky- blky, frm, slty, non- slily calc, micromic, micropyr, occ microglauc


1350.68mMD 1350.57mTVD
Inc 0.72 Az 75.72
MW 1.22sg
PV 19cP
YP 7.7Pa
Gels 2.0/3.0
WOB 1.2-4.9 t
RPM 101-128
SPP 123-136 bar
FLW 2000-2110 lpm
1407.89mMD 1407.80mTVD
Inc 0.46 Az 59.14
1424.82mMD 1424.73mTVD
Inc 0.60 Az 61.37
MW 1.22sg
PV 20cP
YP 7.2Pa
Gels 2.0/2.4
6608/10-13 Fløien reached TD
on the 28th of October 2009
SST: lse clr- trnsl Qtz Grns, also as agg, f- crs, pr- mod srtd, sbang- sbrndd, Qtz cmt
CLST: brnsh gry- lt brnsh gry, occ blk- brnsh blk, sbblky- blky, frm, slty, non- slily calc, micromic, micropyr, occ microglauc
SST: lse clr - trnsl Qtz Grns, also as agg, f- crs, pr- mod srtd, sbang- sbrndd, occ Qtz cmt
Bit#5, 8 1/2"
made 300m in 19.0hrs
159krev, out at 1442mMD
Grade: 1-2-BT-S-X-I-PN-TD
CLST: brnsh gry- lt brnsh gry, occ blk- brnsh blk, sbblky- blky, sft - frm, slty, non- slily calc, micromic, occ micropyr

StatoilHydro

PRESSURE EVALUATION LOG

geoNEXT™

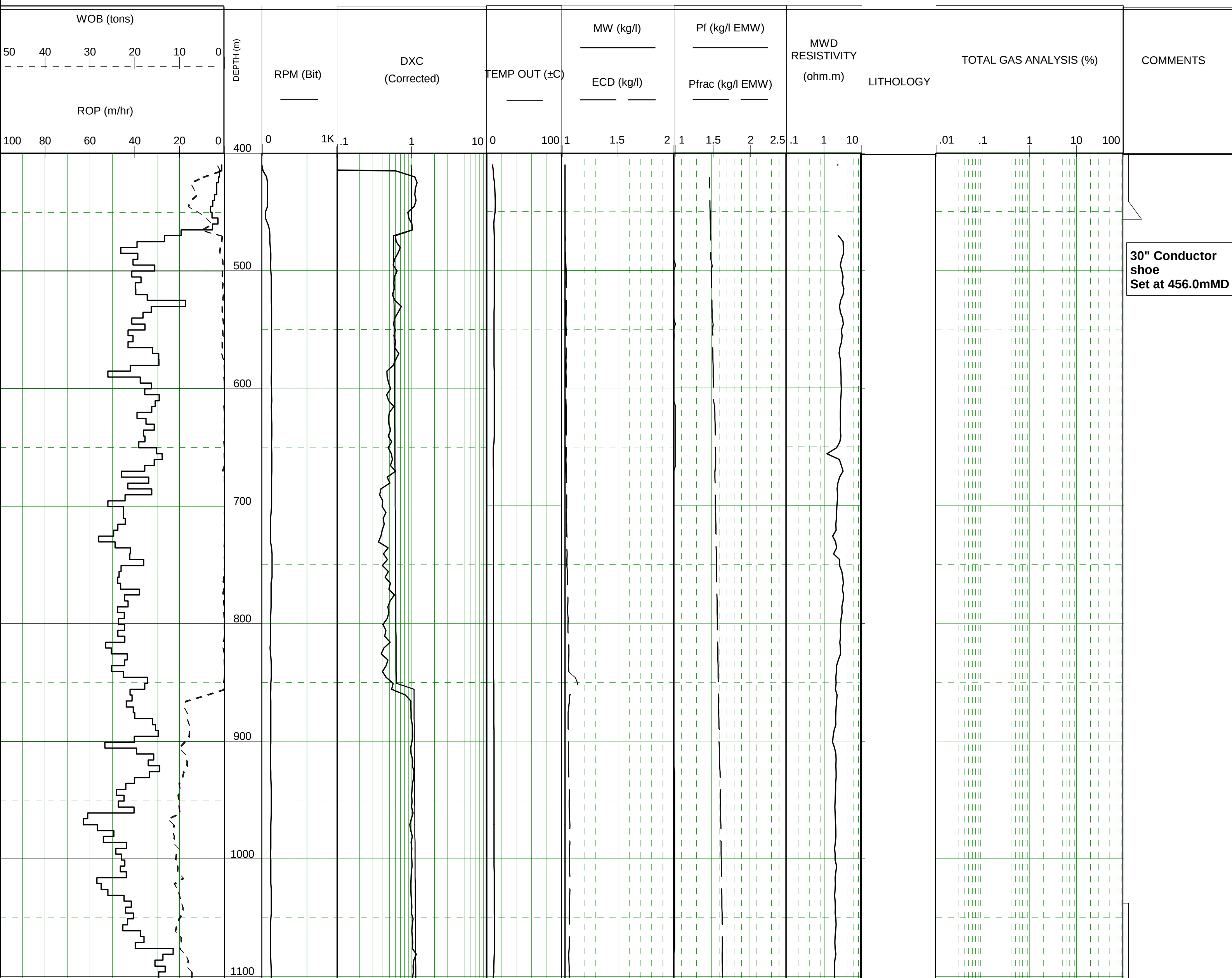
FROM :(m) 400 TO :(m) 1450 SCALE : 1/ 5000

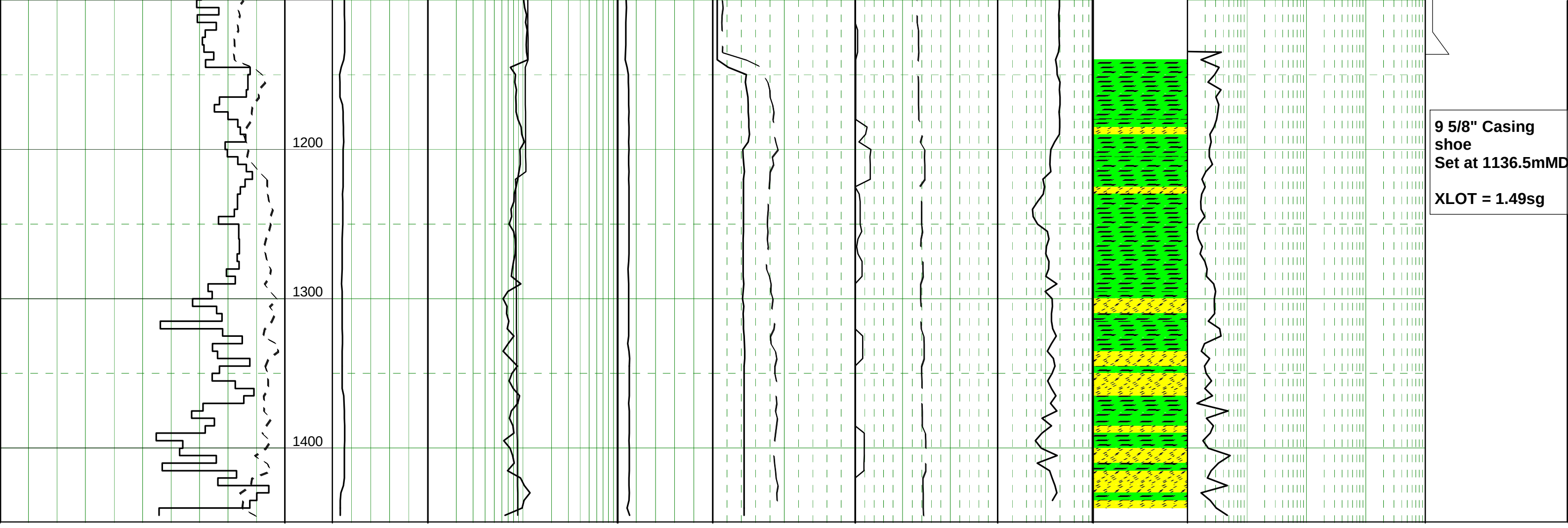
PEL1_N1Q.PR1

Well name	: 6608/10-13	Location	: 08deg 18' 10.2" E 66deg 00' 25.5" N	Rig Name	: OCEAN VANGUARD
Company name	: StatoilHydro	Country	: NORWAY	Rig Type	: Semisubmersible
Reference depth	: ROTARY TABLE (RKB)	Final TD (m)	: 1442.00	UTM coordinate N (m)	: 7321060
Water depth (m)	: 386	Final TVD (m)	: 1441.90	UTM coordinate E (m)	: 468366
Rot table - MSL (m)	: 22	Spud date	: 18th October 2009	Well type	: Exploration
Rot table - seabed (m)	: 408	Reached TD	: 28th October 2009	Area	: North Sea

Generated by ALX Package

Clay-shale	Calcareous clay-shal	Dolomitic clay-shale	Silty clay-shale	Sandy clay-shale	Calc and dol clay-sh	Marl	Dolomitic marl
Dolomite	Calcareous dolomite	Argillaceous dolomit	Sandy dolomite	Argill and calc dolo	Sandy and calc dolom	Argill and sandy dol	Cargneule
Limestone	Dolomitic limestone	Argillaceous limesto	Sandy limestone	Argill and dol lime	Argill and sandy lim	Sandy and dol limest	Chalk
Silt	Fine sand	Medium sand	Coarse sand	Silstone	Fine sandstone	Medium sandstone	Coarse sandstone
Argillaceous sandsto	Calcareous sandstone	Dolomitic sandstone	Calc and dol sandsto	Argill and calc sand	Argill and dol sands	Chert	Glauconitic sandston
Coal-lignite	Halite	Anhydrite	Gypsum	Sylvite	Volcanics	Granit	




1200
1300
1400
9 5/8" Casing
shoe
Set at 1136.5mMD
XLOT = 1.49sg